AWS Solutions Architecture Notes

Your Name

Date:

Contents

Chapter 1

Course Introduction

Pre-Introduction The purpose of these notes are two-fold.

- Prepare for this exam in the future.
- Gain a high level overview of the services offered by one of the main cloud providers.
- Additionally, could be useful for practicing Terraform.

Chapter 2

Simple AWS Architecture

Figure 2.1: Simple EC2 to S3 Architecture

Chapter 3

Identity & Federation

3.1 IAM - What should you know by now

- Users: Long term credentials
- Groups
- Roles: short-term credentials, uses STS
- EC2 Instance Roles: Uses the EC2 metadata service. One role of a time per instance
- Service Roles: API Gateways, CodeDeploy etc
- Cross Account roles
- Policies
- AWS Managed
- Customer Managed
- Inline Policies
- Resource Based Policies (S3 Bucket, SQS queues, etc......)

3.1.1 IAM Policies Deep Dive

- Anatomy of a policy: JSON doc with Effect, Action, Resource, Conditions, Policy Variables
- Explicit DENY has precedence over ALLOW
- Best practice: use least privilege for maximum security
- Access Advisor: See permissions granted and when last accessed
- Access Analyser: Analyse resources that are shared with external entity
- Navigate Examples at:

3.1.2 IAM AWS Managed Policies - Administrator Access example

Listing 3.1: Sample JSON Data

```
{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Action": "*",
"Resource": "*"
}
]
}
```

3.1.3 IAM Policies Conditions

3.1.4 IAM Policies Variables and Tags

3.1.5 IAM Roles vs Resource Based Policies

- Attach a policy to a resource (example: S3 bucket policy) versus attaching of a using a role as a proxy
- When you assume a role (user, application or service), you give up your original permissions and take the permissions assigned to the role.
- When using a resource-based policy, the principal doesn't have to give up any permissions
- Example: User in account A needs to scan a DynamoDB table in Account A and dump it in an S3 bucket in Account B

3.1.6 IAM Permissions Boundaries

- IAM permission boundaries are supported for users and roles (not groups)
- Advanced feature to use a managed policy to set the maximum permissions an IAM entity can get
- Can be used in combinations of AWS organisations SCP

3.1.7 Use cases

- Delegate responsibilities to non administrators within their permission boundaries, for example create create new IAM users
- Allow developers to self-assign policies and managed their own permissions while making sure they can't 'escalate' their privileges (i.e make themselves admin)
- Useful to restrict one specific user (instead of a whole account using Organisations and SCP)

3.1.8 IAM Access Analyser

- Find out which resources are shared externally
- S3 Buckets
- IAM Roles
- KMS Keys
- Lambda Functions and Layers
- SQS queues
- Secrets Manager Secrets
- Define Zone ofTrust = AWS Account or AWS Organisation
- Access outside of zone trusts =¿ findings

3.1.9 IAM Access Analyser

- IAM Access Analyser Policy Validation
- Validates your policy against IAM policy grammar and best practices
- General warnings, security warnings errors suggestions
- Provides actionable recommendations
- IAM Access Analyser Policy Generation
- Generates IAM policy based on access activity
- CloudTrail logs is reviewed to generate the policy with the fine-grained permissions and the appropriate Actions and Services
- Reviews CloudTrail logs for up to 90 days

3.2 STS

3.2.1 Using STS to Assume a Role

- Define an IAM Role within your account or cross-account
- Define which principals can access this IAM role
- Using AWS STS (Security Token Service) to retrieve credentials and impersonate the IAM role you have access to (AssumeRole API)
- Temporary credentials can be valid between 15 to 12 hours.

3.2.2 Assuming a Role with STS

- Provide access for an IAM user in one AWS account that you own to access resources in another account that you own.
- Provide access to IAM users in AWS accounts owned by third parties
- Provide access for services offered by AWS to AWS resources
- Provide access for externally authenticated users (identity federation)
- Ability to revoke active sessions and credentials for a role (by adding a policy using a time statement - AWSRevokeOlderSessions)

When you assume a role (user, application or service), you give up your original permissions and take the permissions assigned to the role.

3.2.3 Providing Access to an IAM User in Yours or Another AWS Account That You Own

- You can grant your IAM users permissions to switch to roles within your AWS account or to roles defined in other AWS accounts that you own.
- Benefits:
- You must explicitly grant your users permission to assume the role
- Your users must actively switch to the role using the AWS management console or assume the role using the AWS CLI or AWS API
- You can add multi-factor authentication (MFA) protection to the role so that only users who sign in with an MFA device can assume the role.
- Least Privilege + auditing using CloudTrail

3.2.4 Providing Access to AWS Accounts Owned by Third Parties

- Zone of trust = accounts, organisations that you own
- Outside Zone ofTrust = 3rd parties
- Use IAM Access Analyser to find out which resources are exposed
- For granting access to a thrid party:
- The third part AWS account ID
- An External ID (secret between you and the third party)
- To uniquely associate with the role between you and 3rd party
- Must be provided when defining the trust and when assuming the role
- Must be chosen by the third party
- Define permissions in the IAM policy

Session Tags in STS

- Tags that you pass when you assume an IAM Role or federate user in STS
- aws:PrincipalTag Condition
- Compares the tags attached to the principal making the request with the tag you specified in the policy.
- Example: Allow a principal to pass session tags only if the principal making the reqeust has the specified tags.

STS Important APIs

- AssumeRole: access a role within your account or cross-account
- AssumeRoleWithSAML: return credentials for users logged with SAML
- AssumeRoleWithWebIdentity: return creds for users logged with an IdP
- Example providers include Amazon Cognito, Login with Amazon, Facebook, Google or any OpenID Connect-compatible identity provider.
- AWS Recommends using Cognito
- GetSessionToken: for MFA from a user or AWS account root user
- GetFederationToken: obtain temporary creds for a federated user, usually a proxy app that will give the creds for a federated user, usually a proxy app that will give the creds to a distributed app inside a corporate network.

3.2.5 Identity Federation in AWS

- Give users outside of AWS permissions to access AWS resources in your account
- You don't need to create IAM Users (user management is outside AWS)
- Use cases:
  - A corporate has its own identity system (e.g Active Directory)
  - Web / Mobile application that needs access to AWS resources
  - Identity Federation can have many flavours:
  - SAML 2.0
  - Custom Identity Broker
  - Web Identity Federation With(out) Amazon Cognito
  - IAM Identity Centre

SAML 2.0 Federation

- Security Assertion Markup Language 2.0 (SAML 2.0)
- Open standard used by many identity providers (e.g ADFS)
- Supports integration with Microsoft Active Directory Federations Services (ADFS)
- Or any SAML 2.0 - compatible IdPs with AWS
- Access the AWS Console, AWS CLI or AWS API using temporary credentials
- No need to create IAM Users for each of your employees
- Need to setup a trust between AWS IAM and SAML 2.0 Identity Provider (both ways)
- Under-the-hood: Uses the STS API AssumeRoleWithSAML
- SAML 2.0 Federation is the ”old way”, IAM Identity Center Federation is the new managed and simpler way

SAML 2.0 Federation - AWS API Access

SAML 2.0 Federation - AWS Console Access

SAML 2.0 Federation - Active Directory DS (ADFS)

Custom Identity Broker Application

- Use only if identity provider is NOT compatible with SAML 2.0
- The identity broker is NOT compatible with SAML 2.0
- The identity broker must determine the appropriate IAM Role
- Uses the STS API AssumeRole or GetFederationToken

Web Identity Federation - Without Cognito

- Not recommended by AWS - use cognito instead

Web Identity Federation - With Cognito

- Preferred over for Web Identity Federation
- Create IAM Roles using Cognito with the least privilege needed
- Build trust between the OIDC IdP and AWS
- Cognito benefits:
- Supports anonymous users
- Supports MFA
- Data Synchronisation
- Cognito replaces a Token Vending Machine (TVM)

Web Identity Federation - IAM Policy

- After being authenticated with Web Identity Federation, you can identify the user with an IAM policy variable
- Example:

3.2.6 AWS Directory Services (AD)?

- Found on any Windows Server with AD Domain Services
- Database of objects: User Accounts, Computers, Printers, File Shares, Security Groups
- Centralised security management, create account, assign permissions
- Objects are organised in trees
- A group of trees

What is ADFS (AS Federation Services)?

- ADFS provides Single Sign-On across applications
- SAML across 3rd party: AWS Console, Dropbox, Officer365, etc.....

AWS Directory Services

- AWS Managed Microsoft AD
- Create your own AD in AWS, manage users, locally supports MFA
- Establish ”trust” connection with your own on premises
- AD Connector
- Directory Gateway (proxy) to redirect to on-premises AD, supports MFA
- Users are managed on the on-premise AD
- Simple AD
- AD-compatible managed directory on AWS
- Cannot be joined with on-premise AD

AWS Directory Services and AWS Managed Microsoft AD

- Managed Service: Microsoft AD in your AWS VPC
- EC2 Windows Instances:
- EC2 Windows instances can join the domain and run traditional AD applications (sharepoint, etc)
- Seamlessly Domain Join Amazon EC2 Instances from Multiple Accounts and VPCs
- Integrations
- RDS for SQL Server, AWS Workspaces, Quicksight.......
- AWS SSO to provide access to third party applications
- Standalone repository in AWS or jointed to on-premises AD
- Multi AZ deployment of AD in 2 AZ, of DC (Domain Controllers) can be increased for scaling
- Automated backups
- Automated Multi-Region replication of your directory

AWS Microsoft Managed AD - Integrations

Connect to on-premise AD

- Ability to connect your on-premise Active Directory to AWS Managed Microsoft AD
- Must establish a Direct Connection (DX) or VPN connection
- Can setup three kinds of forest trust
- One-way trust: AWS -¿ On-Premise
- One-way trust: On-Premise -¿ AWS
- Two-way forest: trust - AWS -¿ On-Premise
- Forest trust is different than synchronisation (replication is not supported)

Solution Architecture: Active Directory Replication

- You may want to create a replica of your AD on EC2 in the cloud to minimise latency of in case DX or VPN goes down
- Establish trust between the AWS Manged Microsoft AD and EC2

AWS Directory Services AD Connector

- AD Connector is a directory gateway to redirect director requests to your on premises Microsoft Active Directory
- No caching capability
- Managed users solely on-premise, no possibility of setting up a trust
- VPN or Direct Connect
- Doesn't work with SQL Server, doesn't do seamless joining, can't share director.

AWS Directory Services Simple AD

- Simple AD is an inexpensive Active Directory-compatible service with the common directory features
- Supports joining EC2 instances, manage users and groups
- Does not support MFA, RDS SQL server, AWS SSO
- Small: 500 users, large: 5000 users
- Powered by Samba 4, compatible with Microsoft AD
- lower cost, low scale, basic AD compatible or LDAP compatibility
- No trust relationship

3.2.7 AWS Orgnisations

AWS Organisations - OrgnizationAccountAccessRole

- IAM role which grants full administrator permissions in the Member account to the Management account
- Used to perform admin tasks in the Member accounts (e.g - creating IAM users)
- Could be assumed by IAM users in the Management account
- Automatically added to all new Member account created with AWS organisations
- Must be created manually if you invite an existing Member account

Multi Account Strategies

- Create accounts per department, per cost centre, per dev / test / prod, based on regulatory restrictions (using SCP), for better resource isolation (ex:VPC), to have separate per-account service limits isolated account for logging.
- Multi Account vs. On Account MultiVPC
- Use tagging standards for billing purposes
- Enable CloudTrail on all accounts, send logs to central S3 account
- Send CloudWatch logs to central logging account
- Strategy to create an account for security

Orgnisational Units (OU) - Examples

AWS Organisation - Feature Modes

- Consolidated billing features:
- Consolidated Billing across all accounts - single payment method
- Pricing benefits from aggregated usage (volume discount for EC2, S3.....)

All Features (Default)

- Includes consolidated billing features, SCP
- Invited accounts must approve enabling all features
- Ability to apply an SCP to prevent member accounts from leaving the org
- Can’t switch back to Consolidated Billing Features only

AWS Organisations - Reserved Instances

- For billing purposes, the consolidated billing features of AWS organisations treats all the accounts in the organisation as one account.
- The means that all accounts in the organisation can receive the hourly cost benefit of Reserved Instances that are purchased by any other account.
- The payer account (Management account) of an organisation can turn off Reserved Instance (RI) discount and Saving Plans discount sharing for any accounts in that organisation, including the payer account.
- This means that RIs and Saving Plans discounts aren't shared between any accounts that have sharing turned off.
- To share an RI or Savings Plans discount with an account, both accounts must have sharing turned on.

AWS Organisation - Moving Accounts

- Remove the member account from the AWS organisation
- Send an invite to the member account from the AWS organisation
- Accept the invite to the new Organisation from the member account.

Service Control Policies (SCP)

- Define allowlist or blocklist IAM actions
- Applied at the OU or Account level
- Does not apply to the Management Account
- SCP is applied to all the Users and Roles in the account, including Root user
- The SCP does not affect Service-linked roles
- Service-linked roles enable other AWS services to integrate with AWS organisations and can't be restricted by SCP's
- SCP must have an explicit Allow from the root of each OU in the direct path to the target account (does not allow anything by default)
- Use cases:
- Restrict access to certain services (for example: can't use EMR)
- Enforce PCI compliance by explicitly disabling services.

SCP Hierarchy

- Management Account - Can do anything (no SCP apply)
- Account A
- Can do anything
- EXCEPT S3 (explicit Deny from Sandbox OU)
- EXCEPT EC2 (explicit deny)
- Account B and C
- Can do anything
- EXCEPT S3 (explicit Deny from Sandbox OU)
- Account D
- Can access EC2
- Prod OU and Account E and F

SCP Examples - Blocklist and Allowlist strategies

IAM Policy Evaluation Logic

Restricting Tags with IAM policies

- You can restrict specific tags on AWS resources
- Using the aws:TagKeys Condition Key
- Validate the Tag Keys attached to a resource against the Tag Keys in the IAM Policy
- Example: Allow IAM users to create EBS Volumes only if it has the ”Env” and ”CostCenter” Tags
- Use either ForAllValues (must have all keys) or ForAnyValue (must have any of these keys at a minimum)

Using SCP to restrict creating resources without appropriate tags

- Prevent IAM Users/Roles in the affected member accounts from creating resources if they don’t have a specific Tag

AWS Organisations - Tag Policies

- Helps you standardise tags across resources in an AWS organisation
- Ensure consistent tags, audit tagged resources, maintain proper resources categorisation
- You define Tag keys and their allowed values
- Helps with AWS Cost Allocation Tags and Attribute-based Access Control
- Prevent any non-compliant tagging operations on specified services and resources
- Generate a report that lists all tagged/ non compliant resources
- Use Amazon EventBridge to monitor non-compliant tags

AWS Organisation - AI Service Opt-out Policies

- Certain AWS AI services may use your content for continuous improvement of Amazon AI/ML services
- Example: Amazon Lex, Amazon Comprehend, Amazon Polly.........
- You can opt-out of having your content stored or used by AWS AI services
- Create an Opt-out Policy that enforces this setting across all Member accounts and AWS Regions
- You can opt-out all AI services or selected services
- Can be attached to Organisation Root, specific OU or individual Member account

AWS Organisations - Backup policies

- AWS Backup enables you to create Backup Plans that define how to backup your AWS resources
- JSON Documents that define backup plans across an AWS Organisation
- Gives you granular control over backup up your resources (e.g backup frequency, time window, backup region,.....)
- Can be attached to Organisation Root, specific OU or individual Member account
- Immutable backup plans appear in Member accounts (view only)

Using SCP to Deny a Region aws:ReqeustRegion

3.2.8 AWS IAM Identity Center

AWS IAM Identity Center -successor to AWS Single Sign-On-

- One login (single sign-on) for all your

- AWS accounts in AWS organisations
- Business cloud applications (e.g, Salesforce, Box, Microsoft 365)
- SAML2.0 enabled applications
- EC2 windows instances
- Identity Providers
  - Built-in identity store in IAM identity center
  - 3rd party Active Directory (AD), OneLogin, Okta
- AWS IAM Identity Center - Login Flow

AWS IAM Identity Center

AWS IAM Identity Center - Fine-grained Permissions and Assignments

- Multi-Account Permissions
- Manage access across AWS accounts in your AWS Organisation
- Permission Sets - a collection of one or more IAM policies assigned to users and groups to define AWS access
- Application Assignments
- SSO access to many SAML 2.0 business applications (Salesforce, Box, Microsoft 365)
- Provide required URLs, certificates and metadata
- Attribute-Based Access Control (ABAC)
- Fine-grained permissions based on users' attributes stored in IAM identity center identity store
- Example: Cost center, title, locale
- Use case: Define permission once, then modify AWS access by changing the attributes

AWS Control Tower

- Easy way to setup and govern a secure and compliant multi-account AWS environment based on best practices
- Benefits:
  - Automate the set up of your environment in a few clicks
  - Automate ongoing policy management using guardrails
  - Detect policy violations and remediate them
  - Monitor compliance through an interactive dashboard
  - AWS ControlTower runs on top AWS Organisations:
  - It automatically sets up AWS Organisations to organise accounts and implement SCPs (Service Control Policies)

AWS Controller Tower - Account Factory

- Automates account provisioning and deployments
- Enables you to create pre-approved baselines and configuration options for AWS accounts in your organisation (e.g VPC default configuration, subnets, region, ...)
- Uses AWS service catalog to provision new AWS accounts

AWS Control Tower - Detect and Remediate Policy Violations

- Guardrail
- Provide ongoing governance for your Control Tower environment (AWS Accounts)
- Preventive - using SCPs (e.g Disallow Creation of Access Keys for the Root User)
- Detective - users AWS Config (e.g Detect Whether MFA for the Root User is Enabled)
- Example: identify non-compliant resources (e.g, untagged resources)

AWS Control Tower - Guardrails Levels

- Mandatory
- Automatically enabled and enforced by AWS control tower
- Example: Disallow public Read access to the Log Archive account
- Strongly Recommended
- Based on AWS best practices (optional)
- Example: Enable encryption for EBS volumes attached to EC2 instances
- Elective
- Commonly used by enterprises (optional)
- Examples: Disallow delete actions without MFA in S3 buckets

3.3 AWS Resource Access Manager (RAM)

- Share AWS resources that you own with other AWS accounts
- Share with any account or within your Organisation
- Avoid resource duplication!
- VPC Subnets
- Allow to have all the resources launched in the same subnets
- Must be from the same AWS organisations

- Cannot share security groups and defaultVPC
- Participants can manage their own resources in there
- Participants can't view, modify, delete resources that belong to other participants or the owner
- AWS Transit Gateway
- Route 53 (Resolver Rules, DNS Firewall Rule Groups)
- License Manager Configurations

AWS Resource Access Manager (RAM)

- Aurora DB Clusters
- ACM Private Certificate Authority
- CodeBuild Project
- EC2 (Dedicated Hosts, Capacity Reservation)
- AWS Glue (Catalog, Database, Table)
- AWS Network Firewall Policies
- AWS Resources groups
- Systems Manager Incident Manager (Contacts, Response Plans)
- AWS Outposts (Outpost, Site)

Resource Access Manager - VPC example

- Each account......
- Is responsible for it own resources
- Cannot view modify or delete other resources in other accounts
- Network is shared so.....
- Anything deployed in the VPC can talk to other resources in the VPC
- Applications are accessed easily across accounts, using a private IP
- Security groups from other accounts can be referenced for maximum security
- Use cases
- Applications within the same trust boundaries
- Applications with a high degree of interconnectivity

Resource Access Manager Managed Prefix List

- A set of one or more CIDR blocks
- Makes it easier to configure and maintain Security Groups and Route Tables
- Customer-Managed Prefix List
- Set of CIDRs that you define and manage by you
- Can be shared with other AWS accounts or AWS Organisation
- Modify to update many security groups at once
- AWS-Managed Prefix List
- Set of CIDRs for AWS services
- You can't create, modify, share or delete them.

Resource Access Manger Route 53 Outbound Resolver

- Helps you scale forwarding rules to your DNS in case you have multiple accounts and VPC

3.4 Summary of Identity and Federation

- Users and Accounts all in AWS
- AWS Organisations
- AWS Control Tower to setup secure and compliant multi-account AWS environment (best practices)
- Federation with SAML
- Federation without SAML with a custom IdP (GetFederationToken)
- IAM Identity Center to connect to multiple AWS Accounts (Organisation) and SAML apps
- Web Identity Federation (not recommended)
- Cognito for most web and mobile applications (has anonymous mode, MFA)
- AWS Directory Service:
  - Managed Microsoft AD - standalone or setup trust AD with on-premises, has MFA, seamless joins, RDS integration
  - AD Connector - proxy requests to on-premises
  - Simple AD - standalone and cheap AD-compatible with no MFA, no advanced capabilities
- AWS RAM to share resource (example VPC subnets)

Chapter 4

Security

4.1 CloudTrail

- Provides governance, compliance and audit for your AWS Account
- CloudTrail is enabled by default.
- Get a history of events / API calls made within your AWS Account by;
- Console
- SDK
- CLI
- AWS Services
- Can put logs from CloudTrail into CloudWatch Logs or S3.
- A trail can be applied to All Regions (default) or a single Region.
- If a resource is deleted in AWS, investigate CloudTrail first
- Management Events:
- Operations that are performed on resources in your AWS account
  - Examples:
  - Configuring security (IAM AttachRolePolicy)
  - Configuring rules for routing data (Amazon EC2 CreateSubnet)
  - Setting up logging (AWS CloudTrail CreateTrail)
  - By default, trails are configured to log management events.
  - Can separate Read Events (that don't modify resources) from Write Events (that may modify resources)
- Data Events:
  - By default, data events are not logged (because high volume operations)
  - Amazon S3 object-level activity (ex: GetObject, DeleteObject, PutObject): can separate Read and Write Events
  - AWS Lambda function execution activity (the Invoke API)
- CloudTrail Insights
  - Enable CloudTrail Insights to detect unusual activity in your account:
  - Inaccurate resource provisioning
  - Hitting service limits
  - Burst of AWS IAM actions
  - Gaps in periodic maintenance activity
  - CloudTrail Insights analyses normal management events to create baseline
  - And then continuously analyses write events to detect unusual patterns

  - Anomalies appear in the CLoudTrail console
  - Event is sent to Amazon S3
  - An EventBridge event is generated (for automation needs)

- CloudTrail Events Retention
  - Events are stored for 90 days in CloudTrail
  - To Keep events beyond this period, log them to S3 and use Athena

4.2 CloudTrail - EventBridge Integration

4.3 CloudTrail - SA Pro

4.3.1 S3 Enhancements

:

- Enable Versioning
- MFA Delete Protection
- S3 Lifecycle Policy (S3 IA, Glacier)
- S3 Object Lock
- SSE-S3 or SSE-KMS encryption
- Feature to perform CloudTrail Log File Integrity validation (SHA-256 for hashing and signing)

4.3.2 Observations

- The S3 Bucket policy is necessary for cross-account delivery
- If Account A wants to access its CloudTrail files:
- Option 1: Create a cross-account role and assume the role
- Option 2: edit the bucket policy

- Log filter metrics can be used to detect a high level of API happening - Ex: Count occurrences of EC2 TerminateInstances API - Ex: Count of API calls per user - Ex: Detect high level of Denied API calls

- The Organisational Trail is created in the management account.

4.3.3 CloudTrail: How to react to events the fastest?

- Overall, CloudTrail may take up to 15 minutes to deliver events
- EventBridge:
- Can be triggered for any API call in CloudTrail
- The fastest, most reactive way.
- CloudTrail Delivery in CloudWatch Logs:
- Events are streamed
- Can perform a metric filter to analyse occurrences and detect anomalies
- CloudTrail Delivery in S3
- Events are delivered every 5 minutes
- Possibility of analysing logs integrity, deliver cross account, long-term storage.

4.4 KMS

4.4.1 AWS KMS (Key Management Service)

- Anytime you hear ”encryption” for an AWS service, it is most likely KMS
- Easy way to control access to your data, AWS manages keys for us.
- Fully integrated with IAM for authorisation
- Seamlessly integrated into:
- Amazon EBS: encrypt volumes
- Amazon S3: Server-side encryption of objects
- Amazon Redshift: Encryption of data
- Amazon RDS: Encryption of data - Etc.....
- Can also use the CLI / SDK

4.4.2 KMS - KMS - Key Types

- Symmetric (AES-256 keys)
- First offering of KMS, single encryption key that is used to Encrypt and Decrypt
- AWS services that are integrated with KMS use Symmetric KMS keys
- Necessary for envelope encryption
- You never get access to the KMS key unencrypted (must call KMS API to use)

- Asymmetric (RSA and ECC key pairs)
- Public (Encrypt) and Private Key (Decrypt) pair
- Used for Encrypt/Decrypt or Sign/Verify operations
- The public key is downloadable but you can't access the Private Key unencrypted
- Use case: encryption outside of AWS by users who can't call the KMS API

4.4.3 Types of KMS Keys

- Customer Managed Keys
- Create, manage and use. Can enable or disable
- Possibility of rotation policy (new key generated every year, old key preserved)
- Can add a Key Policy (resource policy) and audit in CloudTrail
- Leverage for envelope encryption.
- AWS Managed Keys
- Used by AWS service (aws/s3, aws/ebs, aws/redshift)
- Managed by AWS (automatically rotated every 1 year)
- View Key Policy and audit in CloudTrail
- AWS Owned Keys
- Create and managed by AWS, use by some AWS services to protect your resources
- Used in multiple AWS accounts but they are no in your AWS account
- You can't view, use, track or audit.

Types of KMS keys

4.4.4 KMS Key Material Origin

- Identifies the source of the key material in the KMS key - Can't be changed after creation
- KMS(AWS$_KMS)-default-AWSKMScreatedandmanagedthekeymaterialinitsownkeystore$
- External (EXTERNAL) - You import the key material into the KMS key - You're responsible for securing and managing this key material outside of AWS
- Custom Key Store (AWS$_CLOUDHSM)-AWSKMScreatesthekeymaterialinacustomkeystore(CloudHSM$

4.4.5 KMS Key Source - Custom Key Store (CloudHSM)

- Integrate KMS with CloudHSM cluster as a Custom Key Store
- Key materials are stored in a CloudHSM cluster that you own and manage
- The cryptographic operations are performed in the HSMs
- Use cases:
- You need direct control over the HSMs
- KMS Keys needs to be stored in a dedicated HSMs

4.4.6 KMS Key Store - External

- Import your own key material into KMS key, Bring Your Own Key (BYOK)
- You're responsible for key material's security, availability and durability outside of AWS
- Supports both Symmetric and Asymmetric KMS keys
- Can't be used with Custom Key Store (CloudHSM)
- Manually rotate your KMS key (Automatic and On-demand Key Rotation are NOT supported)

4.4.7 KMS Multi-Region Keys

- A set of identical KMS keys in different AWS Regions that can be used interchangeably (same KMS key in multiple Regions)
- Encrypt in one Region and decrypt in other Regions (No need to re-encrypt or making cross-Region API calls)
- Multi-Region keys have the same key ID, key material, automatic rotation........
- KMS - Multi-Region are NOT global (Primary + Replicas)
- Each Multi-Region key is manged independently
- Only one primary key at a time, can promote replicas into their own primary
- Use cases: Disaster Recovery, Global Data Management (e.g DynamoDB, Global Tables), Active-Active Applications that span multiple Regions, Distributed Signing applicatoins.

4.5 Parameter Store

4.5.1 SSM Parameter Store

- Secure storage for configuration and secrets
- Optional Seamless Encryption using KMS
- Serverless, scalable, durable easy SDK

- Version tacking of configurations / secrets
- Security through IAM
- Notifications with Amazon EventBridge
- Integration with CloudFormation

SSM Parameter Store Hierarchy
Standard and advanced parameter tiers
Parameter Policies (for advanced parameters)

- Allows to assign a TTL to a parameter (expiration date) to force updating or deleting sensitive data such as passwords.
- Can assign multiple policies at a time.

4.6 Secrets Mangager

4.6.1 AWS Secrets Manager

- Meant for storing secrets (e.g passwords, API keys,......)
- Capability to force rotation of secrets every X days
- Automate generation of secrets on rotation (uses Lambda)
- Natively supports Amazon RDS (all supported DB engines)
- Support other databases and services (custom Lambda function)
- Control access to secrets using Resource-based Policy
- Integration with other AWS services to natively pull secrets from Secrets Manager: CloudFormation, CodeBuild, ECS, EMR, Fargate, EKS, Parameter Store......

SSM Parameter Store vs Secrets Manager

- Secrets Manager (dollah)
- Automatic rotation of secrets with AWS lambda
- Lambda function is provided for RDS, Redshift, DocumentDB
- KMS encryption is mandatory
- Can integrate with CloudFormation

SSM Parameter Store (dollah)

- Simple API
- No secret rotation (can enable rotation using Lambda triggered by EventBridge)
- KMS encryption is optional
- Can integrate with CloudFormation
- Can pull a Secrets Manager secret using the SSM parameter Store API

SSM Parameter Store vs Secrets Manager Rotation

4.7 RDS Security

- KMS encryption at rest for underlying EBS volumes / snapshots
- Transparent Data Encryption (TDE) for Oracle and SQL Server
- SSL Encryption to RDS is possible for all DB (in-flight)
- IAM Authentication for MySQL, PostgreSQL and MariaDB
- Authorisation still happens within RDS (not in IAM)
- Can copy an un-encrypted RDS snapshot into an encrypted one
- CloudTrail cannot be used to track queries made within RDS

4.8 SSL Encryption, SNI, MITM

4.8.1 SSL/TLS - Basics

- SSL refers to Secure Sockets Layer, used to encrypt connections
- TLS refers to Transport Layer Security which is a newer version
- Nowadays, TLS certificates are mainly used but people still refer as SSL
- Public SSL certificates are issued by Certificate Authorities (CA)
- Comodo, Symantec, GoDaddy, GlobalSign, Digicert, Letencrypt, etc
- SSL certificates have an expiration date (you set) and must be renewed

4.8.2 SSL Encryption - How it works

- Asymmetric Encryption is expensive (SSL)
- Symmetric encryption is cheaper
- Asymmetric handshake is used to exchange a per-client random symmetric key.
- Possibility of client sending an SSL certificate as well (two-way certificate)

4.8.3 SSL - Server Name Indication (SNI)

- SNI solves the problem of loading multiple SSL certificates onto one web server (to server multiple websites)
- It's a ”newer” protocol and requires the client to indicate the hostname of the target server in the inisital SSL handshake
- The server will the find the correct certificate or return the default one.

Note:

- Only works for ALB and NLB (newer generation), CloudFront
- Does not work for CLB (older gen)

4.8.4 SSL - Man in the middle attacks

How to prevent

1. Don't use public-facing HTTP, use HTTPS (meaning use SSL/TLS certificates)
2. Use a DNS that has DNSSEC
   - To send a client to a pirate server, a DNS response needs to be ”forged” by a server which intercepts them.
   - It is possible to protect your domain name by configuring DNSSEC
   - Amazon Route 53 supports DNSSEC for domain registration
   - Route 53 supports DNSSEC for DNS service as of December 2020 (using KMS)
   - You could also run a custom DNS server on Amazon EC2 for example (Bind is the most popular, dnsmasq, KnotDNS, PowerDNS)

4.9 AWS Certificate Manager - ACM

4.9.1 AWS Certificate Manager (ACM)

- To host public SSL certificates in AWS, you can
- Buy your own and upload them using the CLI
- Have ACM provision and renew public SSL certificates for you (free of cost)
- ACM loads SSL certificates on the following integrations:
- Load Balancers (including the ones created by EB)
- CloudFront distributions
- APIs on API gateways
- SSL certificates is overall a pain to manually manage, so ACM is great to leverage in your AWS infrastructure

4.9.2 ACM - Good to know

Possibility of creating public certificates
Must verify public DNS
Must be issued by a trusted public certificate authority (CA)
Possibility of creating private certificates
For you internal applications
You create your own private CA
Your applications must trust your private CA
Certificate renewal:
Automatically done if generated provisioned by ACM
Any manually uploaded certificates must be renewed manually and re-uploaded.
ACM is a regional service
To use with a global application (multilpe ALB for example), you need to issue an SSL certificate
You cannot copy certs across regions

4.10 CloudHSM

- KMS =¿ AWS manages the software for encryption
- CloudHSM -¿ AWS provisions encryption hardware
- Dedicated Hardware (HSM = Hardware Security Module)
- You manage your own encryption keys entirely (not AWS)
- HSM device is tamper resistant, FIPS 140-2 Level 3 compliance
- Supports both symmetric and asymmetric encryption (SSL/TLS keys)
- No free tier available
- Must use the CloudHSM Client Software
- Redshift supports CloudHSM for database encryption and key management
- Good option to use with SSE-C encryption

4.10.1 CloudHSM Diagram

- IAM permissions
- CRUD an HSM Cluster
- CloudHSM Software
- Manage the Keys
- Manage the Users

4.10.2 CloudHSM - High Availability

CloudHSM clusters are spread across Multi AZ (HA)

Great for availability

4.10.3 CloudHSM vs KMS

4.11 Solution Architecture - SSL on ELB

- You can offload SSL to CloudHSM (SSL Acceleration)
- Supported by NGINX Apache Web servers and IIS for windows server
- Extra security: the SSL private key never leaves the HSM device
- Must setup a cryptographic user (CU) on the CloudHSM device

4.12 S3 Security

- SSE-S3: encrypts S3 objects using keys handled and managed by AWS
- SSE-KMS: leverage KMS to manage encryption keys
- Key usage appears in CloudTrail
- objects made public can never be read
- On s3:PutObject make the permission kms:GenerateDataKey is allowed
- SSE-C: when you want to manage your own encryption keys
- Client-Side Encryption

Glacier: all data is AES-256 encrypted, key under AWS control

4.12.1 Encryption in transit (SSL/TLS)

- Amazon S3 exposes:

- HTTP endpoint: non encrypted
- HTTPS endpoint: encryption
- You're free to use the endpoint you want, but HTTPS is recommended
- HTTPS is mandatory for SSE-C
- To enforce HTTPS, use a bucket policy wit aws:SecureTransport

4.12.2 Events in S3 Buckets

- S3 Access Logs:
  - Detailed records for the requests that are made to a bucket
  - Might take hours to deliver
  - Might be incomplete (best effort)
- S3 Event Notifications
  - Receive notifications when certain events happen in your bucket
  - E.g: new objects created, object removal, restore objects, replication events.
  - Destinations: SNS, SQS queue, Lambda
  - Typically delivered in seconds but can take minutes, notification for every object if versioning is enabled, else risk of one notification for two same objects write done simultaneously.
- Trusted Advisor:
  - Check the bucket permission (is the bucket public?)
  - Amazon EventBridge
  - Need to enable CloudTrail object level logging on S3 first
  - Target can be Lambda, SQS, SNS, etc.........

4.12.3 S3 Security

- User based
  - IAM policies - which API calls should be allowed for a specific user from IAM console
- Resource Based
  - Bucket Policies - bucket wide rules from the S3 console - allows cross account
  - Object Access Control List (ACL) - finer grain
  - Bucket Access Control List (ACL) - less common

4.12.4 S3 Bucket Policies

- Use S3 bucket for policy to:
- Grant public access to the bucket
- Force objects to be encrypted at upload
- Grant access to another account (Cross Account)
- Optional Conditions on:
- SourceIp: Public IP or Elastic IP — VpcSourceIp: Private IP (through VPC Endpoint)

- Source VPC or Source VPC Endpoint - only workds with VPC Endpoints
- CloudFront Origin Identity
- MFA

4.12.5 S3 pre-signed URLs

- Can generate pre-signed URLs using SDK or CLI
- For downloads (easy can use the CLI)
- For uploads (harder must use the SDK)
- Valid for a default of 3600 seconds, can change timeout with –expires-in[TIME$_{B}Y_{S}ECONDS]argumentUser$
  $signedURLinheritthepermissionsofthepersonwhogeneratedtheURLforGET/PUT$

- Examples:
  - Allow only logged-in users to download a premium video on your S3 bucket
  - Allow an ever changing list of users to download files by generating URLs dynamically
  - Allow temporarily a user to upload a file to a precise location in our bucket

4.12.6 VPC Endpoint Gateway for S3

4.12.7 S3 Object Lock and Glacier Vault Lock

- S3 Object Lock
- Adopt a WORM (Write once read many) model
- Block an object version deletion for a specified amount of time
- Glacier Vault Lock
- Adopt a WORM (Write Once Read Many) model
- Lock the policy for future edits (can no longer be changed)
- Helpful for compliance and data retention

4.13 S3 Access Points

- Access Points simplify security management for S3 buckets
- Each access point has:
- its own DNS name (Internet Origin or VPC origin)
- an access point policy (similar to bucket policy) - manage security at scale

4.13.1 S3 - Access Points - VPC Origin

- We can define the access point ot be accessible only form within the VPC
- You must create a VPC Endpoint to access the Access Point (Gateway or Interface Endpoint)
- The VPC Endpoint Policy must allow access to the target bucket and Access Point

4.14 S3 Multi-Region Access Points

- Provide a global endpoint that spans S3 buckets in multiple AWS regions
- Dynamically route requests to the nearest S3 bucket (lowest latency)
- Bi-directional S3 bucket replication rules are created to keep data in sync across regions
- Failover Controls - allows you to shift request across S3 buckets in different AWS regions within minutes (Active-Active or Active-Passive)

Multi-Region Access Points - Failover Controls

4.15 S3 Multi-Region Access Points - Hands On

4.16 S3 Object Lambda

- Use AWS Lambda Functions to change the object before it is retrieved by the caller application
- Only one S3 bucket is needed on top of which we create S3 Access Point and S3 Object Lambda Access Points
- Use Cases:
- Redacting personally identifiable information for analytics or non production environments
- Converting across data formats, such as converting XML to JSON
- Resizing and watermarking images on the fly using caller-specific details, such as the user who request the object

4.17 DDoS and AWS Shield

What is a DDoS

4.17.1 Types of Attacks on your infrastructure

- Distributed Denial os Service (DDoS)
- When your service is unavailable because it is receiving too many requests
- SYN Flood (Layer 4) - send too many TCP connection requests
- UDP reflection (Layer 4) - get other servers to send many big UDP requests
- DNS flood attack: Overwhelm the DNS so legitimate users can't find the site
- Slow Loris attack: a lot of HTTP connections are opened and maintained (hmmmm consider websocket servers!!)

4.17.2 Application Level Attacks:

- more complex, more specific (HTTP level)
- Cache bursting strategies: overload the backend database by invalidating the cache (clever)

4.17.3 DDoS Protection on AWS

- AWS Shield Standard: Protects against DDoS attack for your website and applications, for all customers at no additional cost
- AWS Shield Advanced: 24/7 premium DDoS protection
- AWS WAF: Filter specific requests based on rules
- CloudFront and Route 53:
- Availability protection using global edge network
- Combined with AWS shield: provides DDoS attack mitigation at the edge
- Be ready to scale - leverage AWS Auto Scaling
- Separate static resources (S3 / CloudFront) form dynamic ones (EC2 / ALB)
- Read the white paper

4.17.4 Sample Reference Architecture

4.17.5 AWS Shield

- AWS Shield Standard
- Free service that is activated for every AWS customer
- Provides protection from attacks such SYN/UDP Floods, Reflection attacks and other layer 3/layer 4 attacks
- AWS Shield Advanced

- Optional DDoS mitigation service (3000 dollahs per month per organisation)
- Protect against more sophisticated attack on Amazon EC2, Elastic Load Balancing (ELB), Amazon CloudFront, AWS Global Accelerator, Route 53
- 24/7 access to AWS DDoS response team (DRP)
- Protect against higher fees during usage spikes due to DDoS

4.18 AWS WAF - Web Application Firewall

- Protects your web applications from common web exploits (Layer 7)
- Deploy on Application Load Balancer (localised rules)
- Deploy on API Gateway (rules running at the regional or edge level)
- Deploy on CloudFront (rules on edge locations)
- Used in front of other solutions: CLB, EC2 Instances, custom origins, S3 websites
- Deploy on AppSync (protect your GraphQL APIs)
- WAF is not for DDoS protection
- Define Web ACL (Web Access Control List)
- Rules can include IP addresses, HTTP headers, HTTP body, or URI strings
- Protects from common attack - SQL injection and Cross-Site scripting (XSS)
- Size constraints, Geo Match
- Rate-based rules (to count occurrences of events)
- Rule Actions: Count — Allow — Block — CAPTCHA — Challenge

4.18.1 AWS WAF - Managed Rules

- Library of over 190 managed rules
- Ready to use rules that are managed by AWS and AWS Marketplace Sellers
- Baseline Rule Groups - general protection from common threats - AWSManagedRulesCommonRuleSet, AWSManagedRulesAdminProtectionRuleSet,.....
- Use-case Specific Rule Groups - Protection for many AWS WAF use cases - AWSManagedRulesSQLLiRuleSet , AWSManagedRulesWindowsRulesSet, AWSMangedRulesPHPRuleSet, AWSManagedRulesWordPressRuleSet
- IP Reputation Rule Groups - block requests based on source (e.g. malicious IPs) - AWSManagedRulesAmazonIpReputationList, AWSManagedRulesAnonymousIpList
- Bot Control Managed Rule Group - block and manage requests from bots - AWSMangedRulesBotControlRuleSet

4.18.2 WAF - Web ACL - Logging

- You can send your logs to an:
- Amazon CloudWatch Logs log group - 5 MB per second
- Amazon Simple Storage Service (Amazon S3) bucket - 5 minutes interval
- Amazon Kinesis Data Firehouse - limited by Firehose quotas

Solution Architecture - Enhance CloudFront Origin Security with AWS WAF and AWS Secrets Manager

4.19 AWS Firewall Manager

- Manage rules in all accounts of an AWS organisation
- Security policy: common set of security rules
- WAF rules (Application Load Balancer, API Gateways, CloudFront)
- AWS Shield Advanced (ALB, CLB, NLB, Elastic IP, CloudFront)
- Security Groups for EC2, Application Load Balancer and ENI resources in VPC
- AWS Network Firewall (VPC Level)
- Amazon Route 53 Resolver DNS Firewall
- Policies are created at the region level

Rules are applied to new resources as they are created (good for compliance) across all and future accounts in your Organisation

4.19.1 WAF vs Firewall Manager vs Shield

- WAF, Shield and Firewall Manager are used together for comprehensive protection
- Define your Web ACL rules in WAF
- For granular protection of your resources, WAF along is the correct choice
- If you want to use AWS WAF across accounts, accelerate WAF configuration, automate the protection of new resources use Firewall Manager with AWS WAF
- Shield Advanced adds additional features on top of AWS WAF, such as dedicated support from the shield response team (SRT) and advanved reporting
- If you're prone to frequent DDoS attacks, consider purchasing Shield Advanved

4.20 Blocking an IP Address

4.20.1 Blocking an IP address

4.20.2 Blocking an IP address - with an ALB

4.20.3 Blocking an IP address - with an NLB

4.20.4 Blocking an IP address - ALB + WAF

4.20.5 Blocking an IP address - ALB, CloudFront and WAF

4.21 Amazon Inspector

4.21.1 Amazon Inspector

- Automated Security Assessments
- For EC2 instances
- Leveraging the AWS System Manager (SSM) agent
- Analyse against unintended network accessibility
- Analyse the running OS against known vulnerabilities
- For container images push to amazon ECR
- Assessment of Container Images as they are pushed
- For Lambda Functions
- Identifies software vulnerabilities in function code and package dependencies
- Assessment of functions as they are deployed
- Reporting and integration with AWS security hub
- Send findings to Amazon Event Bridge

4.21.2 What does Amazon Inspector evaluate?

- Remember: only for EC2 instances, Container Images and Lambda functions
- Continuous scanning of the infrastructure, only when needed
- Package vulnerabilities (EC2, ECR and Lambda) - database of CVE
- Network reachability (EC2)
- A risk score is associated with all vulnerabilities for prioritisation

4.22 AWS Config

4.22.1 AWS Config

- Help with auditing and recording compliance of your AWS resources
- Helps records configuration and changes over time
- AWS Config rules does not prevent actions from happening (no deny)
- Questions that can be solved by AWS config
- Is there unrestricted SSH access to my security groups?
- Do my buckets have any public access?
- How has my ALB configuration changed over time?
- You can receive alerts (SNS notifications) for any changes
- AWS Config is a per-region service
- Can be aggregated across regions and accounts

4.22.2 AWS config resource

- View compliance of a resource over time
- View configuration of a resource over time
- View CloudTrail API calls if enabled

4.22.3 AWS Config Rules

- Can use AWS managed config rules (over 75)
- Can make custom config rules (must be defined in AWS Lambda)
- Evaluate if each EBS disk is of type gp2
- Evaluate if each EC2 instance is t2.micro
- Rules can be evaluated / triggered
- For each config change And / or: at regular time intervals
- Trigger Amazon EventBridge if the rule is non-compliant (chain with Lambda)
- Rules can have auto remediation's through SSM Automations
- If a resource is not compliant you can trigger an auto remediation Ex: remediate security group rules, stop instances with non-approved tags

4.23 AWS Managed Logs

- Load Balancer Access Logs (ALB, NLB, CLB) -¿ S3
- Access logs for your Load Balancers
- CloudTrail Logs -¿ to S3 and CloudWatch logs
- Logs for API calls made within your account
- VPC Flow Logs -¿ to S3, CloudWatch Logs, Kinesis Data Firehose
- Information about IP traffic going to and from network interfaces in your VPC
- Route 53 Access Logs -¿ to CloudWatch logs
- Log information about the queries that Route 53 receives
- S3 Access Logs -¿ to S3
- Server access logging provides detailed records for the requests that are made to a bucket
- CloudFront Access Logs -¿ to S3
- Detailed information about every user request that CloudFront receives
- AWS Config -¿ to S3

4.24 Amazon GuardDuty

- Intelligent Threat discovery to protect your AWS Account
- Using Machine Learning algorithms, anomaly detection, 3rd party data
- One click to enable (30 days trial) no need to install software
- Input data includes:
- CloudTrail Events Logs - unusual API calls, unauthorised deployments
- CloudTrail Management Events - create VPC subnet, create trail
- CloudTrail S3 Data Events - get object, list objects, delete object
- VPC Flow logs - unusual internal traffic, unusual IP address
- DNS Logs - compromised EC2 instances sending encoded data within DNS queries
- Optional Feature - EKS Audit Logs, RDS and Aurora, EBS, Lambda, S3 Data Events.......
- Can setup EventBridge rules to be notified in case of findings
- EventBridge rules can target AWS Lambda or SNS
- Can protect against CryptoCurrency attacks (has a dedicated ”finding” for it)

4.24.1 Amazon GuardDuty

4.24.2 GuardDuty - Delegated Administrator

- AWS organisation member accounts can be designated to be a GuardDuty Delegated Administrator
- Have full permissions to enable and manage GuardDuty for all accounts in the Organisation
- Can be done only using the Organisation Mangement Account

4.25 IAM Advanced Policies

4.25.1 IAM conditions

4.25.2 IAM for S3

- s3:ListBucket permission applies to arn:aws:s3:::test -¿ bucket level permission
- s3:GetObject, s3:PutObject,s3:DeleteObject applies to arn:awn:s3:::test - Object level permissions

4.25.3 Resource Policies and aws:PrincipalOrgID

- aws:PrincipalOrgID can be used in any resource polices to restrict access to accounts that are memeber of an AWS organisation

4.26 EC2 Instance Connect

EC2 Instance Connect (SendSSHPublicKey API)

4.27 AWS Security Hub

- Central security tool to manage security across several AWS accounts and automate security checks
- Integrated dashboards showing current security and compliance status to quickly take actions
- Automatically aggregates alerts in predefined or personal findings formates from various AWS services and AWS partner tools:
- Config
- GuardDuty
- Inspector
- Macie
- IAM Access Analyser

- AWS Systems Manager
- AWS Firewall Manager
- AWS Health
- AWS Partner Network Solutions

Must first enable the AWS config service

4.28 Amazon Detective

- GuardDuty, Macie and Security Hub are used to identify potential security issues or findings.
- Sometimes security findings require deeper analysis to isolate the root cause and take actions - it is a complex process
- Amazon Detective analyses, investigates and quickly identifies the root cause of security issues or suspicious activities (using ML and graphs)
- Automatically collects and processes events from VPC Flow Logs, CloudTrail, GuardDuty and creates a unified view.
- Produces visualisations with details and context to get to the root cause.

Chapter 5

Compute & Load Balancing

5.1 Solution Architecture on AWS

- DNS Layer - Route 53
- Web Layer - CLB, ALB, NLB, API Gateway, Elastic IP
- Compute Layer - EC2, ASG, Lambda, ECS, Fargate, Batch, EMR
- CDN Layer - CloudFront
- Caching / Session Layer - ElastiCache, DAX, DynamoDB, RDS
- Database Layer - RDS, Aurora, DynamoDB, ElastiSearch, S3, Redshift
- Decoupling Orchestration Layer - SQS, SNS, Kinesis, Amazon MQ, Step Functions
- Storage Layer - EBS, EFS, Instance Store
- Static Assets Layer (storage) - S3, Glacier

5.2 EC2

EC2 Instance Types - Main Ones

- R: applications that need a long of RAM - in-memory caches
- C: applications that need good CPU - compute / databases
- M: Applications that are balanced (think ”medium”) - general / web app
- I: Applications that need good local I/O (instance storage) - databases
- G: Applications that need a GPU - video rendering / machine learniing
- T2 / T3: burstable instances (up to a capacity)
- T2 / T3 - unlimited: unlimited burst

See link at https://www.ec2instances.info.
EC2 - Placement Groups

- Control the EC2 instance placement strategy using placement groups
- Group strategies
- Cluster - cluster instances into a low-latency group in a single Availability Zone
- Spread - spreads instances across underlying hardware (max 7 instances per group per AZ) - critical applications
- Partition - spreads instances across many different partitions (which rely on different sets of racks) within an AZ. Scales to 100s of EC2 instances per group (Hadoop, Cassandra, Kafka)
- You can move an instance into or out of a placement group

- You first need to stop it
- You then need to use the CLI (modify-instance placement)
- You can then start your instances

Placement Groups Cluster

- Pros: Great network (10 Gbps bandwidth between instances with Enhanced Networking enabled - recommended)
- Cons: If the rack fails, all instances fails at the same time.
- Use case:
- Big Data job that needs to complete fast
- Applications that needs extremely low latency and high network throughput

Placement Groups Spread

- - Pros
  - Can span across Availability Zones (AZ)
  - Reduced risk of simultaneous failure
  - EC2 Instances are on different physical hardware
- - Cons
  - Limited to 7 instances per AZ per placement group
  - Use case:
  - Application that need to maximise high availability
  - Critical Applications where each instance must be isolated from failure from each other

Placement Groups Partition

- Up to 7 partitions per AZ
- Up to 100s of EC2 instances
- The instances in a partition do not share racks with the instances in the other partitions
- A partition failure can affect many EC2 instances but won't affect other partitions
- EC2 instances get access to the partition information as metadata
- Use cases: HDFS, HBase, Cassandra, Kafka

EC2 Instance Launch Types

- On Demand Instances: Short workload, prefictable pricing, reliable
- Spot Instance: short workloads, for cheap, can lose instances (not reliable)
- Reserved (Minimum 1 year)

- Reserved Instances: Long workloads
- Convertible Reserved Instances: Long workloads with flexible instances
- Highest to lowest discount: All upfront payment, partial upfront payment, no upfront
- Dedicated Instances: No other customers will share your hardware
- Dedicated Hosts: book an entire physical server, control instance placement
- Great for software licenses that operate at the core, or CPU socket level
- Can define host affinity so that instance reboots are kept on the same host

EC2 Graviton

- AWS Graviton Processors deliver the best price performance
- Supports many Linux OS, Amazon Linux 2, RedHat, SUSE, Ubuntu
- Not available for windows instances
- Graviton2 - 40% better price performance over comparable 5th generation x86 based instances
- Graviton3 - Up to 3x better performance compared to Graviton2
- Use cases: app servers, microservices, HPC, CPU-based ML, video encoding, gaming, in-memory caches,.....

EC2 included metrics

- CPU: CPU Utilisation + Credit Usage / Balance
- Network: Network In / Out
- Status Check
- Instance status = check the ECM VM
- System status = check the undelying hardware
- Disk: Read / Write for Ops / Bytes (only for instance store)
- RAM is not included in the AWS EC2 metrics

EC2 Instance Recovery

- Status Check
- Instance Status = check the EC2 VM
- System status = check the underlying hardware
- Recovery: Same Private, Public, Elastic IP, metadata, placement group

5.3 High Performance Computing (HPC)

- Cloud is good for HPC
- Can create a very high number of resources in no time.
- Can speed up time to results by adding more resources
- You can pay only for the systems you have used —- Examples: Genomics, computational chemistry, financial risk modelling, weather prediction, machine learning, deep learning, autonomous driving......
- Services which help HPC?

Data Management % Transfer

- Aws Direct Connect
- Move GBs of data to the cloud over a private secure network
- Snowball
- Move PB of data to the cloud
- AWS DataSync
- Move large amount of data between on-premise and S3, EFS, FSx for Windows

Computer and Networking

EC2 Instances:CPU optimised, GPU optimised Spot Instances / Spot Fleets for cost savings + Auto Scaling EC2 Place Groups: Cluster for good network performance EC2 Enhanced Networking (SR-IOV) Higher Bandwidth, higher PPS (packet per second), lower latency Option 1: Elastic Network Adapted (ENA) up to 100 Gbps Option 2: Intel 82599 up to 10Gbps - Legacy Elastic Fabric Adapter (EFA) Improved ENA for HPC only works for Linux Great for inter-node communications, tightly coupled workloads Leverages Message Passing interface (MPI) standard Bypasses the underlying Linux OS to provide low-latency reliable transport

Storage

- Instance-attached storage
- EBS: scale up to 256000 IOPS with io2 Block Express
- Instance Store: Scal to million of IOPS, linkedin to EC2 instance, low latency.
- Network Storage
- Amazon S3: large blob, not a file system
- Amazon EFS: scale IOPS based on total size or use provisioned IOPS
- Amazon FSx for Lustre:
- HPC optimised distributed file system, millions of IOPS

- Backed by S3

Automation and Orchestration

- AWS Batch
- AWS Batch supports multi-node parallel jobs which enables you to run single jobs that span multiple EC2 instances
- Easily schedule jobs and launch EC2 instances accordingly
- AWS ParallelCluster
- Open source cluster management tool to deploy HPC on AWS
- Configure text files
- Automate creation of VPC, Subnet, cluster type and instance types

5.4 Auto Scaling

Auto Scaling Groups - Dynamic Scaling Policies

- Target Tracking Scaling
- Most simple and easy to setup
- Example: I want the average ASG CPU to stay at aroudnd 40%
- Simple / Step Scaling
- When a CloudWatch alarm is triggered (example CPU ¿ 70%) then add 2 units
- When a CloudWatch alarm is triggered (example CPU ¡ 30%) then remove 1
- Scheduled Actions
- Anticipate a scaling based on known usage patterns
- Example: Increase the min capacity to 10 at 5pm on Fridays

Auto Scaling Groups - Predictive Scaling

- Predictive Scaling: continuously forecast load and schedule scaling ahead

Good metrics to scale on

- CPUUtilisation: Average CPU utilisation across your instances
- RequestCountPerTarget: to make sure the number of requests per EC2 instances is stable
- Average Network In / Out (if you're application is network bound)
- Any custom metric (that you push using CloudWatch)

Auto Scaling - Good to know

- Spot fleet support (mix on Spot and On-Demand instances)
- Lifecycle Hooks
- Perform actions before an instance is in service, or before it is terminated
- Example: Cleanup, log extraction, special health checks
- To upgrade an AMI, must update the launch configuration / template
- Then terminate instances manually (CloudFormation can help)
- Or use EC2 instance refresh for Auto Scaling

Auto Scaling - Instance Refresh

- Goal: Update launch template and then re-creating all EC2 instances
- For this we can use the native feature of Instance Refresh
- Setting of minimum healthy percentage
- Specify warm-up time (how long until the instance is ready to use)

Auto Scaling - Scaling Processes

- Launch: Add a new EC2 to the group, increasing the capacity
- Terminate: Remove an EC2 instance from the group, decreasing the capacity
- HealthCheck: Check the health of the instances
- ReplaceUnhealthy: Terminate unhealthy instances and re-create them
- AZRebalanced: Balancer the number of EC2 instances across AZ
- Alarm Notification: Accept notification from CloudWatch
- SchedulesActions: Performs scheduled actions that you create
- AddToLoadBalancer: Add instances to the load balancer or target group.
- InstanceRefresh: Perform an instance refresh —-¿ These processes can be suspended

Auto Scaling - Health Checks

- Health Checks available:
- EC2 Status Checks
- ELB Health Checks (HTTP)
- Customer Health Checks - send instances health to an ASG using AWS CLI or AWS SDK (set-instnance-health)
- ASG will launch a new instance after terminating an unhealthy one
- Make sure the health check is simple and check ths correct thing

5.5 Auto Scaling Update Strategies

Auto Scaling - Updating an application
Auto Scaling - Solution Architecture

5.6 Spot Instances and Spot Fleet

EC2 Spot Instances

- Can get a discount of up to 90% compared to On-Demand
- Define max spot price and get the instance while current spot price ¡ max
- The hourly spot price varies based on offer and capacity
- If the current spot price ¿ your max price you can choose to stop or terminate your instance with a two minutes grace period
- Used for batch jobs, data analysis or workloads that are resilient to failures. —¿ Not great critical jobs or databases

Spot Fleets - Spot Fleets = set of Spot Instances + (optional) On-Demand Instances - The Spot Fleet will try to meet the target capacity with price constraints - Define possible launch pools: instance type (m5.large), OS, Availability Zone - Can have mulltiple launch pools, so that the flee can choose - Spot fleets stops launching instances when reaching capacity or max cost. - Strategies to allocate spot instances - lowestPrice: form the pool with the lowest price (cost optimisation, short workload) - diversified: distribute across all pools (great for availability, long workloads) - capcityOptimised: pool with optimal capacity for the number of instances - priceCapacityOptimised (recommended): pools with highest capacity available then select the pool with the lowest price (best choice for most workloads) –¿ Spot fleets allow us to automaticaly requests spot instances with the lowest price

5.7 Amazon ECS - Elastic Container Service

What is Docker? - Docker is a software development platfform to deploy appls - Apps are packaged in containers that can be run on any OS - Apps run the saem, regardless of where they are run - Any machine (no compatibilty issues, predictable behaviour) - Less work - Easier to maintain and deploy - Works with any language, any OS, any technology. - Control how much memeory / CPU is aollcated to your continers - Scale containers up and down very quickly (seconds) - More effecient than virtual machines

Docker Containers Management on AWS - To manage containers, we need a container management platform - Amazon Elastic Container Service (Amazon ECS) - Amazon's own container platform - Amazon Elastic Kubernetes Service (Amazon EKS) - Amazon managed Kubernetes (open source) - AWS Fargate - Amazons own Serverless container platform - Works with ECS and with EKS

Amazon ECS - Use cases - Run Microservices - Run multiple Docker containers on the same machine - Easy service discovery features to enhance communication - Direct integration with

Application Load Balancer and Network Load Balancer - Auto Scaling capability - Run Batch Processing / Scheduled Tasks - Scehdule ECS tasks to run on On-Demand / Reserved / Spot Instances - Migrate Applications to the Cloud - Dockerise legacy applications runngin on-premisses - Move docker containers to run on Amazon ECS

Amazon ECS - Concepts - ECS Cluster - logical grouping of EC2 instances - ECS Service - defines how many tasks shoudl run and how they should be run - Task definitions - metadata in JSON form to tell ECS how to run a Docker containers (image naem, CPU, RAM) - ECS Task - an instance of a Task Definition, a running Docker containers - ECS IAM roles - EC2 Instance Profile - used by the EC2 instance (e.g make API calls to ECS, send logs) - ECSTask IAM Role - allow each task to have a specific role (e.g make API calls to S3, DynamoDB)

Amazon ECS - ALB Integration - We get Dynamic Port Mapping - Allows you to run multiple instances of the same application on the same EC2 instance - The ALB finds the right port on your EC2 instances - Use cases: - Increases resiliency even if running on one EC2 instance - Maximise ustilisation of CPU cores - Ability to perform rolling upgrades withouth impacting app uptime

AWS Fargate - Launch Docker containers on AWS - You do not provision the infrastructure (no EC2 instances to manage) - Its all serverless - You create task definitions - AWS run containers for you based on the CPU / RAM you need - To scale, just increase the number of tasks——no more EC2 instances

Amazon ECS - Security and Netoworking - You can inject secrets and configurations as Environment Variable into running Docker containers - Integration with SSM parameter store and secrets manager - ECS Tasks Networking - none - no network connectivity, no port mappings - bridge - uses Docker's virtual container based network - host - bypass Docker's network uses the underlying host network interface - awsvpc - Every task launched on the instance gets it own ENI and a private IP adress - Simplified networking, enhanced security, Security Groups, monitoring, VPC Flow Logs - Default mode for Fargate tasks

Amazon ECS - Service Auto Scaling - Automaticalyly increase / decrease the desired number of tasks - Amazon ECS leverages AWS application auto scaling - CPU and RAM is tracked in CloudWatch as the ECS Service level - Target Tracking - scale based on target value for a specific CloudWatch metrics - Step Scaling - scale based on a specified CloudWatch Alarm - Scheduled Scaling - scale based on a specified date/time (predictable changes) - ECS Service Auto Scaling (task level) $\neq$ EC2 Auto Scaling (EC2 instance level) - Fargate Auto Scaling is much easier to setup (because Serverless)

Amazon ECS - Spot Instances - ECS Classic (EC2 Launch Type) - Can have the underlying EC2 instances as Spot Instances (managed by an ASG) - Instances may go into draining mode to remove running tasks - Good for cost savings, but will impact reliabilty - AWS Fargate - Specify minimum of tasks for on-demand baseline workload - Add tasks running on FARGATE_SPOT for cost-savings (can be reclaimed by AWS) - Regardless of On-demand or Spot, Fargate scales well based on load

5.8 Amazon ECR - Elastic Container Registry

- Store and manage Docker images on AWS - Private and Public repository (Amazon ECR Public Gallery https://gallery.ecr.aws) - Fully integration with ECS - Access is controlled through IAM (permission errors =¿ check policy) - Supports image vulnerabilty scanning versioning, image tags, image lifecycle.....

Amazon ECR - Cross Region Replications - ECR Private registry supports both cross-Region and cross-account replication

Amazon ECR - Image Scanning - Manual Scan or Scan on Push - Basic Scanning - Common CVE - Enhanced Scanning - Leverages Amazon Inspector (OS and Programming Language vulnerabilties) - Scan results can be retrieved from within the AWS console

5.9 Amazon EKS - Elastic Kubernetes Service

- Amazon EKS = Amazon Elastic Kubernete Service - It is a way to launch managed Kubernets clusters on AWS - Kubernetes is an open-source system for automatic deployment, scaling and management of containerised (usually docker) application - It is an alternative to ECS, similar goal but different API - EKS supprots EC2 if you want to deploy worker nodes or Fargate to deploy serverless containers - Use Case: If you company is already using Kubernetes on-premises or in another cloud and wants to migration to AWS using Kubernetes —-¿ Kubernetes is cloud-agnostic (can be used in any cloud - Azure, GCP) - For multiple regions, deploy on EKS cluster per region. - Collects logs and metrics using CloudWatch Container Insights

Amazon EKS - Diagram

Amazon EKS - Node Types - Managed Node Groups - Creates and manages Nodes (EC2 Instances) for you - Nodes are part of an ASG managed by EKS - Supports On-Demand or Spot Instances - Self-Managed Nodes - Nodes created by you an registered to the EKS cluster and managed by an ASH - You can use prebuilt AMI - Amazon EKS optimised AMI - Supports On-Demand or Spot Instances - AWS Fargate - No maintenance required; no nodes managed.

Amazon EKS - Data Volumes - Need to specify StorageClass manifest on your EKS cluster - Leverages a Container Storage Interface (CSI) compliant driver - Support - Amazon EBS - Amazon EFS (works with Fargate) - Amazon FSx for Lustre - Amazon FSx for NetApp ONTAP

5.10 AWS App Runner

- Fully managed service that makes it easy to deploy web applications and APIs at scale - No infratructure experience requried - Start with your source code or container image - Automatically builds and deploy the web app - Automatic scaling highly available, load balancer, encryption - VPC access support - Connect to database, cache and message queue services - Use cases: web apps, APIs, microservices, rapid production deployments

Solution Architecture - App Runner Multi-Region Architecture

5.11 ECS Anywhere and EKS Anywhere

Amazon ECS Anywhere - Easily run containers on Customer-managed infrastructure (on-premises, VMs) - Allows customers to deploy native Amazon ECS tasks in any environment - Fully-manged Amazon ECS Control Plane - ECS Container Agent and SSM Agent needs to be installed - ËXTERNALL̈aunch Type —-¿ Must have a stable connectoin to the AWS region - Use cases: - Meet compliance, regulatory and latency requirements - Run apps outside AWS regions and closer to their other services - On-premises ML, video processing, data processing.

Amazon EKS Anywhere - Create an operate kubernetes clusters created outside AWS - Leverage the Amazon EKS Distro (AWS' bundled release of Kubernetes) - Reduce support costs and avoid maintaining redundant thrid party tools. - Install using the EKS Anywhere installer - Optionally use the EKS Connector to connect the EKS Anywhere clusters to AWS - Full Connected and

Partially Disconnected: you can connect to Amazon EKS Anywhere clusters to AWS and leverage the EKS console - Fully Disconnected: Must install the EKS Distro and leverage open-source tools to manage your clusters.

5.12 AWS Lambda - Part 1

AWS Lambda Integration Main ones - API Gateway - Kinesis - DynamoDB - AWS S3 - Simple Storage Service - AWS IoT - Internet of Things - Amazon EventBridge - CloudWatch Logs - AWS SNS - AWS Cognito - Amazon SQS

AWS Lambda Language Support (runtimes) - node.js (javascript) - Python - Java - C (.NET Core) / Powershell - Ruby - Custom Runtime API (community supported, example Rust or Golang) hmmmmmmmm.......... - Lambda Container Image - The container image must implement the Lambda Runtime API - ECS / Fargate is preferred for running arbitrary Docker images

Lambda - Limits to know – RAM - 128 to 10,240 MB (10 GB) – CPU - is linked to RAM (cannot be set manually) - 2 vCPUs are allocated at 1,769 MB of RAM - 6 vCPUs are alloacted at 10,240 MB of RAM - Timeout - up to 15 minutes - /tmp Storage - 10,240 MB - Deployment Package - 50 MB (zipped), 250 MB (unzipped) including layers - Concurrent Executions - 1000 (soft limit that can be increased) - Container Image Size - 10GB - Invocation Payload (reqeust/ response) - 6 MB (sync), 256KB (async)

Lambda Concurrency and Throttling - Concurrency limit: up to 1000 concurrent executions - Can set a 'reserved concurrency' at the funciton level (=limit) - Each invocation over the concurrency limit will trigger a "Throttle" - Can request a quote increase in AWS Service Quotes

Lambda Concurrency Issue - If you don't reserve (=limit) concurrecny, the following can happen

Lambda and CodeDeploy - CodeDeploy can help you automate traffic shift for Lambda aliases - Feature is integrated within the SAM framework - Linear: grow traffic every N minutes until 100% - Linear10PercentEvery3Minutes - Linear10PercentEvery10Minutes - Canary: try X percent then 100% - Canary10Percent5Mintues - Canary10Percent30Minutes - AllAtOnce: immediate - Can create Pre and Post Traffic Hooks to check the health of the Lambda function

AWS Lambda Logging, Monitoring and Tracing - CloudWatch: - AWS Lamabda execution logs are stored in AWS CloudWatch Logs - AWS Lambda metrics are displayed in AWS CloudWatch Metrics (successful invocations, error rates, latency, timeouts, etc....) - Make sure you AWS Lambda functions has an execution role with an IAM policy that authorises writes to CloudWatch Logs - X-Ray - It's possible to trace Lambda with X-Ray - Enable in Lambda configurations (runs the X-Ray daemon for you) - Use AWS SDK in Code - Ensure Lambda Function has correct IAM Execution Role

5.13 AWS Lambda - Part 2

Lambda in a VPC Note: Lambda: CloudWatch Logs works even withouth endpoint or NAT gateway

Lambda - Fixed Public IP for external comms

Lambda - Synchronous Invocations - Synchronous: CLI, SDK, API Gateway - Results is returned right away - Error handling must happen client side (retries, exponential backoff, etc)

Lambda - Asynchornous Invocation - S3, SNS, Amazon EventBridge - Lambda attempts to retry on errors (3 tries total) - Make sure the processing is idempotent (in case of retries) - Can define a DLQ (dead letter queue) - SNS or SQS for failed processing

Lambda - Architecture Discussion - Start immediately paralell executions - Amazon S3 -¿ Amazon SNS -¿ Lambda - Batched Executions Delay - Amazon S3 -¿ Amazon SNS -¿ Amazon SQS -¿ Lambda

5.14 Elastic Load Balancers - Part 1

Types of load balancer on AWS - AWS has 4 kinds of managed Load Balancers - Classic Load Balancer (v1 - old generation) - 2009 - CLB - HTTP, HTTPS, TCP, SSL (secure TCP) - Application Load Balancer (v2 - new generation) - 2016 - ALB HTTP, HTTPS, WebSocket - Network Load Balancer (v2 - new generation - 2017 - NLB) - TCP, TLS (secure TCP), UDP - Gateway Load Balancer - 2020 - GWLB - Operates a layer 3 (Network Layer) - IP Protocol

- Overall, it is reccomended to use the newer generation load balancers as they provide more features - Some load balancers can be setup as internal (private) or external (public) ELBs

Classic Load Balancers (v1) - Heath Checks can be HTTP (L7) or TCP (L4) based including with SSL - Supports only one SSL certificate - The SSL certificate can have many SAN (Subject Alternate Name), but the SSL cerficate must be changed anytime a SAN is added / edited / removed - Better to use ALB with SNI (Server Name Indication) is possible - Can use multiple CLB if you want distinct SSL certifcates - TCP -¿ TCP passes all the traffic to the EC2 instance - Only way to use 2-way SSL authetication

Application Load Balancer (v2) - Application Load Balancers is Layer 7 (HTTP) - Load balancing to multiple HTTP applications across machines (target groups) - Load balancing to multiple applications on the same machine (ex: containers) - great fit with ECS and has dynamic port mapping - Support for HTTP/2 and WebSocket - Support redirects (from HTTP to HTTPS for example) - Routing Rules for path, headers, query string.

Application Load Balancer (v2) - HTTP Based Traffic

Application Load Balancer (v2) - Target Groups - EC2 Instances (can be managed by an Auto Scaling Group) - HTTP - ECS tasks (managed by ECS itself) - HTTP - Lambda functions - HTTP request is translated into a JSON event - IP Addresses - must be private IPs - ALB can route to multiple target groups - Health Checks are at the target group level

Network Load Balancer (v2) - Network load balancers (Layer 4) allow to: - Forward TCP and UDP traffic to your instances - Handles milliions of request per seconds - Less latency (100 ms) (vs 400ms for ALB)

- NLB has one static IP per AZ and supports assigning Elastic IP (helpful for whitelising specific IP) - NLB are used for extreme performance, TCP or UDP traffic - Not included in the AWS free tier

Network Load Balancer - Target Groups - EC2 instances - IP Addresses - must be private IPs - Application Load Balancer

Network Load Balancer – Zonal DNS Name - Resolving Regional NLB DNS name returns the IP addresses for all NLB nodes in all enabled AZs - Zonal DNS Name - NLB has DNS names for each of its nodes - Use to determine the IP address of each node - Used to minimies latency and data transfer costs - You need to implement app specific logic

Gateway Load Balancer - Deploy, scale and manage a fleet of third party network virtual appliances in AS - Example: Firewalls, Intrusion Detection and Prevention Systems, Deep Packet Inspection Systems, payload manipulation.... - Operates at Layer 3 (Network Layer) - IP Packets - Combines the following functions - Transparent Network Gateway - single entry/exit for all traffic - Load Balancer - distributes traffic to you virtual appliances - Uses the GENEVE protocol on port 6081

Gateway Load Balancer - Target Groups - EC2 Instances - IP Addresses - must be private IPs

5.15 Elastic Load Balancers - Part 2

Cross-Zone Load Balancing

- With cross zone load balancing - each load balancer instance distributes evenly across all registered instances in all AZ - Without Cross Zone Load Balancing - Requests are distributed in the instances of the node of the Elastic Load Balancer

- Classic Load Balancer - Disabled by default - No charges for inter AZ data if enabled - Application Load Balancer - Always on (can't be disabled) - No charges for inter AZ data - Network Load Balancer - Disabled by default - You pay charges for inter AZ data if enabled - Gateway Load Balancer - Disabled by default - You pay charges for inter AZ data if enabled

Sticky Sessions (Session Affinity) - It is possible to implement stickiness so that the same client is always redirected to the same instance behind a load balancer - This works for Classic Load Balancers and Application Load Balancers - The "cookie" used for stickiness has an expiration data you control - Use case: make sure the user doesn't lose his session data - Enabled stickiness may bring imbalance to the load over the backend EC2 instances.

Request Routing Algorithms - Least Outstanding Requests - The next instances to receive the requst is the instance that has the lowest number of pending / unfinished requests - Works with Application Load Balancer and Classic Load Balancer (HTTP / HTTPS)

Request Routing Algorithms - Round Robin - Equally choose the targets from the target group - Works with Application Load Balncer and Class Load Balancer

Request Routing Algorithsm - Flow Hash - Selects a target based on the protocol, source / destination IP address source/ destination port and TCP sequence number - Each TCP/ UDP connection is routed to a single target for the life of the connection - Works with Network Load Balancer

5.16 API Gateway

API Gateway - Overview - Helps expose Lambda, HTTP and AWS Services as an API - API versioning, authorisation, traffic management (API Keys, throttles), huge scale, serverless, req/resp transformations, OpenAPI spec, CORS - Limits to know - 29 Seconds timeout - 10 MB max payload size

API Gateway - Deployment Stages - API changes are deployed to "stages" (as many as you want) - Use the naming you like for stages (dev, test, prod) - Stages can be rolled back as a history of deployments is kept

API Gateway - Integrations - HTTP - Expose HTTP endpoints in the backend - Example: internal HTTP API on premise, Application Load Balancer - Why? Add rate limiting, caching, user authentication, API keys, etc..... - Lambda Function - Invoke Lambda function - Easy way to expose REST API backed by AWS lambda - AWS Service - Expose any AWS API through the API Gateway? - Example: Start an AWS Step Function workflow, post a message to SQS - Why? Add authentication, deploy publicly. rate control.......

Solution Architecture Discussion: API Gateway in front of S3

API Gateway - Endpoint Types - Edge-Optimised (default): For gobal clients - Reqeusts are routed through the CloudFront Edge Locations (improves latency) - The API Gateway still lives in only one region - Regional: - For clients within the same region - Could manually combine with

CloudFront (more control over the caching strategies and the distribution) - Private: - Can only be access from your VPC using an interface VPC endpoint (ENI) - Use a resource policy to define access

Caching API responses - Caching reduces the number of calls made to the backend - Default TTL is 300 seconds (min 0s, max is 3600s) - Caches are defined per stage - Possible to override cache settings —¿ per method - Clients can invalidate the cache with header: Cache-Control:max-age=0 (with proper IAM authorisation) - Able to flush the entire cache (invalidate it) immediately. - Cache encryption option - Cache capacity between 0.5GB to 237GB

API Gateway - Errors - 4xx means client errors - 400: Bad Requst - 403: Access Denied: WAF filtered - 429: Quote exceeded, Throttle - 5xx means Server errors - 502: Bad Gateway Exception, usually for an incompatible output returned from a Lambda proxy integration backend and occasionally for out-of-order invocations due to heavy loads. - 503: Service Unavailable Exception - 504: Integration Failure - ex Endpoint Request Timed-out Exception —¿ API Gateway requests time out after 29 seconds maximum

API Gateway - Security - Load SSL certificates and use Route53 to defined a CNAME - Resource Policy (S3 Bucket Policy) - control who can access the API - Users from AWS accounts, IP or CIDR blocks, VPC or VPC endpoints - IAM Execution Roles for API gateway at the API level - To invoke a Lambda Function, an AWS services.............. - CORS (Cross-origin resources sharing) - Browser based security - Control which domains can call your API

API Gateway - Authentication - IAM based access (AWS$_IAM)-Goodforprovidingaccesswithinyourinfrastru$
$PassIAMcredentialinheadersthroughSigV4-LambdaAuthoriser(formerlyCustomerAuthoriser)-$
$UseLambdatoverfifyacustomOAuth/SAML/thridpartyauthentication-Cognitouserpools-Clientauthenticate$
$ClientpassesthetokentotheAPIGateway-APIGatewayknowout-of-the-boxhowtoverifytotoken$

API Gateway - Logging, Monitoring, Tracing - CloudWatch logs: - Enabled CloudWatch logging at the Stage level (with Log Level - ERROR, INFO) - Can log full reqeusts / responses data - Can send API Gateway Access Logs (customisable) - Can send logs directly into Kinesis Data Firehose (as an alternative to CW logs) - CloudWatch Metrics - Metrics are by stage, possibility to enable detailed metrics - IntegrationLatency, Latency, CacheHitCount, CacheMissCount - X-Ray: - Enable tracing to get extra information about requests min API Gateway - X-Ray API Gateway + AWS Lambda give you the full picture.

5.17 API Gateway - Part 2

AWS Gateway - Usage Plans and API Keys - If you wan to make an API available as an offering to your customers - Usage PLan: - Who can access on or more deployed API stages and methods - how much and how fast they can access them - uses API keys to identify API clients and meter accesss - configure throttling limits and quota limits that are enforced on individual clients - API Keys: - alphanumeric string values to distribute to you customers - API key here...... - Can use with usage plans to control access - Throtling limits are applied to the API keys - Quotes limits is the overall number of maximum reqeusts - 429 Too Many Requests - Acount level throttling across all APIs in a region - Clients must implement retry mechanisms

API Gateway - WebSocket API - Overview - What is a WebSocket? - Two-way interactive communication between a uses's browser and a server - Server can push information to the client - This enables stateful application use cases - WebSocket APIs are often used in real-time applications such as chat applications, collaboration platforms, multiplayer games and financial trading platforms. - Works with AWS services (Lambda, DynamoDB) or HTTP endpoints.

Server to Client Messaging - @conenctions used for replies to clients

API Gateway - Private APIs - Can only be accessed from your VPC by using a VPC Interface Endpoint - Each VPC Interface Endpoint can be used to access multiple Private APIs

- API Gateway Resource Policy - Allow or deny access to API from selected VPCs and VPC Endpoints, including across AWS accounts - aws:SourceVpc and aws:SourceVpce

5.18 AWS AppSync

- AppSync is a managed service that uses GraphQL - GraphQL makes it easy for applications to get exactly the data they need. - This includes combining data from one or more sources - NoSQL data stores, Relational databases, HTTP APIs....... - Integrates with DynamoDB, Aurora, ElasticSeach and others - Customer sources with AWS Lambda - Retrieve data in real-time with WebSocket or MQTT or WebSocket - For mobile apps: local data access and data synchronisation - It all start with uploading a GraphQL Schema

AppSync Diagram

AppSync - Cognito Integration - Perform authorisation on Cognito users based on the groups they belong to. - In the GraphQL schema, you can specify the security for Cognito groups

5.19 Route 53 - Part 1

Route 53 - Record Types - A - maps a hostname to IPv4 - AAAA - maps a hostname to IPv6 - CNAME - maps a hostname to another hostname - The target is a domain name which must has an A or AAAA record - Can't create a CNAME record for the top node of a DNS namespace (Zone Apex) - Example: can't create example.com but you can create www.example.com - NS - Name Servers for the Hosted Zone - Control how traffic is routed for a domain

Route 53 - CNAME vs. Alias - AWS Resources (Load Balancer, CloudFront...) expose an AWS hostname: - CNAME - Points a hostname to any other hostname ——¿ Only for non root domain!!! - Alias - Points a hostname to an AWS resource —-¿ Works for root domain and non root domain - Free of charge - Native health check

Route 53 - Alias Records Targets - Elastic Load Balancers - CloudFront Distributions - API Gateway - Elastic Beanstalk environments - S3 Websites - VPC Interface Endpoitns - Global Accelerator accelerator - Route 53 record in the same hosted zone —-¿ You cannot set an ALIAS record for an EC2 DNS name

Route 53 - Records TTL (Time To Live) - High TTL - e.g 24hr - Less traffic on Route 53 - Possibly outdated records - Low TTL - e.g 60 seconds - More traffic on Route 53 –¿ more cost - Records are outdated for less time - Easy to change records - Except for alias records TTL is mandatory for each DNS record

Routing Policies - Simple - Typically route traffic to a single resource - Can't be associated with Health Checks - Can specify multiple values in the same record - If multiple values are returned, a random one is chose by the client.

Routing Policies - Weighted - Control the % of the requests that go to each specific resource - Can be associated with health checks - Use cases: load balancing between regions, testing new application versions.

Routing Policies - Latency based - Redirect the resource that has the least latency close to us - Super helpful when latency for users is a priority - Latency is based on traffic betwen users and AWS regions — Germany users may be directed to the US (if that is the lowest latency) — Can be associated with Health Checks (has failover capbility)

Routing Policies - Failover (Active-Passive)

Routing Policies - Geolocation - Different from Latency-based - This routing is based on user location - Specify location by Continent, Country or by US State (if theres overlapping, most precise location selected) - Should create a ”default” record (in case theres no match on location) - Use cases: website localisation, restrict content distribution, load balancing...... - Can be associate with Health Checks

Routing Policies - Geoproximity - Route traffic to your resources based on the geographic location of users and resources - Ability to shift more traffic to resources based on the defined bias - To change the size of the geographic region specify bias values: - To expand (1 to 99) - more traffic to the resource - To shrink (-1 to -99) - less traffic to the resource - Resources can be: - AWS resources (specify AWS region) - Non-AWS resources (specify Lattitude and Longitude) - You must use Route 53 Traffic Flow to use this feature

Route 53 - Traffic Flow - Simplify the process of creating and maintaining records in large an complex configurations - Visual editor to manage complex routing decision trees - Configurations can be saved as Traffic Flow Policy - Can be applied to different Route 53 Hosted Zones (different domain names) - Support versioning

Routing Policies - Multi-Value - Use when routing traffic to multiple resources - Route 53 return multiple values/resources - Can be associated with Health Checks (return only values for healthy resources) - Up to 8 healthy records are returned for each Multi-Value query - Multi-value is not a substitute for having a ELB

Routing Policies - IP- Based Routing - Routing is based on clients IP addresses - You provide a list of CIDRs for you clients and the corresponding endpoints/locations (user-IP-to-endpoing-mappings) - Use cases: Optimise performance, reduce network costs...... - Example: route end users from a particular ISP to a specific endpoint

5.20 Route 53 - Part 2

Route 53 - Hosted Zones - A container for records that define how to route traffic to a domain and its subdomain - Public Hosted Zones - contains records that specify how to route traffic on the Internet (public domain names) - Private Hosted Zone - contains records that specify how you route traffic within one or more VPCs (private domain names)

Route 53 - Public vs. Private Hosted Zones

Route 53 - Good to Know - For internal private DNS (Private Hosted Zone), you must enable the VPC settings enableDNSHostnames and enableDNSSupport - DNS Security Extensions (DNSSEC) - A protocol for securing DNS traffic, verifies DNS data integrity and origin - Protects agains Man in the Middle (MITM) attacks - Route 53 supports both DNSSEC for Domain Registration and DNSSEC Signing - Works only with Public Hosted Zones - Route 53 with third registrar - you can buy the domain out of AWS and use Route 53 as the DNS provider - Update the NS records on the thrid party Registrar

Route 53 - Health Checks - HTTP Health Checks are only for public resources - Health Check -¿ Automated DNS Failover: - Health checks that monitor an endpoint (application, server, other AWS resource) - Health checks that monitor other health checks (Calculated Health Checks) - Health checks that monitor CloudWatch Alarms – e.g throttles DynamoDB, alarms on RDS, customer metrics (useful for private resources) - Health Checks are integrated with CW metrics

Route 53 - Calculated Health Checks - Combine the results of multiple Health Checks into a single health check - You can use OR, AND or NOT - Can monitor up to 256 Child Health Checks

- Specify how many of the health checks need to pass to make the parent pass. - Usage: perform maintenance to you website withouth causing all health checks to fail.

Health Checks - Monitor an Endpoint - About 15 global health checkers wiill check the endpoint health - Health Checks pass only when the endpoint responds with the 2xx and 3xx status codes. - Health Checks can be setup to pass / fail based on the text in the first 5120 bytes of the response.

Health Checks - Private hosted Zones - Route 53 health checkers are outside the VPC - They can't access private endpoints (private VPC or on-premises resource) - You can create a CloudWatch Metric and associate a CloudWatch Alarm then create a Health Check that checks the alarm itself.

Health Checks Solutions Architecture - RDS multi-region fail over

5.21 Route 53 - Resolvers and Hybrid DNS

- By default, Route 53 Resolver automatically answers DNS queries for: - Local domain names for EC2 instances - Records in Private Hosted Zones - Records in public Name Servers - Hybrid DNS - resolving DNS queries between VPC (Route 53 Resolver) and your networks (other DNS Resolvers) - Networks can be: - VPC itself / Peered VPC - On-premises Network (connected through Direct Connect or AWS VPN)

Route 53 - Resolver Endpoints - Inbound Endpoint - DNS Resolvers on your network can forward DNS queries to Route 53 Resolver - Allows you DNS Resovlers to resolve domain names for AWS resources (e.g. EC2 instances) and records in Route 53 Private Hosted Zones - Outbound Endpoint - Route 53 Resolver conditionally forwards DNS queries to you DNS Resolvers - Use Resolver Rules to forward DNS queries to you DNS Resolvers - Associated with one or more VPC's in the same AWS Region - Create in two AZs for high availability - Each Endpoint support 10,000 queries per second per IP address

Route 53 - Resolver Inbound Endpoints

Route 53 - Resolver Outbound Endpoints

Route 53 - Resolver Rules - Control which DNS queries are forwarded to DNS Resolvers on your network - Conditional Forwarding Rules (Forwarding Rules) - Forward DNS queries for specified domain and all its subdomains to target IP addresses - System Rules - Selectively overriding the behaviour defined in Forwarding Rules (e.g don't forward DNS queries for a subdomain acme.example.com) - Auto-Defined System Rules - Defines how DNS queries for selected domains are resovled (e.g AWS internal domain names, Private Hosted Zones) - If multiple rules matched, Route 53 Resolver chooses the most specific match. - Resolver Rules can be shared across accounts using AWS "RAM" - Manage them centrally in one account - Send DNS queries from multiple VPC to the target IP defined in the rule

5.22 AWS Global Accelerator

- Leverage the AWS internal network to route to your application - 2 Any cast IP are created for you application - The Anycast IP send traffic directly to edge locations - The edge locations send the traffic to your application - Works with Elastic IP, EC2 Instances, ALB, NLB, public or private - Support Client IP Address Preservation except EIPs endpoints - Consistent Performance - Intelligent routing to lowest latency and fast regional failover - No issue with client cache (because the IP doesn't change) - Internal AWS network - Health Checks - Global Accelerator performas a health check of your applications - Helps make your application global (failover less than 1 minute for unhealthy) - Great for disaster recovery (thanks to the health checks) - Security - Only 2 external IP need to be whitelisted - DDoS protection thanks to AWS Shield

AWS Global Accelerator vs CloudFront - They both use the AWS global network and its edge locations around the world - Both services integrate with AWS sheild for DDoS protection - CloudFront - Improves performance for both cacheable content (such as images and videos) - Dynamic content (such as API acceleration and dynamic site delivery) - Content is server at the edge - Global Accelerator - Improves perfromance for a wide range of applications over TCP or UDP - Proxying packets at the edge to applications running in one or more AWS regions - Good fit for non-HTTP uses cases, such as gaming (UDP), IoT (MQTT) or Voice over IP - Good for HTTP use cases that require static IP addresses - Good for HTTP use cases that required deterministic, fast regional failover

5.23 Comparison of Solution Architecture

- EC2 on its own with Elastic IP - EC2 with Route53 - ALB + ASG - ALB + ECS on EC2 - ALB + ECS on Fargate - ALB + Lambda - API Gateway + Lambda - API Gateway + AWS Service - API Gateway + HTTP backend (ex: ALB)

EC2 with Elastic IP - Quick failover - The client should not see the change happen - Helpful if the client needs to resolve by static Public IP address - Does no scale - Cheap

Stateless web app - scaling horizontally - DNS-based load balancing - Ability to use multiple instances - Route53 TTL implies client may get outdated information - Clients must have logic to deal with hostname resolution failures - Adding an instance may not receive full traffic right away due to DNS TTL

ALB + ASG - Scales well, classic architecture - New instances are in service right away - Users are not sent to instances that are out of service - Time to scale is slow (EC2 instance startup + bootstrap) - An AMI can help. - ALB is elastic but can't handle sudden huge peak of demand (pre-warm) - Could lose a few requests if instances are overloaded - CloudWatch used for scaling - Cross-Zone balancing for even traffic distribution - Target utilisation should be between 40% and 70%

ALB + ECS on EC2 (backed by ASG) - Same properties as ALB + ASG - Application is run on Docker - ASG + ECS allows to have dynamic port mappings - Tough to orchestrate ECS service auto-scaling + ASG auto-scaling

ALB + ECS on fargate - Application is Run on Docker - Service Auto Scaling is easy - Time to be in-service is quick (no need to launch an EC2 instance in advance) - Still limited by the ALB in case of suddent peaks - "serverless" application tier - "managed" load balancer

ALB + Lambda - Limited to Lambdas runtimes - Seamless scaling thanks to Lambda - Simple way to expose Lambda functions as HTTP/S without all the features from API Gateway - Can combine with WAF (Web Applicatoin Firewall) - Good for hybrid microservices - Example: Use ECS for some reqeusts, use Lambda for others

API Gateway + Lambda - Pay per request, seamless scaling, fully serverless - Soft limits: 10000/s API Gateway, 1000 concurrent Lambda - API Gateway feature: authentication, rate limiting, caching, etc...... - Lambda Cold Start time may increase latency for some reqeusts - Full integrated with X-Ray

API Gateway + AWS Service (as a proxy) - Lower latency, cheaper - Not using Lambda concurrent capacity, no custom code - Expose AWS APIs securely through API Gateway - SQS, SNS, Step Functions......... - Remember API Gateway has a payload limit of 10MB (can be a problem for S3 proxy)

API Gateway + HTTP backend (ex: ALB) - Use API Gateway features on top of custom HTTP backend (authentication, rate control, API keys, caching....) - Can connect to.... - on-premises service - Application Load Balancer - Third party HTTP service

5.24 AWS Outposts

- Hybrid Cloud: business that keep an on-premises infrastructure alongside a cloud infrastructure - Therefore two ways of dealing with IT systems: - One for the AWS cloud (using the AWS console, CLI and AWS APIs) - One for their on-premises infrastructure - AWS Outposts are ”sever racks” that offers the same AWS infrastructure, services, APIs and tools to build your own applications on-premises just as in the cloud. - AWS will setup and manage ”Outposts Racks” within your on-premises infrastructure and you can start leveraging AWS services on-premises —–¿ You are responsible for the Outposts Rack physical security.

- Benefits - Low-latency access to on-premises systems - Local data processing - Data residency - Easier migration from on-premises to the cloud - Fully managed service - Some service that work on Outposts: - Amazon EC2 - Amazon EBS - Amazon S3 - Amazon EKS - Amazon ECS - Amazon RDS - Amazon EMR

S3 on AWS Outposts - Use S3 APIs to store and retrieve data locally on AWS Outposts - Keeping data close to on-premises applications - Reduce data transfer to AWS regions - S3 Storage Class named S3 Outposts - Default encryption using SSE-S3

5.25 AWS WaveLength

- WaveLength Zones are infrastructure deployments, embedded within the telecommunications providers datacentres at the edge of 5G networks - Brings AWS services to the edge of the 5G networks - Example: EC2, EBS, VPC - Ultra-low latency applications through 5G networks - Traffic doesn’t leave the Communication Service Provider’s (CSP) network - High-bandwidth and secure connection to the parent AWS region - No additional charges or service agreements - Use cases: Smart Cities, ML-assisted diagnostics, Connected Vehicles, Interactive Live Video Streams, AR/VR Real-time Gaming........

5.26 AWS Local Zones

- Places AWS compute, storage, database and other selected AWS services closer to end users to run latency-sensitive applications. - Extend your VPC to more locations ”Extentsion of an AWS regions” - Compatible with EC2, RDS, ECS, EBS, ElastiCache, Direct Connect......... - Example: - AWS Region: N.Virginia (us-east-1) - AWS Local Zone: Boston, Chicago, Dallas, Houston, Miami

Chapter 6

Storage

6.1 EBS and Local Instance Store

6.1.1 EBS

- Network drive you attach to one instance only.
- Linked to a specific availability zone (transfer: snapshot -¿ restore)
- Volumes can be resized
- Make sure to choose an instance type tha tis EBS optimised to enjoy maximum throughput

6.1.2 EBS Volume Types

- EBS Volumes come in 6 types
  - gp2/gp3 (SSD): General purpose SSD volume that balances price and performance for a wide variety of workloads
  - io1/io2 Block Express - Highest-performance SSD volume for mission critical low latency or high throughput workloads
  - st1 (HDD) low cost HDD volume designed for frequently access, throughput intesive workloads
  - sc1 (HDD): Lowest cost HDD volume designed for less frequently accessed workloads
- EBS volumes are characterised in Size / throughput / IOPS (I/O Ops Per Sec) — See AWS Docs when required
- Only gp2/gp3 and io1/io2 can be used as boot volumes

6.1.3 EBS Snapshots

- Incremental - only backup changed blocks
- EBS backups use IO and you shouldn't run them while your application is handling a lot of traffic
- Snapshots will be stored in S3 (you won't directly see them)
- Not necessary to detach volume to do snapshot, but recommended
- Can copy snapshots across region (for DR)
- Can make Image (AMI) from snapshot
- EBS volumes restored by snapshots need to be pre-warmed (use the Fast Snapshot Restore FSR feature or fio/dd command to read the entire volume)

6.1.4 Amazon Data Lifecycle Manager

- Automate the creation, retention and deletion of EBS snapshots and EBS-backed AMIs
- Schedule backups, cross-account snapshot copies delete outdated backups
- Uses resource tags to identify the resources (EC2 instances, EBS volumes)
- Can't be used to manage snapshots/AMIs created outside Data Life Cycle manager
- Can't be used to manage instance-store backed AMIs

6.1.5 Amazon Data Lifecycle Manger vs AWS Backup

- Use Data Lifecycle Manager - when you want to automate the creations, retention and deletion of EBS snapshots
- Use AWS backup - to manage and monitor backups across the AWS services you use, including EBS volumes from single place

6.1.6 EBS Encryption - Account level setting

- New Amazon EBS volumes aren't encrypted by default
- There's an account-level setting to encrypt automatically EBS volumes and Snapshots
- The setting needs to be enabled on a per-region basis

6.1.7 EBS Multi-Attach - io1/io2 family

- Attach the same EBS volume to multiple EC2 instances in the same AZ
- Each instance has full read and write permissions to the volume
- Use case:
  - Achieve higher application availability in clustered in clustered Linux applications (ex: Teradata)
  - Applications must manage concurrent write operations
  - Must use a file system that's cluster-aware (not XFS, EX4, etc ...)

6.1.8 Local EC2 instance store

- Physical disk attached to the physical server where your EC2 is
- Very High IOPS (because its physical)
- Disks up to 7.5 TiB (can change over time), stripped to reach 60 TiB (can change over time)
- Block Storage (just like EBS)
- Cannot be increased in size
- Risk of data loss if hardware fails

6.1.9 Instance Store vs EBS

- Instance store is physically attached to the machine (ephemeral storage)
- EBS is a network drive (persistent)
- Pros
  - Better I/O performance (EBS gp2 has a max IOPS of 16000, io1 of 6400, io2 Block Express of 256000)
  - Good for buffer / cache / scratch data / temporary content
  - Data survives reboots
- Cons
  - On stop or termination , the instance store is lost
  - You can't resize the instance store
  - Backups must be operated by the user

6.2 Amazon EFS

- Managed NFS (network file system) that can be mounted on many EC2
- EFS works with EC2 instances in multi-AZ and on-premises (DX and VPN)
- Highly available, scalable, expensive (3x gp2), pay per GB used

6.2.1 EFS - Elastic File System

- Use cases: content management, web serving, data sharing, WordPress
- Compatible with Linux based AMI (not Windows), POSIX-compliant
- Uses NFSv4.1 protocol
- Uses security group to control access to EFS
- Encryption at rest using KMS
- POSIX file system (roughly equal to Linux) that has a standard file API
- File system scala automatically, pay per-use, no capcity planning

6.2.2 EFS - Performance and Storage Classes

EFS Scale

- 1000s of concurrent EFS clients, 10 GB+ /s throughput
- Grow to Petabyte-scala network file system, automatically
- Performance Mode (set as EFS creation time)

- General Purpose (default) - latency-sensitive use cases (web server, CMS, etc.....)
- Max I/O - higher latency, throughput, highly parallel (big data, media processing)
- Throughput Mode
  - Bursting - 1 TB = 50 MiB/s + burst of up to 100MiB/s
  - Provisioned - set your throughput regardless of storage size ex: 1GiB/s for 1 TB storage
  - Elastic - automatically scales throughput up or down based on your workloads
  - Up to 3GiB/s for reads and 1GiB/s for writes - Used for unpredictable workloads

6.2.3 EFS - Storage Classses

- Storage Tiers (Lifecycle management feature - move file after N days)
  - Standard: for frequently accessed files
  - Infrequent access (EFS-1A): cost to retrieve files, lower price to store
  - Archive: rarely accessed data (few times each year), 50% cheaper
  - Implement lifecycle policies to move files between storage tiers
- Availability and durability
  - Standard: Multi-AZ great for prod
  - One Zone: One AZ great for dev, backup enabled by default, compatible with IA (EFS One Zone-IA)
- Over 90% in cost savings

6.2.4 EFS - On-premises and VPC peering

6.2.5 EFS - Access Points

- Easily manage applications access to NFS environments
- Enforce a POSIX user and group to use when accessing the file system
- Restrict access to a directory within the file system and optionally specify a different root directory
- Can restrict access from NFS clients using IAM policies

6.2.6 EFS - File System Policies

- Resource - based policy to control access to EFS file systems (same as S3 bucket policy)
- By default, it grants full access to all clients

6.2.7 EFS - Cross Region Replication

- Replicate objects in an EFS file system to another AWS regions
- Setup for new or existing EFS file systems
- Provides RPO and RTO of minutes
- Doesn't affect the provisioned throughput of the EFS file system
- Use cases: meet you compliance and business continuity goals

6.3 Amazon S3

6.3.1 S3 - Overview

- Object storage, serverless, unlimited storage, pay-as-you-go
- Good to store static content (image, video files)
- Access object by key, no indexing facility
- Not a filesystem, cannot be mounted natively on EC2
- Anti patterns:
  - Lots of small files
  - POSIX file system (use EFS instead), file locks
  - Search features, queries, rapidly changing data
  - Website with dynamic content

6.3.2 S3 Storage Classes Comparison

- You can transition object between tiers (or delete) using S3 Lifecycle policies –¿ See URL

6.3.3 S3 - Replication (Versioning Enabled)

- Cross Region Replication (CRR)
- Same Region Replication (SRR)
- Combine with Lifecycle rules
- Helpful to reduce latency, disaster recovery, security
- S3 Replication Time Control (S3 RTC)
  - Replicates most objects that you upload to Amazon S3 in seconds, and 99.99% of those objets within 15 minutes
  - Helpful for compliance, DR etc

6.3.4 S3 Event Notifications

- S3:ObjectCreated, S3:ObjectRemoved, S3:ObjectRestore, S3:Replication,......
- Object name filtering possible (*.jpg)
- Use case: generate thumbnails of images uploaded to S3
- Can create as many "S3 events" as desired
- S3 event notifications typically deliver events in seconds but can sometimes take a minute or longer

6.3.5 S3 Event Notification with Amazon EventBridge

- Advanced filtering options with JSON rules (metadata, object size, name.......)
- Multiple Destinations - ex Step Functions, Kinesis Streams / Firehose ......
- EventBridge Capabilities - Archive, Replay Events, Reliable Delivery

6.3.6 S3 - Baseline Performance

- Amazon S3 automatically scales to high request rates, latency 100-200ms
- Your application can achieve at least 3500 PUT/COPY/POST/DELETE or 5,500 GET/HEAD reqeusts per second per prefix in a bucket
- There are limits to the number of prefixes in a bucket
- Example (object path =¿ prefix)
  - bucket/folder1/sub1/file =¿ /folder1/sub1/
  - bucket/folder1/sub2/file =¿ /folder1/sub2/
  - bucket/1/file =¿ /1/
  - bucket/2/file =¿ /2/
- If you spread reads across all four prefixes evenly, you can achieve 22,000 requests per second for GET and HEAD

6.3.7 S3 Performance

- Multi-Part upload:
  - recommended for files ¿ 100MB, must use for files ¿ 5GB
  - can help parallelize uploads (speed up transfer)
- S3 Transfer Acceleration
  - Increase transfer speed by transferring file to an AWS edge location which will forward the data to the S3 bucket in the target region
  - Compatible with multi-part upload

6.3.8 S3 Performance - S3 Byte-Range Fetches

- Parallelize GETs by requesting specific byte ranges
- Better resilience in case of failures

Can be used to speed up downloads
Can be used to retrieve only partial data (for example the head of a file)

6.3.9 S3 Multi-Part - Remove Incomplete Parts

6.4 Amazon S3 - Storage Class Analysis

- May be seen as ”storage class analysis” at the exam
- Help you decide when to transition objects to the right storage class
- Recommendations for Standard and Standard IA
  - Does NOT work for One-Zone IA or Glacier
  - Report is updated daily
  - 24 to 48 hours to start seeing data analysis
  - Visualise data in Amazon QuickSight
  - Good first step to put together Lifecycle Rules (or improve them!)

6.5 Amazon S3 - Storage Lens

- Understand, analyse and optimise storage across entire AWS Organisation
- Discover anomalies, identify cost efficiencies and apply data protection best practices across entire AWS Organisation (30 days usage and activity metrics)
- Aggregate data for Organisations, specific accounts, regions, buckets or prefixes
- Default dashboard or create your own dashboards
- Can be configured to export metrics daily to an S3 bucket (CSV, Parquet)

6.5.1 Storage Lens - Default Dashboard

- Visualise summarised insights and trends for both free and advanced metrics
- Default dashboard shows Multi-Region and Multi-Account data
- Preconfigured by Amazon S3
- Can’t be deleted, but can be disabled

6.5.2 Storage Lens - Metrics

- Summary Metrics
  - General insights about your S3 storage
  - StorageBytes, ObjectCount.....
  - Use cases: identify the fastest-growing (or no used) buckets and prefixes
- Cost-Optimised Metrics
  - Provide insights to managed and optimise your storage costs
  - NonCurrentVersionStorageBytes, IncompleteMultipartUploadStorageBytes...
  - Use cases: identify buckets with incomplete multipart uploaded older than 7 days, Identify which objects could be transitioned to lower-cost storage class
- Data-Protection Metrics
  - Provide insights for data protection features
  - VersioningEnabledBucketCount, MFADeleteEnabledBucketCount, SSEKMSEnabledBucketCount, CrossRegionReplicationRuleCount...
  - Use cases: identify buckets that aren't following data-protection best practices
- Access-management Metrics
  - Provide insights for S3 Object Ownership
  - ObjectOwnershipBucketOwnerEnforcedBucketCount...
  - Use cases: identify which Object Ownership settings your buckets use
- Event Metrics
  - Provide insights for S3 Event Notifications
  - EventNotificationEnabledBucketCount (identify which buckets have S3 Event Notifications)
- Performance Metrics
  - Provide insights for S3 Transfer Acceleration
  - TransferAccelerationEnabledBucketCount (identify which buckets have S3 Transfer Acceleration enabled)
- Activity Metrics
  - Provide insights about how your storage is requested
  - AllRequests, GetRequests, PutRequests, ListRequests, BytesDownloaded
- Detailed Status Code Metrics
  - Provide insights for HTTP status codes
  - 200OKStatusCount, 403ForbiddenErrorCount, 404NotFoundErrorCount...

6.5.3 Storage Lens - Free vs Paid

- Free Metrics
  - Automatically available for all customers
  - Contain around 28 usage metrics
  - Data is available for queries for 14 days
- Advanced Metrics and Recommendations
  - Additional paid metrics and features
  - Advanced Metrics - Activity, Advanced Cost Optimisation, Advanced Data Protection, Status code
  - CloudWatch Publishing - Access Metrics in CloudWatch without additional charges
  - Prefix-Aggregation - Collect metrics at the prefix level
  - Data is available for queries for 15 months

6.6 S3 Solution Architecture

6.6.1 S3 Solution Architecture Indexing objects in Dyanmo DB

—- Notes for DynamoDB table

API for object metadata

- Search by date
- Total storage used by a customer
- List of all objects with certain attributes
- Find all objects uploaded within a date range

6.6.2 Solution Architecture on AWS Dynamic vs Static Content

6.7 Amazon FSx

6.7.1 Amazon FSx - Overview

- Launch 3rd party high-performance file systems on AWS
- Fully managed service

6.7.2 Amazon FSx for Windows (File Server)

- FSx for Windows is fully managed Windows file system share drive
- Supports SMB protocol and Windows NTFS
- Microsoft Active Directory integration, ACLs, user quotas
- Can be mounted on Linux EC2 instances
- Supports Microsoft's Distributed File System (DFS) Namespaces (group files across multiple FS)
- Scale up to 10s of GB/s, millions of IOPS, 100s PB of data
- Storage Options:
  - - SSD - latency sensitive workloads (databases, media processing, data analytics....)
  - - HDD - broad spectrum of workloads (home directory, CMS,....)
- Can be accessed from your on-premises infrastructure (VPN or Direct Connect)
- Can be configured to be Multi-AZ (high availability)
- Data is backed-up daily to S3

6.7.3 Amazon FSx for Lustre

- Lustre is a type of parallel distributed file system, for large-scale computing
- The name Lustre is derived from "Linux" and "cluster"
- Machine Learning, High Performance Computing (HPC)
- Video Processing, Financial Modeling, Electronic Design Automation
- Scales up to 100s GB/s, millions IOPS, sub-ms latencies
- Storage Options:
  - SSD - low-latency, IOPS intensive workloads, small and random file operations
  - HDD - throughput-intensive workloads, large and sequential file operations
- Seamless integrations with S3
  - Can "read S3" as a file system (through FSx)
  - Can write the output of the computations back to S3 (through FSx)
- Can be used from on-premises servers (VPN or Direct Connect)

6.7.4 FSx Lustre - File System Deployment Options

- Scratch File System
  - Temporary storage
  - Data is not replicated (doesn't persist if file server fails)
  - High burst (6x faster, 200MBps per TiB)
  - Usage: short-term processing, optimised costs
- Persistent File System
  - Long-term storage
  - Data is replicated within same AZ
  - Replace failed files within minutes
  - Usage: long-term processing, sensitive data

6.7.5 Amazon FSx for NetApp ONTAP

- Managed NetApp ONTAP on AWS
- File System compatible with NFS, SMB, iSCSI protocol
- Move workloads running on ONTAP or NAS to AWS
- Work with:
  - Linux
  - Windows
  - MacOS
  - VMWare Cloud on AWS
  - Amazon Workspaces and AppStream 2.0
  - Amazon EC2, ECS and EKS
- Storage shrinks or grows automatically
- Snapshots, replication, low-cost, compression and data de-duplication
- Point-in-time instantaneous cloning (helpful for testing new workloads)

6.7.6 Amazon FSx for OpenZFS

- Managed OpenZFS file system on AWS
- File System compatible with NFS (v3, v4, v4.1, v4.2)
- Move workloads running on ZFS to AWS
- Works with:
  - Linux

  - Windows
  - MacOS
  - VMware Cloud on AWS
  - Amazon Workspaces and AppStream 2.0
  - Amazon EC2, ECS and EKS

- Up to 1,000,000 IOPS with ¡ 0.5ms latency
- Snapshots, compression and low-cost
- Point-in-time instantaneous cloning (helpful for testing new workloads)

6.8 Amazon FSx - Solution Architecture

6.8.1 FSx - Solution Architecture Decrese FSx Volume Size

- If you take a backup, you can only restore to a same size
- You can only increase the amount of storage capacity for a file system; you cannot decrease storage capacity
- Instead, create a new FSx (smaller), use DataSync to sync data and then migrate your app over

6.8.2 FSx for Lustre - Data Lazy Loading

- Any data processing job on Lustre with S3 as an input data source can started withouth Lustre doing a full download of the data set first
- Data is lazy loaded: only the data this is actually processed is loaded, meaning you can decrease your costs and latency
- Data is also loaded only once, therefore you reduce your requests on Amazon S3

6.9 AWS DataSync

- Move large amount of data to and from
  - On-premises / other cloud to AWS (NFS, SMB, HDFS, S3 API...) - needs agent
  - AWS to AWS (different storage services) - no agent needed
- Can synchronise to:
  - Amazon S3 (any storage classes, including glacier)
  - Amazon EFS
  - Amazon FSx (Windows, Lustre, NetApp, OpenZFS)
- Replication tasks can be scheduled hourly, daily, weekly
- File permissions and metadata are preserved (NFS POSIX, SMB...)
- One agent task can use 10Gbps, can setup a bandwidth limit

6.10 AWS DataSync - Solution Architecture

6.10.1 AWS DataSync NFS / SMB to AWS (S3, EFS, FSx)

6.10.2 AWS Datasync Transfer between AWS storage services

6.10.3 AWS DataSync Private VIF through Direct Connect

6.11 AWS Data Exchange

- Find, subscribe to and use third - party data in the cloud
  - Reuters, who curate data from over 2.2 million unique news stories per year in multiple languages
  - Change healthcare, who process and anonymize more that 14 billion healthcare transactions and $1 trillion in claims annually
  - Dun and Bradsreet, who maintain a database of more than 330 million global business records;
  - Foursquare, whose location data is derived from 220 million unique consumers and includes more 60 million global commercial venues
- Once subscribed to a data product, you can use the AWS data Exchange API to load data directly into Amazon S3 and then analyze it with wide variety to AWS analytics and machine learning serivces

6.11.1 AWS Data Exchange - Other Products

- AWS Data Exchange for Redshift
  - Find and subscribe to third-party data in AWS Data Exchange that you can query in an Amazon Redshift data warehouse in minutes
  - Easily license your data in Amazon Redshift through AWS Data Exchange
- AWS Data Exchange for APIs
  - Find and subscribe to third-party APIs with a consistent access using AWS SDKs
  - Consistent AWS-native authentication and governance

6.12 AWS Transfer Family

A fully-managed service for file transfers into and out of Amazon S3 or Amazon EFS using the FTP protocol

Supported Protocols

- AWS Transfer for FTP (File Transfer Protocol (FTP))
- AWS Transfer for FTPS (File Transfer Protocol over SSL (FTPS))
- AWS Transfer for SFTP (Secure File Transfer Protocol (SFTP))

Managed infrastructure, Scalable, Reliable, Highly Available (multi-AZ)

Pay per provisioned endpoint per hour + data transfers in GB

Store and manage users' credentials within the service

Integrate with existing authentication systems (Microsoft Active Directory, LDAP, Okta, Amazon Cognito, custom)

Usage: sharing files, public datasets, CRM, ERP,.....

6.12.1 AWS Transfer Family - Endpoint Types

Public Endpoint

- IPs managed by AWS (subject to change, use DNS names)
- Can't setup allow lists by source IP addresses

VPC Endpoint with Internal Access

- Static private IPs
- Setup allow lists (SGs and NACLs)

VPC Endpoint with Internet-facing Access

- Static private IPs
- Static public IPs (EIPs)
- Setup security groups

6.13 AWS Storage Services Price Comparison

Chapter 7

Caching

7.1 Cloudfront - Part 1

- Content Delivery Network (CDN)
- Improves read performance, content is cached at the edge
- 225+ Point of Presence globally (215+ Edge Locations and 13 Regional Edge Caches)
- Protect against Network and Application layer attacks (e.g DDoS attacks)
- Integration with AWS Shield, AWS WAF and route 53
- Can expose external HTTPS and can talk to internal HTTPS backends
- Supports WebSocket protocol

7.1.1 CloudFront - Origins

- S3 Bucket
- For distributing files
- For uploading to S3 (using CloudFront as an ingres)
- Enhanced security with CloudFront Origin Access Control (OAC)
- MediaStore Container and MediaPackage Endpoint
- To deliverVideo on Demand (VOD) or live streaming video using AWS Media Services
- VPC Origin
- For applications hosted in VPC private subnets
- Application Load Balance / Network Load Balancer / EC2 Instances
- Customer Origin (HTTP)
- API Gateway (for more control.... otherwise use API Gateway Edge)
- S3 Bucket configured as a website (enable Static Website hosting)
- Any HTTP backend you want

7.1.2 CloudFront - S3 as an Origin

7.1.3 CloudFront vs S3 Cross Region Replication

- CloudFront:
  - Global Edge network
  - Files are cached for a TTL (maybe a day)
  - Great for static content that must be available everywhere
- S3 Cross Region Replication

  -
  - Must be setup for each region you want replication to happen
  - Files are updated in near real-time
  - Read only
  - Great for dynamic content that needs to be available at low-latency in few regions

7.1.4 CloudFront - ALB or EC2 as an origin Using VPC Origins

- Allows you to deliver content from you applications hosted in your VPC private subnets (no need to expose them on the Internet)
- Deliver traffic to private
- Application Load Balancer
- Network Load Balancer
- EC2 Instances

7.1.5 CloudFront - EC2 or ALB as an origin

7.1.6 CloudFront - Restrict Access to Application Load Balancers and Custom Origins

- Prevent direct access to you ALB or Custom Origins (only access through CloudFront)
- First, configurations CloudFront to add a CustomHTTPHeader to requests it sends to the ALB
- Second, configure the ALB to only forward requests that contain that Customer HTTP Header
- Keep the custom header name and value secret!

7.1.7 CloudFront - Origin Groups

- To increase high-availability and do failover
- Origin Group: one primary and on secondary
- If the primary origin fail; the second on is used
- Origins can be cross AWS regions

7.2 Cloudfront - Part 2

7.2.1 CloudFront Geo Restrictions

- - You can restrict who can access your distribution
- Allow list: Allow you users to access your content only if they're in one of the countries on a list of approved

- Block list: Prevent your users from accessing your content if they're in one of the countries on a blacklist of banned countries..... lol
- The "country" is determined using a third party Geo-IP database
- Use case: Copyright Laws to control access to content
- Note: the geo header CloudFront-Viewer-Country is in Lambda at Edge

7.2.2 CloudFront - Pricing

- CloudFront Edge locations are all around the world
- The cost of data out per edge location varies

7.2.3 CloudFront - Price Classes

- You can reduce the number of edge locations for cost reductions
- Three price classes
- Price Class All: all regions - best performance
- Price Class 200: most regions, but excludes the most expensive regions
- Price Class 100: only the least expensive regions

7.2.4 CloudFront Signed URL Diagram

- Signed URL with expiration to control access to content in CloudFront
- The Signed URL are generated by an API call into CloudFront as a trusted signer

7.2.5 CloudFront Signed URL vs S3 Pre-Signed URL

- CloudFront Signed URL:
- Allow access to a path, no matter the origin
- Account wide key-pair, only the root can manage
- Can filter by IP, path, date, expiration
- Can leverage caching features
- - S3 Pre-Signed URL:
- Issue a request as the person who pre-signed URL
- Uses the IAM key of the signed IAM principal
- Limited lifetime

7.2.6 CloudFront - Custom Error Pages

Return an object to the viewer (e.g html) when your origin returns an HTTP 4xx or 5xx status code to CloudFront

Use Error Caching Minimum TTL to specify how long CloudFront caches the customer error pages

7.3 Lambda at Edge and CloudFront Functions

7.3.1 CloudFront - Customisation at the edge

- Many modern application execute some form of the logic at the edge
- Edge function:
- A code that you write an attach to CloudFront distributions
- Runs close to your users to minimise latency
- Doesn't have any cache, only to change requests/ responses
- CloudFront provides two types: CloudFront Functions and Lambda and Edge

Use cases:

- Manipulate HTTP requests and responses
- Implement request filtering before reaching your application
- User authentication and authorisation
- Generate HTTP resources at the edge
- A/B Testing
- Bot mitigation
- You don't have to manage any servers, deployed globally

7.3.2 CloudFront Functions and Lamabda at Edge

7.3.3 CloudFront - CloudFront Functions

- Lightweight functions written in JavaScript
- For highlatency-sensitive CDN customisation
- Sub-ms startup times, million of requests per second
- RuN at Edge Locations
- Processed-Based isolation
- Used to changeViewer requests and responses:

- Viewer Requests: after CloudFront recevives a reqeusts from a viewer
- Viewer Response: before CloudFront forwards the response to the viewer
- Native feature of CloudFront (manage code entirely within CloudFront)

7.3.4 CloudFront - Lambda at Edge

- - Lambda functions written in NodeJS or Python
- Scala to 1000s of reqeust/ second
- Runs at the nearest Regional Edge Cache
- VM-based isolation
- Used to change CloudFront requests and responses
- Viewer Request - after CloudFront recieves a reqeust from a viewer
- Origin Request - before CloudFront forwards the response to the origin
- Orign response - after CloudFront recieves the response from the origin
- Viewer Response - before CloudFront forwards the response to the viewer
- Author your functions in one AWS Region (some region) then CloudFront replicats to its locations

7.3.5 CloudFront Functions with Lambda at Edge

CloudFront Functions and Lambda at Edge can be used together Note: You can't combine CloudFront Functions and Lambda at Edge in viewer events (viewer reqeust and viewer response)

7.3.6 Using Lambda at Edge only

Use when you need some of the capabilities of Lambda at Edge that aren't available with CloudFront Functions (e.g longer execution time, network access,....)

7.3.7 CloudFront Funtions vs. Lamda at Edge

7.3.8 CloudFront Functions vs Lambda@Edge Use Cases

- CloudFront Functions
  - Cache key normalization
  - Transform request attributes (headers, cookies, query strings, URLs) to create an optimal cache key
  - Header manipulation
  - Insert, modify, or delete HTTP headers in requests or responses
  - URL rewrites and redirects

  - Request authentication and authorization
  - Create and validate user-generated tokens (e.g., JWTs) to allow or deny requests

- Lambda@Edge
  - Longer execution time (up to several seconds)
  - Adjustable CPU and memory allocation
  - Support for third-party libraries and dependencies
  - Access to AWS services through the AWS SDK
  - Network access to external services for processing
  - Access to the request body
  - File system access via the Lambda execution environment
  - Suitable for complex request and response processing

7.3.9 CloudFront Functions vs. Lambda at Edge - Authentication and Authorisation

CloudFront Functions
Lambda at Edge

7.3.10 Lambda@Edge: Loading content based on User-Agent

7.3.11 Lambda at Edge - Global Application

7.4 Lambda at Edge Reduce Latency

7.4.1 Lambda at Edge - Route to different origin

7.5 Amazon ElastiCache

7.5.1 Amazon ElasticCache Overview

- The same way RDS is to get managed Relational Databases.....
- ElastiCache is to get managed Redis or Memcached
- Caches are in memory databases with really high performance and low latency
- Helps reduce load off of databases for read intensive workloads
- Helps make you application stateless
- AWS takes care of OS maintenance / patching, optimisations, setup, configuration, monitoring failure recovery and backups.
- **Using ElastiCache involves heavy application code changes**

7.5.2 ElastiCache Solution Architecture - DB Cache

- Applications queries ElastiCache, if not available, get from RDS and store in ElasticCache
- Helps relive load in RDS
- Cache must have an invalidation strategy to make sure only the most current data is used there.

7.5.3 ElastiCache Solution Architecture - User Session Store

- User logs into any of the application
- The application writes the session data into ElastiCache
- The user hits another instance of our application
- The instance retrieves the data the user is already logged in

7.5.4 ElastiCache - Redis vs Memcached

- - Redis
- Multi AZ with Auto-Failover
- Read Replicas to scale reads and have high availability
- Persistent, Data Durability: Append Only File (AOF), backup and restore features
- Memcached
- Multi-node for partitioning of data (sharding)
- Non Persistent
- Backup and restore (Serverless)
- Multi-threaded architecture

7.6 Handling Extreme Rates

Chapter 8

Databases

8.1 DyanamoDB

8.1.1 DynamoDB - in short

- NoSQL database, fully managed, massive scale (1,000,000 requests per second)
- Similar to Apache Cassandra (can migrate to DynamoDB)
- No disk space to provision, max object size is 400kb
- Capacity: provisioned (WCU, RCU and Auto Scaling) or on-demand
- Support CRUD (Create, Read, Update, Delete)
- Read: Eventually or strong consistency
- Supports transaction across multiple tables (ACID support)
- Backups available, point in time recovery
- Table classes: Standard and Infrequent Access (IA)

8.1.2 DynamoDB - Basics

- DynamoDB is made of tables
- Each table has a primary key (must be decided at creation time)
- Each table can have aan infinite number of times (= rows)
- Each item has attributes (can be added over time - can be null)
- Maximum size of a item is a 400kb
- Data types supported are:
  - ScalarTypes: String, Number, Binary, Boolean, Null
  - Document Types: List, Map
  - Set Types: String Set, Number Set, Binary Set

8.1.3 DynamoDB - Primary Keys

- Option 1: Partition Key only (HASH)
- Partition key must be unique for each item
- Partition key must be "diverse" so that the data is distributed

- Example: user$_{i}dforauserstable$

- Option 2: Partition Key + Sort Key
- The combination must be unique

  - Data is grouped by partition key
  - Sort key == range key
  - Example: user-games table
- user$_idforthepartitionkeygame_idforthesortkey$
- **Example good sort key: timestamp**

8.1.4 DynamoDB - Indexes

  - Object = partitioon key + optional sort key + attributes
  - LSI - Local Secondary Index
  - Keep the same parition key
  - Select an alternative key
  - Must be defined at table creation time
  - GSI - Global Secondary Index
  - Change the partition key and optionally the sort key
  - Can be defined after the table is created
- **You can only query by PK + sort key on the main table and indexes** $\neq RDS$

8.1.5 Amazon Kinesis Data Streams for DynamoDB

  - You can use Kinesis Data Streams to capture item-level changes in DynamoDB
  - Customer and longer data retention period (¿ 24 hours in DynamoDB Streams)

8.1.6 DynamoDB Solution Architecture Indexing Objects in DynamoDB

API for object metadataSearch by date Total storage used by a customer List of all object with certain attributes Find all objects uploaded within a date range

8.1.7 DynamoDB - DAX

  - DAX = DynamoDB Accelerator
  - Seamless cache for DynamoDB, no application re-write
  - Writes go through DAX to DynamoDB
  - Micro second latency for cached reads and queries
  - Solives the Hot Key problem (too many reads)
  - 5 minutes TTL for cache by default

- Up to 10 nodes in the cluster
- Multi AZ (3 nodes minimuum reccommended for production)
- Secure (Encryption at rest with KMS, VPC, IAM, CloudTrail....)

8.1.8 DynamoDB - DAX vs ElastiCache

8.2 Amazon OpenSearch

8.2.1 Amazon OpenSearch (ex ElasticSearch)

- New name is Amazon OpenSearch
- ElasticSearch =¿ OpenSearch
- Kibana =¿ OpenSearch Dashboards
- Managed versions of OpenSearch (open-source project, fork of ElastiSearch)
- Two modes: Managed cluster or Serverless cluster
- Use cases:
- Log Analytics
- Real time application monitoring
- Security Analytics
- Full Text Search
- Clickstream Analytics
- Indexing

8.2.2 OpenSearch + OS Dashboards + Logstash

- OpenSearch (ex ElasticSearch): Provide search and indexing capability
- OpenSearch Dashboard (ex Kibana)
- Provide real-time dashboards on top of the data that sits in OpenSearch
- Alternative to CloudWatch dashboards (more advanced capabilities)
- Logstash:
- Log ingestion mechanism, use the ”Logstash Agent”
- Alternative to CloudWatch Log (you decide on retention and granularity)

8.2.3 OpenSearch patterns DynamoDB

8.2.4 OpenSearch patterns CloudWatch Logs

8.3 RDS

- Engines: PostgreSQL, MySQL, MariaDB, IBM DB2, Oracle, SQL Server
- Managed DB: provisioning, backups, patching, monitoring
- Launched within a VPC, usually in a private subnet, control network access using security groups (important when using Lambda)
- Storage by EBS, can increase volume size with auto-scaling
- Backups: automated with point-in-time recovery. Backups expire
- Snapshots: manual, can make copies of snapshots cross region
- RDS Events: get notified via SNS for events (operations, outages.....)

8.3.1 RDS - Multi AZ and Read Replicas

Multi-AZ

- Standby instance for failover in case of outage

Read Replicas

- Increase read, throughput. Eventual consistency. Can be cross-region

8.3.2 RDS - Distributing Reads across Replicas

8.3.3 RDS - Security (reminder)

- KMS encryption at rest for underlying EBS volumes / snapshots
- Transparent Data Encryption (TDE) for Oracle and SQL Server
- SSL encryption to RDS is possible for all DB (in - flight)
- IAM authentication for MySQL, PostgreSQL and MariaDB
- Authorisation still happens within RDS (not in IAM)
- Can copy an un-encrypted RDS snapshot into an encrypted one
- CloudTrail cannot be used to track queries made within RDS

8.3.4 RDS - IAM Authetication

- IAM database authentication works with MariaDB, MySQL and PostgreSQL
- You don't need a password, just an authentication token obtained through IAM and RDS API calls
- Auth token has a lifetime of 15 minutes
- Benefits
- Network in/out must be encrypted using SSL
- IAM to centrally manage users instead of DB
- Can leverage IAM Roles and EC2 instance profiles for easy integration

8.3.5 About RDS for Oracle - Exam Tips

- Backups
- Use RDS backups for backups and restore to Amazon RDS for Oracle
- Use Oracle RMAN (Recovery Manager) for backups and restore to-non RDS (RDS not supported)
- Real Application Clusters (RAC)
- RDS for Oracle does NOT support RAC
- RAC is working on Oracle on EC2 instances because you have full control
- RDS for Oracle supports Transparent Data Encryption (TDE) to encrypt data before it's written to storage
- DMS work on Oracle RDS

8.3.6 About RDS for MySQL

- You can use the native mysqldump to migrate a MySQL RDS DB to non-RDS
- The external MySQL database can run either on-premises in your data center, or on an Amazon EC2 instance

8.3.7 RDS Proxy for AWS Lambda

- When using Lambda functions with RDS, it opens and maintains a database connection
- This can result in a ”TooManyConnections” exception
- With RDS Proxy, you no longer need code that handles cleaning up idle connections and managing connection pools
- Supports IAM authentication or DB authentication, auto-scaling
- The lambda function must be deployed in you VPC, because RDS Proxy is never publicly accessible

8.3.8 RDS Solution Architecture Cross Region Failover

8.4 Aurora - Part 1

- DB Engines: PostgreSQL-compatible and MySQL- compatible
- Storage: Automatically grows up to 256TB, 6 copies of data, multi-AZ
- Read Replicas: up to 15 RR, reader endpoint to access them all
- Cross Region RR: entire database is copied (not select tables)
- Load / Offload data directly from / to S3: efficient uuse of resources
- Backup, Snapshots and Restore: same as RDS

8.4.1 Aurora High Availability and Read Scaling

- 6 copies of your data across 3 AZ
- 4 copies out of 6 needed for writess
- 3 copies out of 6 need for reads
- self healing with peer-to-peer replication
- storage is striped across 100s of volumes
- Automated failover for master for less that 30 seconds
- Master + up to 15 Aurora Read Replicas serve reads
- Support for Cross Region Replication

8.4.2 Aurora DB Cluster

8.4.3 Aurora Endpoints

- Endpoints = Host Address + Port
- Cluster Endpoint (Writer Endpoint)
- Connects to the current primary DB instance in the Aurora cluster
- Used for all write operations in the DB cluster (inserts, updates, deletes and queries)
- Reader Endpoint
- Provides load-balancing for read only connection to all Aurora Replicas in the Aurora cluster
- Used only for read operations (queries)
- Custom Endpoint
- Represent a set on DB instances that you choose in the Aurora cluster

- Used when you want to connect to different subset of DB instances with different capacities and configurations (e.g, different DB parameter group)
- Instance Endpoint
- Connects to specific DB instances in the Aurora cluster
- Used when you want to diagnosis and fine tune a specific DB instance

8.4.4 Aurora - Customer Endpoints

- Define a subset of Aurora Instances as a Customer Endpoint
- Example: Run analytical queries on specific replicas
- The reader endpoint is generally not used after defining Customer Endpoints

8.4.5 Aurora Logs

- You can monitor the following types of Aurora MySQL log files:
- Error log
- Slow query log
- General log
- The audit log
- These log files are either downloaded or published to CloudWatch Logs

8.4.6 Troubleshooting RDS and Aurora Performance

- Performance Insights: find issues by waits, SQL statements, hosts and users
- CloudWatch Metrics: CPU, Memory, Swap Usage
- Enhanced Monitoring Metrics: at host level, process view, per second metric
- Slow Query Logs

8.4.7 Aurora Serverless

- Automated database instantiation and auto-scaling based on actual usage
- Good for infrequent, intermittent or unpredictable workloads
- No capacity planing needing
- Pay per second, can be most cost-effective

8.4.8 Aurora Serverless - Data API

- Access Aurora Serverless DB with a simple API endpoint (no JDBC connection needed)
- Secure HTTPS endpoint to run SQL statements
- No persistent DB connections management
- Users must be granted permissions to Data API and Secrets Manager (where credentials are checked)

8.4.9 RDS Proxy for Aurora

- Ability to create an additional readendpoint that connects to Aurora Read Replicas only

8.4.10 Global Aurora

- Aurora Cross Region Read Replicas
  - Useful for disaster recovery
  - Simple to put in place
- Aurora Global Database (recommended)
  - 1 Primary Region (read/write)
  - Up to 10 secondary (read-only) regions; replication lag is less than 1 second
  - Up to 16 Read Replicas per secondary region
  - Helps decrease latency
  - Promoting another region (for disaster recovery) has an RTO of less than 1 minute
  - Ability to manage the RPO in Aurora PostgreSQL

8.4.11 Aurora Global - Write Forwarding

- Enables Secondary DB Clusters to forward SQL statements that perform write operations to the Primary DB Cluster
- Data is always changed first on the Primary DB Cluster, then replicated to the Secondary DB Clusters
- Primary DB Cluster always has an up-to-date copy of all data
- Reduce the number of endpoints

8.4.12 Covert RDS to Aurora

Chapter 9

Service Communication

9.1 Step Functions

- Build serverless virtual workflow to orchestrate you Lambda functions
- Represent flow as a JSON state machine
- Features: sequence, parallel, conditions, timeouts, error handling........
- Maximum execution time of 1 year
- Possibility to implement human approval feature
- If you chain Lambda functions using Step functions, be mindful of the added latency to pass the calls

9.1.1 Visual Work flow in Step Functions

9.1.2 Step Function Integrations

- Optimised Integrations
  - Can invoke a Lambda function
  - Run an AWS batch job
  - Run an ECS task and wait for it to complete
  - Insert an item from DynamoDB
  - Publish message to SNS, SQS
  - Launch an EMR, Glue or SageMaker jobs
  - Launch another Step Function workflow
- AWS SDK Integrations
  - Access 200+ AWS services from your State machine
  - Works like a standard AWS SDK API call

9.1.3 Step Functions Workflow Triggeers

You can invoke a step function workflow (State Machine) using:

- AWS Management Console
- AWS SDK (StartExecution API call)
- AWS CLI (start-execution)
- AWS Lambda (StartExecution API call)
- API Gateway
- EventBridge
- CodePipeline
- Step Functions

9.1.4 Step Funcitons - Sample Projects

https://console.aws.amazon.com/states/home?region=us-east- 1/sampleProjects

9.1.5 Step Functions - Tasks

- Lambda Tasks:
  - Invoke a Lambda function
- Activity Tasks:
  - Activity work (HTTP), EC2 Instances, mobile device, on premise DC
  - They poll the step functions service
- Service Tasks:
  - Connect to a supported AWS service
  - Lambda function, ECS Task, Fargate, DyanamoDB, Batch job SNS topic, SQS queue
- Wait Task
  - To wait for a duration or until a timestamp
- **Note: Step functions does not integrate natively with AWS Mechanical Turk**

9.1.6 Step Functions - Standard vs. Express

- Standard Workflows Maximum duration - 1 year Supported execution start rate - over 2000 per second Supported state transition rate - over 4000 per second per account Pricing - Priced per state transition. A state trasition is counted each time a step in your execution is complete (move expensive) Execution - Executions can be listed and described with Step Functions APIs, and visually debugged through the console They can also be inspected in CloudWatch logs by enabling loggging on your state machine Execution semantics - Exactly once workflow execution
- Express Workflows Maximum duration - 5 minutes Supported execution state rate - over 100,000 per second Supported state transition rate - Nearly unlimited Pricing - Priced by the number of executions you run, their duration and memeory consumption. (cheaper) Execution history - Executions can be inspected in CloudWatch logs by enabling logging on your state machine Execution semantics - At-least once workflow execution

9.1.7 Step Functions - Express Workflow - Synchronous vs. Asynchoronous

- Synchronous Express Workflows
  - Wait until the Workflow completes, then return the result
  - Examples: orchestrate microservices, handler errors, retries, parallel tasks

  - Asynchronous Express Workflows
  - Doesn’t wait? for the Workflow to complete
  - Examples: Workflow that don’t require immediate response messaging

9.1.8 Step Functions - Error handling

- You can enable error handling, retries and add alerting to Step Function State Machine
- Example: set up EventBridge to alert via email if a State Machine execution failes

9.1.9 Step Functions - Solution Architecture

9.2 SQS

- Serverless, managed queue, integrated with IAM
- Can handle extreme scale, no provisioning required
- Used to decouple services
- Message size of max 1024KB (use a pointer to S3 for large message)
- Can be read from EC2 (optional ASG), Lambda
- SQS could be used as a write bugger for DynamoDB
- SQS FIFO
- receive message in order they were sent
- 300 messages/s without batching , 3000/s with batching

9.2.1 Amazon SQS - Dead Letter Queue (DLQ)

- If a consumer fails to process a message within the visibility timeout....the message goes back to the queue
- We can set a threshold of how many times a message can go back to the queue
- After the MaximumReceives threshold is exceeded, the message goes into a Dead Letter Queue (DLQ)
- Useful for debugging
- DLQ of a FIFO queue must also be a FIFO queue
- DLQ of standard queue must also be a standard queue
- Make sure to process the messages in the DLQ before they expire
- Good to set a retention of 14 days in the DLQ

9.2.2 SQS DLQ - Redrive to source

- Feature to help consume messages in the DLQ to understand what is wrong with them
- When our code is fixed we can redrive the message from the DLQ back into the source queue (or any other queueu) in batch without writing customer code

9.2.3 SQS - Solution Architecture Idempotency

- Messages can be processed twice by consumer (in case of failures, timeouts, etc)
- To hedge against that problem, implement idempotency at the consumer level
- Means the same action done twice by the consumer won't duplicate the effect

9.2.4 Lambda - Event Source Mapping SQS and SQS FIFO

- Event Source Mapping will poll SQS (Long polling)
- Specify batch size (1-10 messages)
- Recommended: Set the queue visibility timeout to 6x the timeout of your Lambda function
- To use a DLQ
  - Set-up on the SQS queue not Lambda (DLQ for Lambda is only for async invocations)
  - Or use a Lambda destination for failures

9.2.5 SQS - Solution Architecture Request / Response queue (async)

Decoupling

Fault Tolerance

Load Balancing

9.3 Amazon MQ

- SQS, SNS are "cloud" services: proprietary protocol from AWS
- Traditional applications running from on-premises may use open protocols such as: MQTT, AMQP, STOMP, Openwire, WSS
- When migrating to the cloud, instead of re-engineering the application to use SQS and SNS we can use amazon MQ
- Amazon MQ is a managed message broker service for rabbit MQ/ Active MQ
- Amazon MQ doesn't "scale" as much as SQS / SNS
- Amazon MQ runs on servers, can run in Multi-AZ with failover
- Amazon MQ has both queue feature (roughly equal to SQS) and topic features (roughly equal to SQS)

9.3.1 Amazon MQ - Re-platform

IBM MQ, TIBCO EMS, Rabbit MQ and Apache ActiveMQ can be migrated to Amazon MQ

9.4 Amazon SNS

9.4.1 Amazon SNS

- What if you want to send one message to many receivers?
- The ”event producer” only sends message to one SNS topic
- As many ”event receivers” (subscriptions) as we want to listen to the SNS topic notifications
- Each subscriber to the topic will get all the message (note: new feature to filter messages)
- Up do 12,500,000 subscriptions per topic
- 100,000 topics limit

9.4.2 SNS integrates with a lot of AWS serivces

- Many AWS services can send data directly to SNS for notifications

9.4.3 Amazon SNS - How to publish

- Topic Publish (using the SDK)
  - Create a topic
  - create a subscription
  - publish to the topic
- Direct Publish (for mobile apps SDK)
  - Create a platform appilcation
  - Create a platform endpoint
  - Publish to the platform endpoint
  - Works with Google GCM, Apple APNS, Amazon ADM

9.4.4 Amazon SNS - Security

- Encryption
  - In-flight encryption using HTTPS API
  - At-rest encryption using KMS keys
  - Client-side encryption if the client wants to perform encryption / decryption itself
- Access control: IAM policies to refulate access to the SNS API
  - SNS Access Policies (similar to S3 bucket policies)
  - Useful for cross-account access to SNS topics
  - Useful for allowing other services (S3....) to write to an SNS topic

9.5 Amazon SNS - SQS Fan Out Pattern

9.5.1 SNS + SQS: Fan out

- Push once in SNS, recieve in all SQS queue that are subscribers
- Full decoupled, no data loss
- SQS allows for dataa: persistence, delayed processing and retries of work
- Ability to add more SQS subscribers over time
- Make sure your SQS queue access policy allows for SNS to write
- Cross-Region Delivery: works with SQS queues in other regions

9.5.2 Application: S3 Events to multiple queues

- For the same combination of: event type (e.g object create) and prefix (e.g. images/) you can only have one S3 event rule
- If you want to send the same S3 event to many SQS queues, use fan-out

9.5.3 Application: SNS to Amazon S3 through Kinesis Data Firehose

SNS can send to Kinesis and therefore we can have the following solutions architecture

9.5.4 Amazon SNS - FIFO Topic

- FIFO - First in First Out (ordering of messages in the topic)
- Similar features as SQS FIFO:
- Ordering by Message Group ID (all messages in the same group are ordered)
- Deduplication using a Deduplication ID or Content Based Deduplication
  - Can have SQS Standard and FIFO queues as subscribers
  - Limited throughput (same throughput as SQS FIFO)

9.5.5 SNS FIFO + SQS FIFO: Fan Out

In case you need fan out + ordering + deduplication

9.5.6 SNS - Message Filtering

JSON policy used to filter messages sent to SNS topics subscriptions. If a subscription doesn't have a filter policy, it receives every message

9.6 Amazon SNS - Message Delivery Retries

9.6.1 SNS - Message Delivery Retries

When a message is delivered to an SNS subscriber, case of service-side errors, a delivery policy is applied

AWS Managed endpoints

- Delivery protocols - Amazon Kinesis Data Firehose, AWS Lambda, Amazon SQS
- Immediate retry (no delay) phase - 3 times withouth delay
- Pre-backoff phase - 2 times, 1 second apart
- Backoff - 10 times with exponential backoff, from 1 seconds to 20 seconds
- Post-backoff phase - 100,000 times, 20 seconds apart
- Total attempts - 100,015 over 23 days

Customer managed endpoints

- Delivery protocols - SMTP
- Immediate retry (no delay) phase - 0 times without delay
- Pre-backoff - 2 times, 10 seconds apart
- Backoff phase - 10 times with exponential backoff from 10 seconds to 600 seconds (10 minutes)
- Post-backoff - 38 times, 600 seconds (10 minutes) apart
- Total attempts - 50 attempts, over 6 hours

9.6.2 SNS Customer Delivery Policies

Only HTTP/S supports custom policies

9.6.3 SNS - Dead Lette Queues

- After exhausting the delivery policy, messages that haven't been delivered are discarded unless you set a DLQ (Dead Letter Queue)
- DLQ are Amazon SQS queues or Amazon SQS FIFO queues (for SNS FIFO)
- Dead Letter Queues are attached to a subscription (rather than a topic)

Chapter 10

Data Engineering

10.1 Amazon Kinesis Data Streams

- Collect and store streaming data in real-time
- Producers - Applications, Kinesis Agent
- Amazon Kinesis Data Streams
- Consumers - Application, Lambda, Amazon Data Firehose, Managed Service for Apache Flink

Kinesis Data Streams

- Retention between up to 365 days.
- Ability to reprocess (replay) data by consumers
- Data can't be deleted from Kinesis (until it expires)
- Data up to 1MB (typical use case is a lot of "small" real-time data)
- Data ordering guarantee from data with the same "Partition ID"
- At-rest KMS encryption, in flight HTTPS encryption
- Kinesis Producer Library (KPL) to write an optimiser producer application
- Kinesis Client Library (KCL) to write an optimised consumer application

Kinesis Data Streams - Capacity Modes

- Provisioned mode:
- Choose number of shards
- Each shard gets 1MB/s in (or 1000 records per second)
- Each shard gets 2MB/s out
- Scale manually to increase or decrease the number of shards
- You pay per shard provisioned per hour
- On-demand mode:
- No need to provision or manage the capacity
- Default capacity provisioned (4 MB/s in or 4000 records per second)
- Scales automatically based on observed throughput peak during the last 30 days
- Pay per stream per hour and data in/out per GB

10.2 Amazon Data Firehose

10.2.1 AWS Kinesis Data Firehose

- Full Managed Service, no administration
- Near Real Time (Buffer based on time and size, optionally can be disabled)
- Load data into Redshift / Amazon S3 / OpenSearch / Splunk
- Automatic scaling
- Supports many data formats
- Data Conversion from JSON to Parquet / ORC (only for S3)
- Data Transformation through AWS Lambda (ex: CSV =¿ JSON)
- Supports compression when target is Amazon S3 (GZIP, ZIP and SNAPPY)
- Only GZIP is the data is further loaded into Redshift
- Pay for the amount of data going through Firehose
- Spack / KCL do not read from KDF

10.2.2 Kinesis Data Firehose Delivery Diagram

10.2.3 Firehose Buffer Sizing

- Firehose accumulates records in a buffer
- The buffer is flushed based on time and size rules
- Buffer Size (ex: 32MB): if that buffer size is reached, it's flushed
- Buffer Time (ex: 2 Minutes): if that time is reached, it's flushed
- Firehose can automatically increase the buffer size to increase throughput
- High throughput =¿ Buffer Size will be hit
- Low throughput =¿ Buffer Time will be hit

10.2.4 Kinesis Data Streams vs Firehose

- Streams
- Going to write custom code (producer / consumer)
- Real Time (200ms latency for classic, 70ms latency for enhanced fan-out)
- Must manage scaling (shard splitting / merging)
- Data storage for 1 to 365 days, replay capability, multi consumers

- Use with Lambda to insert data in real-time to OpenSearch (for example)
- Firehose
- Full managed, send to S3, Splunk, Redshift, OpenSearch
- Serverless data transformation with Lambda
- Near real time
- Automated Scaling
- No data storage

10.3 Amazon Managed Service for Apache Flink

- Previously named: Kinesis Data Analytics for Apache Flink
- Flink (Java, Scala or SQL) is a framework for processing data streams
- Run any Apache Flink application on a managed cluster on AWS
- Provisioned compute resources, parallel computation, automatic scaling.
- Application backups (implemented as checkpoint and snapshots)
- Use any Apache Flink programming features to transform data
- Important: Flink does not read from Amazon Data Firehose

10.4 Streaming Architectures

10.4.1 Full Data Engineering Pipeline - Real-Time Layer

10.4.2 Streaming Architectures - 3000 messages of 1kb per second

- Kinesis
- 3 shard: 3MB/s in
- 3 * 0.015 = 32.4/mth
- Must use KDF for output to S3
- DynamoDB + Streams
- 3000 WCU = 3 MB/s
- 1450.90 / month
- Storage in DynamoDB

10.4.3 Comparison Charts

10.5 Amazon MSK

10.5.1 Amazon Managed Streaming for Apache Kafka (Amazon MSK)

- Alternative to Amazon Kinesis
- Fully managed Apache Kafka on AWS
- Allow you to create, update, delete clusters
- MSK creates and manages Kafka brokers nodes and Zoopkeeper nodes for you
- Deploy the MSK cluster in your VPC, multi-AZ (up to 3 for HA)
- Automatic recovery from common Apache Kafka failures
- Data is stored on EBS volumes for as long as you want
- MSK Serverless
- Run Apache Kafka on MSK without managing the capacity
- MSK automatically provisions resources and scales compute and storage

10.5.2 Apache Kafka at a high level

10.5.3 Kinesis Data Streams vs. Amazon MSK

- Kinesis Data Streams
- 1MB message size limit
- Data Streams with Shards
- Shard Splitting and Merging
- TLS In-flight encryption
- KMS at-rest encryption
- Amazon MSK
- 1MB default, configure or higher (ex: 10MB)
- Kafka Topics with Partitions
- Can only add partitions to a topic
- PLAINTEXT or TLS In-Flight Encryption
- KMS at-rest encryption

10.5.4 Amazon MSK Consumers

- Kinesis Data Analytics for Apache Flink
- AWS Glue - Streaming ETL Jobs - Powered by Apache Spark Streaming
- Lambda
- Applications running on Amazon EC2, ECS, EKS

10.6 AWS Batch

- Run batch jobs as Docker images
- Two options:
- Run on AWS Fargate (full serverless offering)
- Dynamic provisioning of the instances (EC2 and Spot Instances) - in VPC
- Optimal quantity and type based on volume and requirements
- No need to manage clusters, fully serverless
- You just pay for the underlying resources used.
- Example: batch process of images, running thousands of concurrent jobs
- Schedule Batch Jobs using Amazon EventBridge
- Orchestrate Batch Jobs using AWS Step Functions

10.6.1 AWS Batch - Solution Architecture

10.6.2 Batch vs. Lambda

- Lambda
- Time limit
- Limited runtimes (built in runtimes, or Docker images built for Lambda)
- Limited temporary disk space
- Serverless
- Batch
- No time limit
- Any runtime as long as it's packaged as a Docker image
- Rely on EBS / instance store for disk space
- Relies on EC2 (can be managed by AWS) or AWS Fargate

10.6.3 AWS Batch - Compute Environments

- Managed Compute Environments
- AWS Batch managed the capacity and instances types within the environment
- You can choose EC2 On-Demand or Spot Instances
- You can choose Fargate On-Demand or Fargate Spot Instances
- You can set a maximum price for Spot Instances
- Launched within your own VPC
- If you launch within your own private subnet, make sure it has access to the ECS services
- Either using a NAT gateway / instance or using VPC Endpoints for ECS
- Unmanaged Computer Environment
- You control and manage EC2 instance configuration, provisioning and scaling

10.6.4 AWS Batch - Managed Compute Environment

10.6.5 AWS Batch - Multi Node Mode

- Multi Node: large scale, good for HPC
- Leverages multiple EC2 / ECS instances at the same time.
- Good for tightly coupled workloads
- Represents a single job and specified how many nodes to create for the job
- 1 mean node and many child notes.
- Does not work with Spot Instances
- Works better if your EC2 launch mode is a placement group "cluster"

10.7 Amazon EMR

- EMR stands for "Elastic MapReduce"
- EMR helps creating Hadoop clusters (Big Data) to analyse and process vast amounts of data
- The clusters can be made of hundreds of EC2 instances
- Also support Apache Spark, HBase, Presto, Flink
- EMR takes care of all the provisioning and configuration of EC2
- Auto-scaling with CloudWatch
- Use cases: data processing, machine learning web indexing big data..........

10.7.1 EMR - Integrations

10.7.2 Amazon EMR - Node types and purchasing

- Master Node: Mange the cluster, coordinates, manage health - long running
- Core Node: Run tasks and store data - long running
- Task Node (optional): Just to run tasks - usually Spot
- Purchasing options:
- On-demand: reliable, predictable, won't be terminated
- Reserved (min 1 year): cost savings (EMR will automatically use if available)
- Spot Instances: cheaper can be terminated, less reliable
- Can have long-running cluster or transient (temporary) cluster.

10.7.3 Amazon EMR - Instance Configuration

- Uniform instance groups: select a single instance type and purchasing option for each node (has auto scaling)
- Instance fleet: select target capacity, mix instance types and purchasing options (no Auto Scaling)

10.8 Running Jobs on AWS

Strategy 1: Provision EC2 Instance (long running - CRON jobs) Strategy 2: Amazon EventBridge + Lambda (cron) Strategy 3: Reactive Workflow - EventBridge, S3 Events, API Gateway, SQS, SNS etc. Strategy 4: use AWS batch - EventBridge, Batch Strategy 5: Use fargate - EventBridge -¿ Fargate Strategy 6: Use EMR (step execution or cluster)

10.9 AWS Glue

- Managed extract, transform and load (ETL) service
- Useful to prepare and transform data for analytics
- Fully serverless service

10.9.1 Glue Data Catalog

- Glue Data Catalog: catalog of datasets

10.10 Redshift

- Redshift is based on PostgreSQL but it's not used for OLTP
- Its OLAP - online analytical processing (analytics and data warehousing)
- 10x better performance that other data warehouses, scale to PBs of data
- Columnar storage of data (instead of row based)
- Massively Parallel Query Execution (MPP)
- Two modes: Provisioned cluster or Serverless cluster
- Has a SQL interface for performing the queries
- BI tools such as AWS Quicksight or Tableau integrate with it

10.10.1 Redshift Continued.......

- Data is loaded from S3, Kinesis Firehose, DynamoDB, DMS......
- Based on node type: up to 100+ nodes, up to 16TB of space per node
- Can provision multiple nodes, with Multi-AZ only for some clusters
- Leader node: for query planning, results aggregation
- Computer node: for performing the queries, send result to leader
- Backup and Restore, Security VPC / IAM / KMS, Monitoring
- Redshift Enhanced VPC Routing: COPY / UNLOAD goes through VPC
- Redshift is provisioned, so it's worth it when you have a sustained usage (use Athena if the queries are sporadic instead)

10.10.2 Redshift - Snapshots and DR

- Snapshots are point-in-time backups of a cluster, stored internally in S3
- Snapshots are incremental (only what hass changed is saved)
- You can restore a snapshot into a new cluster
- Automated: every 8 hours, every 5 GB, or on a scheduled. Set retention.
- Manual: snapshot is retained until you delete it.
- You can configure Amazon Redshift to automatically copy snapshots (automated or manual) of a cluster to another AWS region

10.10.3 Cross-Region Snapshot Copy for an KMS-Encrypted Redshift

- Enables Redshift to perform encryption operations in the destination Region

- Redshift Spectrum
- Query data the is already in S3 without loading it.
- Must have a Redshift cluster available to start the query
- The query is then submitted to thousands of Redshift Spectrum nodes.

10.10.4 Redshift Workload Management (WLM)

- Enables you to flexibly manage queries priorities within workloads
- Example: Prevent short, fast-running queries from getting stuck behind long-running queries
- Define multiple query queues (Superuser queue, User-defined queues)
- Route queries to the appropriate queues at runtime
- Automatic WLM - queues and resources managed by Redshift
- Manual WLM - queue and resources managed by you

10.10.5 Redshift Concurrency Scaling

- Enables you to provide consistently fast performance with virtually unlimited concurrent users and queries
- Redshift automatically adds additional cluster capacity (Concurrency-Scaling Cluster) to process an increase in requests.
- Ability to decide which queries sent to the concurrency-Scaling Cluster using WLM
- Charged per second

10.11 Amazon DocumentDB

- Aurora is an "AWS-implementation" of PostgreSQL / MySQL
- DocumentDB is the same for MongoDB (which is a NoSQL database
- MongoDB is used to store query and index JSON data
- Similar "deployment concepts" as Aurora
- Fully Manged, highly available with replication across 3 AZ
- DocumentDB storage automatically grows in increments of 10GB
- Automatically scales to workloads with millions of requests per seconds

10.11.1 DocumentDB - Pricing

- Pay for what you use, no upfront costs
- On-demand Instances (per second with minimum of 10 minutes)
- Database I/O - amount of I/O used when read and write (per million I/Os)
- Database Storage (per GB/month)
- Backup storage (per GB/month)

10.12 Amazon Timestream

- Fully managed, fast, scalable, serverless time series database
- Automatically scales up/down to adjust capacity
- Store and analyse trillions of events per day
- 1000s times faster and 1/10th the cost of relational databases
- Schedules queries, multi-measure records, SQL compatibility
- Data storage tiering, recent data kept in memory and historical data kept in a cost-optimised storage
- Built-in time series analytics functions (helps you identify patterns in your data in near real-time)
- Encryption in transit and at rest
- Use cases: IoT apps, operational applications, real-time analytics.........
- Amazon Time-stream - Architecture

10.13 Amazon Athena

- Serverless query service to analyse data stored in Amazon S3
- Uses standard SQL language to query the files (built on Presto)
- Supports CSV, JSON, ORC, Avro and Parquet
- Pricing: 5.00 per TB of data scanned
- Commonly used with Amazon Quicksight for reporting/dashboards
- Use cases: Business intelligence / analytics / reporting, analyse and query VPC Flow Logs, ELB Logs, CloudTrail trails, etc....... —–¿ Exam Tip: analyse data in S3 usign serverless SQL, use Athena

10.13.1 Amazon Athena - Performance Improvement

- Using columnar data for cost-savings (less scans)
- Apache Parquet or ORC is recommended
- Huge performance improvement
- Use Glue to convert your data to Parquet or ORC
- Compress data for smaller retrievals (bzip2 gzipm lz4, snappy, zlip, zstd)
- Partition datasets in S3 for easy querying on virtual columns
- Use larger files (¿ 128 MB) to minise overhead

10.13.2 Amazon Athena - Federated Query

- Allows you to run SQL queries across data stored in relational, non-relational object and custom data sources (AWS or on-premises)
- Uses Data Source Connectors that run on AWS Lambda to run Federated Queries (e.g CloudWatch logs, DynamoDB, RDS)
- Store the results back in Amazon S3

10.14 Amazon QuickSight

- Serverless machine learning-powered business intelligence service to create interactive dashboards
- Fast automatically scalable, embeddable, with per-session pricing
- Use cases:
- Business analytics
- Building visualisation
- Perform ad-hoc analysis
- Get business insights using data
- Integrated with RDS, Aurora, Athena, Redshift, S3......
- In-memory computation using SPICE engine if data is imported into QuickSight
- Enterprise edition: Possibility to setup Column-Level security (CLS)

10.14.1 QuickSight Integrations

10.14.2 QuickSight - Dashboard and Analysis

- Define Users (standard versions) and groups (enterprise version)
- These users and groups only exist within QuickSight not IAM!!
- A dashboard...
- Is a read-only snapshot of an analysis that you can share
- preserves the configuration of the analysis (filtering, parameters, controls, sort)
- You can share the analysis or the dashboard with Users or Groups
- To share a dashboard, you must first publish it.
- Users who see the dashboard can also see the underlying data

10.15 Big Data Architecture

10.15.1 Full Data Engineering Pipeline - Analytics layer

- Amazon EMR - Hadoop / Spark / Hive
- Redshift / Redshift Spectrum —- Data Warehousing
- Amazon Athena Serverless
- Visualisation Amazon QuickSight

10.15.2 Big Data Ingestion Pipeline

- IoT devices
- Amazon Kinesis Data Streams
- Amazon Kinesis Data Firehose
- Amazon Simple Storage Service (S3)
- Amazon Simple Queue Service (Optional)
- AWS Lambda
- Amazon Athena
- Reporting Bucket - Amazon Simple Storage (S3)
- Amazon QuickSight ¡—-¿ Amazon Redshift

10.15.3 Comparison of warehousing technologies

- EMR
- Need to use Big Data tools such as Apache Hive, Spark
- One long-running cluster, many jobs, with auto-scaling or one cluster per job?
- Purchasing options - Spot, On Demand, Reserved Instances
- Can access data in DynamoDb and / or S3
- Scratch data on EBS disks (HDFS) and long term storage in S3 (EMRFS)
- Athena
- Simple queries and aggregations, data must live in S3
- Serverless, simple SQL queries out-of-the-box queries for many services (cost and billing)
- Audit queries through CloudTrail
- Redshift
- Advanced SQL queries must provision servers
- Can leverage Redshift Spectrum for serverless queries on S3

Chapter 11

Monitoring

11.1 CloudWatch

11.1.1 CloudWatch Metrics

- Provided by many AWS services
- EC2 standard: 5 minute, detailed monitoring: 1 minute
- EC2 RAM is not a built-in metric
- Can create customer metrics: standard resolution 1 minute, high resolution 1 sec

11.1.2 CloudWatch Alarms

- Can trigger actions: EC2 action (reboot, stop, terminate, recover), Auto Scaling, SNS
- Alarm events can be intercepted by Amazon EventBridge

11.1.3 CloudWatch Dashboards

- Display metrics and alarms
- Can show metrics of multiple regions

11.1.4 CloudWatch Alarms Integrations

11.1.5 CloudWatch Syntethics Canary

- Configurable script that monitors your APIs, URL, Websites,......
- Reproduce what your customers do programmatically to find issues before customers are impacted
- Check the availability and latency of your endpoints and can store load time data and screen shots of the UI
- Integration with CloudWatch Alarms
- Scripts written in Node.js or Python
- Programmatic access to a headless Google Chrome browser
- Can run once or on a regular schedule

11.1.6 CloudWatch Synthetics Canary Blueprints

- Heartbeat Monitor - load URL, store screenshot and an HTTP archive files
- API Canary - test basic read and write functions of REST APIs
- Broken Link Checker - check all links inside the URL that you are testing
- Visual Monitoring - compare a screenshot taken during a canary run with a baseline screen shot

- Canary Recorder - used with CloudWatch Synthetics Recorder (record your actions on a website and automatically generates a script for that)
- GUI Workflow Builder - verifies that action can be taken on your webpage (e.g test a webpage with a login form)

11.2 CloudWatch Logs

11.2.1 AWS Cloudwatch Logs - Sources

- SDK, CloudWatch Logs Agent, CloudWatch Unified Agent
- Elastic Beanstalk: collection of logs from application
- ECS: collection from containers
- AWS Lambda: Collection from function logs
- VPC Flow Logs: VPC specific logs
- API Gateway
- CloudTrail based on filter
- CloudWatch log agents: for examples on EC2 machines
- Route53: Log DNS queries

11.2.2 CloudWatch Logs

- Log groups: arbitrary name, usually representing an application
- Log stream: instances within application / log files / containers
- Can define log expiration policies (never expire, 30 days, etc.......)
- Optional KMS encryption
- CloudWatch Logs can send logs to:
  - Amazon S3 (exports)
  - Kinesis Data Streams
  - Kinesis Data Firehose
  - AWS Lambda
  - ElasticSearch

11.2.3 CloudWatch Logs Metric Filter and Insights

- CloudWatch Logs can use filter expressions
- For example, find a specific IP inside of a log
- Or count occurrences of "ERROR" in your logs
- Metric filters can be used to trigger alarms

z CloudWatch Logs Insights can be used to query logs and add queries to CloudWatch Dashboards

11.2.4 CloudWatch Logs - S3 Export

- S3 buckets must be encrypted with AES-256 (SSE-S3) or SSE-KMS
- Log data can take up to 12 hours to become available for export
- The API call is CreateExportTask
- Not near-real time or real-time..... use Logs Subscriptions instead

11.2.5 CloudWatch Logs Subscription

11.2.6 CloudWatch Logs Aggregation - Multi-Account and Multi Region

11.2.7 CloudWatch Agent - Integration with SSM

- Install CW Agent using SSM Run Command
- Install CW agent using SSM State Manager
- Configure CW Agent by storing config in SSM Parameter Store

11.3 Amazon EventBridge

11.3.1 Amazon EventBridge - formerly cloudwatch events

- Schedule: Cron jobs (scheduled scripts)
- Event pattern: Event rules to react to a service doing something
- Trigger Lambda functions, send SQS/ SNS messages

11.3.2 Amazon EventBridge Rules

11.3.3 Amazon EventBridge

- Event buses can be accessed by other AWS accounts using Resource-based Policies
- You can archive events (all/filter) sent to an event bus (indefinitely or set period)
- Ability to replay archived events

11.3.4 Amazon EventBridge - Schema Registry

- EventBridge can analyse the events in your bus and infer the schema
- The Schema Registry allows you to generate code for you application, that will know in advance how data is structured in the event bus
- Schema can be versioned

11.3.5 Amazon EventBridge - Resource-based Policy

- Manage permissions for specific Event Bus
- Example: allow/deny events from another AWS account or AWS region
- Use case: aggregate all events from your AWS Organisation in a single AWS account or AWS region

11.4 X-Ray

11.4.1 Visual analysis of our application

11.4.2 X-Ray

- Tracing requests across your microservices (distributed systems)
- Integrations with:
  - EC2 - install the X-Ray agent
  - ECS - install the X-Ray agent or Docker container
  - Lambda
  - Beanstalk - agent is automatically installed
- The X-Ray agent or services need IAM permissions to X-Ray

11.5 AWS Personal Health Dashboard

- Global service
- Show how AWS outages directly impact you
- Show the impact on your resources
- List issues and actions you can do to remediate them
- Will show **maintenance** events from AWS
- Programmatically accessible through the AWS Health API
- Aggregations across multiple accounts of an AWS Organisation
- https://phd.aws.amazon.com/

11.5.1 Health Event Notifications

- Use EventBridge (CloudWatch events) to react to changes for AWS Health events in your AWS account
- Examples: receive email notifications when EC2 instances in your AWS account are scheduled for updates
- Can't be used to return public events from the Service Health Dashboard
- Use cases: send notifications, capture event information, take corrective action,..........

Chapter 12

Deployment and Instance Management

12.1 Elastic Beanstalk

12.1.1 AWS Elastic Beanstalk Overview

- Elastic Beanstalk is a developer centric view of deploying an application on AWS
- It uses all the components we've seen before: EC2, Auto Scaling Group, Elastic Load Balancers, RDS, etc.......
- We still have full control over the configuration of each component
- Beanstalk is free but you pay for the underlying instances

12.1.2 Elastic Beanstalk

- Support for many platforms:
  - Go
  - Java SE
  - Java with Tomcat
  - .NET on Window Server with IIS
  - Node.js
  - PHP
  - Python
  - Ruby
  - Packer Builder
  - Single Container Docker
  - Multi-container Docker
  - Preconfigured Docker

- Beanstalk is great to "platform" your application from on-premises to the cloud

- Managed service
  - Instance configuration / OS is handled by Beanstalk
  - Deployment strategy is configurable by performed by Elastic Beanstalk
- Just the application code is the responsibility of the developer
- Three architecture models:
  - Single instance deployment: good for developers
  - LB + ASG: great for production or pre-production web applications
  - ASG only: great for non-web apps in production (workers. etc......)

12.1.3 Beanstalk Environments

12.1.4 Web Server vs Worker Environment

- If your application performs tasks that are long to complete, offload these tasks to a dedicated worker environment
- Decoupling you application into two tiers is common
- Example: processing a video, generating a zip file, etc
- You can define periodic task in a file cron.yaml

12.1.5 Elastic Beanstalk Deployment Blue / Green

- Not a ”direct feature” of Elastic Beanstalk
- Zero downtime and release facility
- Create a new ”stage” environment and deploy v2 there
- The new environment (green) can be validated independently and roll back if there are issues
- Route 53 can be setup using weighted policies to redirect a little bit of traffic to the stage environment
- Using Beanstalk ”swap URLs” (DNS swap) when done with the environment test

12.2 CodeDeploy

12.2.1 AWS CodeDeploy

- We want to deploy out application automatically to many EC2 instances
- These instances are no managed by Elastic Beanstalk
- There are several ways to handle deployment using open source tools (Ansible, Terraform, Chef, Puppet, etc......)
- We can use the managed Service AWS CodeDeploy
- CodeDeploy can deploy to: EC2, ASG, ECS and Lamabda

12.2.2 CodeDeploy to EC2

- Define how to deploy the application using appspec.yml + deployment strategy
- Will do in-place update your fleet of EC2 instances
- Can use hooks to verify the deployment after each deployment phase

12.2.3 CodeDeploy to ASG

- In place updates:
  - Updated current existing EC2 instances
  - Instances newly created by an ASG will also get automated deployments
- Blue / green deployment:
  - A new auto-scaling group is created (settings are copied)
  - Choose how long to keep the old instances
  - Must be using an ELB

12.2.4 CodeDeploy to AWS Lambda

- Traffic Shifting feature
- Pre and Post traffic hooks features to validate deployment (before the traffic shift starts and after it ends)
- Easy and automated rollback using CloudWatch Alarms
- SAM framework natively uses CodeDeploy

12.2.5 CodeDeploy to ECS

- Support for Blue/Green deployments for Amazon ECS and AWS Fargate
- Setup is done within the ECS service
- A new task set is created and traffic is re-routed to the new task test
- Then if everything is stable for X minutes, the old task set is terminated (so you have time to notice the issues)
- Supports Canary deployments (Canary10Percent5Minute)

12.3 CloudFormation

- Infrastructure as code (IaC) in AWS
- Portability of stacks across multiple accounts and regions
- Backbone of the elastic beanstalk service
- Backbone of the Service Catalog service
- Backbone of the SAM (Serverless Application Model) framework
- Must-know service as a developer / sysops / devops

12.3.1 Retaining Data on Deletes

- You can put a DeletionPolicy on any resource to control what happens when the CloudFormation template is deleted
- DeletionPolicy=Retain
  - Specify on resources to preserve / backup in case of CloudFormation deleted
  - To keep a resource, specify Retain (works for any resource / nested stack)
- DeletionPolicy=Snapshot:
  - EBS Volume, Elasticache Cluster, ElastiCache ReplicationGroup
  - RDS DBInstance, RDS DBCluster, Redshift Cluster
- DeletePolicy=Delete (default behaviour)
  - Note: for AWS::RDS::DBCluster resources, the default policy is Snapshot
  - Note: to delete an S3 bucket, you first empty the bucket of its content

12.3.2 CloudFormation Customer Resources (Lambda)

- You can define a Custom Resource in CloudFormation to address any of these use cases
- An AWS resource is not yet supported (new service for example)
- An on-premises resource
- Emptying an S3 bucket before being deleted
- Fetch an AMI id

12.3.3 CloudFormation - StackSets

- Create, update or delete stacks across multiple accounts and regions with a single operation
- Administrator account to create StackSets
- Trusted accounts to create, update, delete stack instances from StackSets
- When you updated a stack set, all associated stack instances are updated through all accounts and regions
- Enabled Automatic Deployment feature to autmatically deploy to accounts in AWS Organisations or OUs

12.3.4 CloudFormation Drift Overview

- CloudFormation allows you to create infrastructure
- But it doesn't protect you against manual configuration changes
- How do we know if our resources have drifted? -¿ Use cloud formation drift
- Detect drift on an entire stack or on individual resource within a stack

12.3.5 CloudFormation - Integration with Secrets Manager

12.3.6 CloudFormation - Resource Import

- Import existing resources into existing / new stack s
- You don't need to delete and re-create the resources as part of a CloudFormation stack
- During import operation, you'll ned
- A template that describes the entire stack (original stack resources and target resources to import)
- A Unique identifier for each target resource (ex BucketName for S3 buckets)
- Each resource to import must have a DeletionPolicy attribute (any value) and Identifier
- Can't import the same resource into multiple stacks

12.4 Service Catalog

12.4.1 AWS Service Catalog

- Users that are new to AWS have too many options and may create stacks that are not compliant / in line with the rest of the organisation
- Some users just want a quick self-service portal to launch a set of authorised products pre-defined by admins
- Includes: virtual machines, databases, storage options, etc......... —¿ AWS Service Catalog use case

12.4.2 Service Catalog Diagram

12.4.3 AWS Service Catalog

- Create and manage catalogs of IT services that are approved on AWS
- The ”products” are CloudFormation templates
- Ex:Virtual machine images, Servers, Software, Databases, Regions, IP address ranges
- CloudFormation helps ensure consistency and standardisation by Admins
- They are assigned to Portfolios (teams)
- Team are presented a self-service portal where they can launch the products
- All the deployed products are centrally managed deployed services
- Help with governance, compliance and consistency
- Can give user access to launching products withouth requiring deep AWS knowledge
- Integrations with ”self-service portals” such as ServiceNow

12.5 SAM - Serverless Application Model

- SAM = Serverless Application Model
- Framework for developing and deploying serverless applications
- All the configuration is YAML code. Examples:
- Lambda Functions (AWS::Serverless::Function)
- DynamoDB tables (AWS::ServerLess::SimpleTable)
- API Gateway (AWS::Serverless::API)
- StepFunction - State Machine (AWS::Serverless::StateMachine)
- SAM can help you to run Lambda, API Gateway, DynamoDB locally
- SAM can use CodeDeploy to deploy Lambda functions (traffic shifting)
- Leverages CloudFormation in the backend

12.5.1 CICD Architecture for SAM

12.6 AWS CDK - Cloud Development Kit

- Define your cloud infrastructure using a familiar language:
- JavaScript/ TypeScript, Python, Java and .NET
- The code is ”compiled” into a CloudFormation template (JSON/YAML)
- You can therefore deploy infrastructure and application runtime code together
- Great for Lambda functions
- Great for Docker containers in ECS / EKS

12.6.1 CDK Example

12.7 AWS Systems Manger - SSM

12.7.1 AWS Systems Manager Overview

- Helps you manage your EC2 and on-premises system at scal
- Get operational insights about the state of your infrastructure
- Easily detect problems
- Patching automation for enhancd compliance
- Works for both Windows and Linux OS

- Integrated with CloudWatch metrics / dashboards
- Integrated with AWS Config
- Free service

12.7.2 How Systems Manger Works

- We need to install the SSM agent onto the systems we controll
- Installed by default on Amazon Linux AMI and some Ubuntu AMI
- If an instance can't be controlled with System Manger, it's probably an issue with the SSM agent!
- Make sure the EC2 instances have a proper IAM role to allow System Manager actions

12.7.3 AWS Systems Manager Run Command

- Execute a document (script) or just run a command
- Run command across multiple instances (using resources groups)
- Rate Control / Error Control
- Integrated with IAM and CloudTrail
- No need for SSH
- Results in the console

12.7.4 AWS Systems Manager - Send Command before an ASG Instance is Terminated

- Perform any action before the ASG terminates an EC2 instance
- Create a ASG lifecycle hook that puts the instance in Terminating:Wait state
- Monitor the Terminating:Wait state using EventBridge
- Trigger a SSM automation document to perform the actions on the instance before termination

12.7.5 Systems Manager Patch Managers - Steps

1. Define a patch baseline to use (or multiple if you have multiple environments)
2. Define patch groups: define based on tags, example different environments (dev, test, prod) — use tag Patch Group
3. Define Maintenance Windows (schedule, duration, registered targets/ patch groups and registered tasks)

4. Add the AWS-RunPatchBaseline Run Command as part of the registered tasks of the Maintenance Window (works cross platform Linux and Windows)
5. Define Rate control (concurrency and error threshold) for the task
6. Monitor Patch Compliance using SSM inventory

12.7.6 Systems Manager / Session Manager

- Allows you to start a secure shell on your EC2 and on-premises servers
- Access through AWS Console, AWS CII or Session Manager SDK
- Does not need SSH access, bastion hosts or SSH keys
- Supports Linux, macOS and Windows
- Log connections to your instances and executed commands
- Session log data can be sent to S3 or CloudWatch Logs
- CloudTrail can intercept StartSession events

12.7.7 Systems Manager OpsCenter

- Resolve Operational Issues (OpsItems) related to AWS resources
- OpsItems - issues, events and alerts
- Aggregates information to resolve issue on each OpsItems such as:
- AWS Config changes and relationships
- CloudTrail Logs
- CloudWatch Alarms
- CloudFormation Stack Information
- Provides Automation Runbook that you can resolve issues
- EventBridge or CloudWatch Alarms can create OpsItems

12.8 AWS Cloud Map

12.8.1 AWS Cloud Map

- A fully managed resource discovery service
- Creates a map of the backend services / resources that your application depend on
- You register your application components, their locations, attributes and health status with AWS Cloud Map
- Integrated health checking (stop sending traffic to unhealthy endpoints)
- Your applications can query AWS Cloud Map using AWS SDK, API or DNS

Chapter 13

Cost Control

13.1 Cost Allocation Tags

- With Tags we can track resources that relate to each other
- With Cost Allocation Tags we can enable detailed costing reports
- Just like Tags, but they show up as columns in Reports
- AWS Generated Cost Allocation Tags
  - Automatically applied to the resource you create
  - Start with Prefix aws: (e.g. aws: createdBy)
  - They're not applied to resources created before the activation
- User tags
  - Defined by the user
  - Start with Prefix user:
- Cost Allocation Tags just appear in the Billing Console
- Takes up to 24 hours for the tags to show up in the report

13.2 AWS Tag Editor

- Allows you to managed tags of multiple resources at once
- You can add/update/delete tags
- Search tagged/untagged resources in all AWS Regions

13.3 Trusted Advisor

- No need to install anything - high level AWS account assessment
- Analyse your AWS accounts and provides recommendation
- Cost Optimisation
- Performance
- Security
- Fault Tolerance
- Service Limits
- Operational Excellence - Core Checks and recommendations - all customers
- Can enable weekly email notification from the console
- Full Trusted Advisor - Available for Business and Enterprise support plans

- Ability to set CloudWatch alarms when reaching limits
- Programmatic Access using AWS support API

Column1	Basic Support
AWS Trusted Advisor Best Practice Checks	7 Core Checks
Enhanced Technical Support	24x7 customer service, documentation, whitepapers and support
Case Severity / Response Times	Data12
Data16	Data17

Table 13.1: Example Table with 4 Rows and 5 Columns

Trusted Advisor - Good to Know

- Can check if an S3 bucket is made public
- But cannot check for S3 objects that are public inside of your bucket
- Use Amazon EventBridge / S3 Events instead / AWS Config Rules
- Service Limits
  - Limits can only be monitored in Trusted Advisor (cannot be changed)
  - Cases must be created manually in AWS Support Centre to increase limits
  - OR use the AWS Service quotas service

13.4 AWS Service Quotas

- Notify you when you're close to a service quota value threshold
- Create CloudWatch Alarms on the Service Quotas console
- Example: Lambda concurrent executions
- Helps you know if you need to request a quota increase or shutdown resources before limit is reached

13.5 EC2 Launch Types and Savings Plans

- On Demand Instances - short workload, predictable pricing, reliable.
- Spot Instances - short workloads for check, can lose instances (not reliable)
- Reserved: (Minimum 1 year)
- Reserved Instances - long workloads
- Convertible Reserved Instances - long workloads with flexible instances

- Dedicated Instances: no other customers will share you hardware
- Dedicated Hosts: book an entire physical server, control instance placement
- Great for software licenses that operate at the core, or socket level
- Can define host affinity so that instance reboots are kept on the same host

13.5.1 AWS Savings Plan

- New pricing model to get a discount based on long-term usage
- Commit to a certain type of usage: ex $10 per hour for 1 to 3 years
- Any usage beyond the savings plan is billed at the on-demand price
- EC2 Instance Savings plan (72% - same discount as Standard RIs)
- Select instance family and locked to a specific region
- Flexible across size, OS (Windows to Linux) tenancy. (dedicated or default)
- Compute Savings Plan
- Ability to move between instance family, region, compute type and OS and tenancy
- SageMaker Savings plan (up to 64% off)

13.6 S3 Cost Savings

13.6.1 S3 Storage Classes

- Amazon S3 Standard - General Purpose
- Amazon S3 Standard-Infrequent Access (IA)
- Amazon S3 One Zone-Infrequent Access
- Amazon S3 Glacier Instant Retrieval
- Amazon S3 Glacier Flexible Retrieval
- Amazon S3 Glacier Deep Archive
- Amazon S3 Intelligent Tiering

Can move between classes manually or using S3 lifecycle configurations

13.6.2 S3 - Other Cost Savings

- S3 Lifecycle Rules: transition objects between tiers
- Compress Objects - to save space
- S3 Requester Pays:
- In general, bucket owners pay for all Amazon S3 storage and data transfer costs associated with their bucket
- With Requester Pays buckets, the requester instead of the bucket owner pays the cost of the request and the data downloaded from the bucket
- The bucket owner always pays the cost of storing data
- Helpful when you want to share large datasets with other accounts
- If an IAM role is assumed the owner account of that role pays for the request

13.7 S3 Storage Classes - Reminder

13.7.1 S3 Storage Classes

  - Amazon S3 Standard - General Purpose
  - Amazon S3 Standard-Infrequent Access (IA)
  - Amazon S3 One Zone-Infrequent Access
  - Amazon S3 Glacier Instant Retrieval
  - Amazon S3 Glacier Flexible Retrieval
  - Amazon S3 Glacier Deep Archive
  - Amazon S3 Intelligent Tiering

Can move between classes manually or using S3 lifecycle configurations

13.7.2 S3 Durability and Availalbility

- Durability:

  - High Durability (99.9999999999, 11 nines) of objects across multiple AZ
  - If you store 10,000,000 object with Amazon S3, you can on average expect to incur a loss of a single object once every 10,000 years (nice....but basic math)
  - Same for all storage classes

- Availability

  - Measures how readily available a service is
  - Varies depending on storage class
  - Example: S3 standard has 99.99% availability = not available 53 minutes a year

S3 Standard - General Purpose

- 99.99% Availability
- Used for frequently accessed data
- Low latency and high throughput
- Sustain 2 concurrent facility faiures
- Use Cases: Big Data analytics, mobile and gaming applications, content and distribution

S3 Storage Classes - Infrequent Access

- For data that is less frequently accessed, but requires rapid access when needed
- Lower cost than S3 standard

  Amazon S3 Standard-Infrequent Access (S3 Standard-IA)99.9% Availability Use cases: Disaster Recovery, backups

Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)

- High durability (99.999999999) in a single AZ; data lost when AZ is destroyed
- 99.5% Availability
- Use Cases: Storing secondary backup copies of on-premise data or data you can recreate

Amazon S3 Glacier Storage Classes

- Low-cost object storage meant for archiving / backup
- Pricing: price for storage + object retrieval cost

- Amazon S3 Glacier Instant Retrieval

- Millisecond retrieval, great for data accessed once a quarter
- Minimum storage duration of 90 days
- Amazon S3 Glacier Flexible Retrieval (Formerly Amazon S3 Glacier)
- Expedited (1 to 5 minutes), Standard (3 to 5 hours), Bulk (5 to 12 hours) - free
- Minimum storage duration of 90 days
- Amazon S3 Glacier Deep Archive - for long term storage
- Standard (12 hours), Bulk (48 Hours)
- Minimum Storage duration of 180 days

13.7.3 S3 Intelligent Tiering

- Small monthly monitoring and auto-tiering fee
- Moves objects automatically between Access Tiers based on usage - There are no retrieval charges in S3 Intelligent - Tiering

- Frequent Access tier (automatic): default tier

- Infrequent Access Tier (automatic): objects not accessed for 30 days
- Archive Instant Access tier (automatic): objects not accessed for 90 days
- Archive Access tier (optional): configurable from 90 days to 700+ days
- Deep Archive Access tier (optional): config from 180 days to 700+ days

S3 Storage Classes Comparision

S3 Storage Classes - Price Comparison

13.8 AWS Budgets and Cost Explorer

13.8.1 AWS Budgets

- Create budget and send alarms when costs exceeds the budget
- 4 Types of budgets: Usage, Cost, Reservation, Savings Plans
- For Reserved Instances (RI)
- Track utilisation
- Supports EC2, ElastiCache, RDS, Redshift
- Up to 5 SNS notifications per budget
- Can filter by: Service, Linkedin Account, Tag, Purchase Option, Instance Type, Region, Availability Zone, API Operations, etc......
- Same options as AWS Cost Explorer
- 2 budgets are free than $0.002 / day / budget

13.8.2 Budget Actions

- Run actions on your behalf when a budget exceeds a certain cost or usage threshold
- Supports 3 actions types
- Applying an IAM Policy to a user, group or IAM role
- Applying Service Control Policy (SCP) to an OU
- Stop EC2 or RDS Instances
- Actions can be executed automatically or require a workflow approval process
- Reduced unintentional overspending in your account.

13.8.3 Centralised Budget Management

13.8.4 DeCentralised Budget Management

13.8.5 Cost Explorer

- Visualise, understand and manage your AWS costs and usage over time
- Create custom reports that analyse cost and usage data
- Analyse you data at a high level: total costs and usage across all accounts
- Or Monthly, hourly, resource level granularity
- Choose an optimal Savings PLan (to lower prices on your bill)
- Forecast usage up to 12 months based on previous usage

13.9 AWS Compute Optimiser

- Reduce costs and improve performance by recommending optimal AWS resources for you workloads
- Helps you choose optimal configurations and right-size your workloads (over/under provisioned)
- Uses Machine Learning to analyse your resources configurations and their utilisation CloudWatch metrics
- Supported resources
- EC2 Instances
- EC2 Auto Scaling Groups
- EBS volumes
- Lambda functions
- Lower your costs by up to 25%
- Recommendations can be exported to S3

13.9.1 Computer Optimiser - CloudWatch Agent

- Needed to analyse Memory Utilisation
- Not needed for CPU, NetworkIn/Out, DiskReadOps, DiskWriteOps

13.10 EC2 Reserved Instance

- Reserved Instances in an AWS Organisation
- All accounts share the Reserved Instances and Savings Plan
- The payer account (Management account) of an organisation can turn off Reserved Instance (RI) discount and Savings Plans discount sharing for any accounts in that organisation, including the payer account.
- Renewal of Reserved Instances
- You can queue (schedule or reserve ahead of time) you reserved instances
- To renew a RI, just queue an RI purchase whenever the previous one expires

Chapter 14

Migration

14.1 Cloud Migration Strategies - The 7Rs

https://aws.amazon.com/blogs/enterprise-strategy/new-possibilities-seven-strategies-to-accelerate-your-application-migration-to-aws/

- Retire
  - Turn off things you don't need (maybe as a result of Re-architecting)
  - Helps with reducing the surface for attacks (more security)
  - Save cost, maybe up to 10% to 20%
  - Focus your attention on resource that must be maintained
- Retain
  - Do nothing for now (its still a decision to make in a Cloud Migration)
  - Security, data compliance, performance, unresolved dependencies
  - No business value to migrate, mainframe or mid-range and non-x86 Unix apps
- Relocate
  - Move apps from on-premises to its Cloud version
  - Move EC2 instances to a different VPC, AWS account or AWS region
  - Example: Transfer servers from VMWare Software-defined Data Centre (SSDC) to VMware Cloud on AWS
- Rehost ”lift and shift”
  - Simple migrations by re-hosting on AWS (applications, databases, data......)
  - Migrate machines (physical, virtual, another cloud) to AWS cloud
  - No cloud optimisations being done, applications is migrated as is
  - Cloud save as much as 30% on cost
  - Example: Migrate using AWS application migration service
- Replatform ”lift and reshape”
  - Example: migrate your database to RDS
  - Example: migrate your application to Elastic Beanstalk
  - Not changing the core architecture but leverage some Cloud optimisations
  - Save time and money by moving to a fully manged service or Serverless
- Repurchase ”drop and shop”
  - Moving to a different product while moving to the Cloud
  - Often you move to SaaS platform
  - Expensive in the short term, but quick to deploy
  - Example: CRM to Salesforce.com, HR to Workday, CMS to Drupal
- Refactor / Re-architect

- Reimagining how the application is architected using Cloud Native features
- Driven by the need of the business to add features and improve scalability, performance, security and agility
- Move from a monolithic application to micro-services
- Example: move an application to Serverless architectures, use AWS S3

14.2 Storage Gateway

- Bridge between on-premises data and cloud data
- Use cases:
  - disaster recovery
  - backup and restore
  - tiered storage
  - on-premises cache and low-latency file access
- Types of Storage Gateway
  - S3 File Gateway
  - Volume Gateway
  - Tape Gateway

Chapter 15

VPC

15.1 VPC - Basics

15.1.1 VPC

- A VPC must have a defined list of CIDR blocks, that cannot be changed
- Each CIDR within VPC: min size is /28, max size is /16 (65536 IP addresses)
- VPC is private, so only Private IP CIDR ranges are allowed

15.1.2 Subnets

- Within a VPC, defined as a CIDR that is a subset of the VPC CIDR
- AAll instances within subnets get a private IP
- First 4 IP and last one in every subnet is reserved by AWS

15.1.3 Route Tables

- Used to control where the network traffic is directed to
- Can be associated with specific subnets
- The ”most specific” routing rule is always followed (192.168.0.1/24 beats 0.0.0.0/0)

15.1.4 Internet Gateways IGW

- Helps a VPC connect to the internet, HA scales horizontally
- Acts as a NAT for instances that have a public IPv4 or public IPv6

15.1.5 Public Subnets

- Has a route table that send 0.0.0.0/0 to an IGW
- Instances must have a public IPv4 to talk the internet

15.1.6 Private Subnets

- Access internet with a NAT instance or NAT Gateway setup in a public subnet
- Must edit routes so that 0.0.0.0/0 routes traffic to the NAT

15.1.7 VPC Basics - NAT Instance

- EC2 instance you deploy in a public subnet
- Edit the route in your private subnet to route 0.0.0.0/0 to your NAT instance
- Not resilient to failure, limited bandwidth based on instance type, cheap
- Must manage failover yourself
- **Must disable Source/Destination Check (EC2 setting)**

15.1.8 VPC Basics - NAT Gateway

- Managed NAT solution, bandwidth scales automatically
- Resilient to failure within a single AZ
- Must deploy multiple NAT Gateways in multiple AZ for HA
- Has an Elastic IP, external services see the IP of the NAT gateway as the source

15.1.9 Network ACL (NACL)

- Stateless firewall defined at the subnet level, applies to all instances within
- Support for allow and deny rules
- Stateless = return traffic must be explicitly allowed by rules
- Helpful to quickly and cheaply block specific IP addresses

15.1.10 Security groups

- Applied at the instance level, only support for allow rules, no deny rules
- Stateful = return traffic is automatically allowed, regardless of rules
- Can reference other security groups in the same region (peered VPC, cross-account)

15.1.11 VPC Flow Logs

- Log internet traffic going through your VPC
- Can be defined at the VPC level, Subnet level or ENI-level
- Helpful to capture "denied internet traffic"
- Can be sent to CloudWatch Logs and Amazon S3

15.1.12 Bastion Hosts

- SSH into private EC2 instances through a public EC2 instance (bastion host)
- You must manage these instances yourself (failover, recovery)
- SSM Session Manager is a more secure way to remote control without SSH

15.1.13 IPv6 in short

- All IPv6 addresses are public, total 3.4 x $10^38addresses(vs4.3billionIPv4)ExampleCIDR : 2600 : LfL8 : 80c : a900 :: /56$

- Addresses are "random" and can't be scanned online (because too many)

15.1.14 VPC support for IPv6

- Create an IPv6 CIDR for VPC and use and IGW (support IPv6)
- Public subnet:
  - Create an instance with IPv6 support
  - Create a route table entry to ::/0 (IPv6 ”all”) to the IGW
  - Public subnet (instances cannot be reached by IPv6 but can reach IPv6)
  - Create an Egree-Only Internet Gateway in the VPC
  - Add a route table entry for the private subnet from ::/0 to the Egress-Only IGW

15.2 VPC - Peering

- Connect two VPC, privately using the AWS’ network
- Make them behave as if they were in the same network
- Must not have overlapping CIDR
- VPC Peering connection is not transitive (must be established for each VPC that need to communicate with on another)
- You can do VPC peering with another AWS account
- **You must update route tables in each VPCs subnet to ensure instances can communicate**

15.2.1 VPC Peering - Good to know

- VPC peering can work inter-region, cross account
- You can reference a security group of a peered VPC (works cross account)

15.2.2 VPC Peering - Longest Prefix Match

- VPC uses the longest prefix match to select the most specific route
- Here the longest prefix for 10.0.0.77 is 10.0.0.77/32 (route table VPC A) - (other way of saying it is ”most specific route”)

15.2.3 VPC Peering - Invalid Configurations

Overlapping CIDR for IPv4

No Transitive VPC Peering

No Edge to Edge Routing

15.3 Transit Gateway

- For having transitive peering between thousands of VPCs and on-premises, hub-and-spoke (star) connection
- Regional resource, can work cross-region
- Share cross-account using Resource Access Manager (RAM)
- You can peer Transit Gateways across regions
- Route Tables: limit which VPC can talk with other VPCs
- Works with Direct Connect Gateway, VPC connections
- Support IP Multicast (not supported by any other AWS service)
- Instances in a VPC can access a NAT gateway, NLB, PrivateLink and EFS in others VPCs attached to the AWS Transit Gateway

15.3.1 Transit Gateway - Central NAT Gateway

- The NAT gateway is shared in the Egress-VPC
- The private App VPC can access internet through the TGW
- In this example: the App VPCs cannot communicate with each other based on the TGW route table

15.3.2 Transit Gateway - Sharing through RAM

- You can use AWS resource access manager (RAM) to share a transit gateway for VPC attachments across account or across AWS Organisation

15.3.3 Transit Gatway - Use Different Route Tables to Prevent VPC from Communicating

15.3.4 Transit Gateway to Direct Connect Gateway

15.3.5 Transit Gateway - Inter and Intra Region Peering

- Billed hourly for each peering attachment, no data processing charges - If data goes cross-region through the attachment, standard charges - Specify the Transit Gateway ID

15.4 VPC Endpoints

- Endpoints allow you to connect to AWS Services using a private network instead of the public "www" network
- The scale horizontally and are redundant
- No more IGW, NAT, etc... to access AWS Services
- VPC Endpoint Gateway (S3 and DynamoDB)
- VPC Endpoint Interface (all services)
- In case of issues:
- Check DNS Setting Resolution in your VPC
- Check Route Tables

15.4.1 VPC Endpoint Gateway

- Only works for S3 and DynamoDB, must create one gateway per VPC
- Must update route tables entries
- Gateway is defined at the VPC level
- DNS resolution must be enabled in the VPC
- The same public hostname for S3 can be used
- Gateway endpoint cannot be extended out of a VPC (VPN. DX, TGW, peering)

15.4.2 VPC Endpoints Interface

- Provision an ENI that will have a private endpoint interface hostname
- Leverage security groups security
- Private DNS (setting when you create the endpoint)
- The public hostname of a service will resolve the private Endpoint interface hostname
- VPC Setting: "Enabled DNS hostnames" and "Enable DNS Support" must be "true"
- Example for Athena:
- Interface can be accessed from Direct Connect and Site-to-Site VPN

15.4.3 VPC Endpoint Policies

- Endpoint policies are JSON documents to control access to services
- Does not override or replace IAM user policies or service-specific policies (such as S3 bucket policies)
- Note: the IAM user can still use other SQS API from outside the VPC endpoint
- You could add an SQS queue policy to deny any action not done through the VPC endpoint

15.4.4 VPC Endpoint Policy and S3 bucket policy

- VPC Endpoint Policy to restrict access to bucket "my$_{s}ecure_{b}ucket$"

- VPC Endpoint to allow access to Amazon Linux 2 repositories
- S3 bucket policy may have
  - Condition: "aws:sourceVpce":"vpce-la2b3c4d" to Deny any traffic that doesn't come from a specific VPC endpoint (more secure)
  - Condition: "aws:sourceVpc":"vpc-111bbb22" for a specific VPC
  - The aws:sourceVpc condition only works for VPC endpoints in case you have multiple endpoints and want to manage access to your S3 buckets for all yyou endpoints
  - The S3 bucket policies policies can restrict access only from a specific public IP address or an elastic IP address. **You can't restrict based on private IP**
  - Therefore aws:Source condition doesn't apply for VPC endpoints

Example S3 bucket policies

- S3 bucket policy to restrict to specific VPC endpoint
- S3 bucket policy to restrict to an entire VPC (multiple VPC Endpoints)

15.4.5 VPC Endpoint Policies for S3 Troubleshooting

15.5 PrivateLink

- Most secure and scalable way to expose a service to 1000s of VPC (own or other accounts)
- Does not require VPC peering, internet gateway NAT, route tables...
- Requires a network load balancer (Service VPC) and ENI (CustomerVPC)
- If the Network Load Balancer (NLB) is in multiple AZ and the ENI in multiple AZ the solution is fault tolerant (good)

15.5.1 PrivateLink for Amazon S3 with Direct Connect

15.5.2 VPC Endpoints / PrivateLink and VPC peering

15.5.3 Site to Site VPN (AWS Managed VPN)

- on-premises
  - setup a software of hardware VPN appliance to your on-premises network
  - The on-premises VPN should be accessible using a public IP
- AWS-side
  - Setup a Virtual Private Gateway (VGW) and attach to you VPC
  - Setup a Customer Gateway to point the on-premises VPN appliance
  - Two VPN connections (tunnels) are created for redundancy,encrypted using IPSec
  - Can optionally accelerate it using Global Accelerator (for worldwide networks)

15.5.4 Route Propagation in Site-to-Site VPN

Static Routing

Create static route in corporate data centre for 10.0.0.1/24 through the CGW

Create static route in AWS for 10.3.0.0/20 through the VGW

Dynamic Routing (BGP)

Using BGP (Border Gateway Protocol) to share routes automatically (eBGP for internet)

We don't need to update the routing tables, it will be done for us dynamically

Just need to specify the ASN (Autonomous System Number) of the CGW and VGW

15.5.5 Site to Site VPN and Internet Access

- Not Okay (blocked by NAT Gateway restrictions)
- Okay (self manage NAT instance - more control)
- Okay (alternative to NAT instances / gateway)

15.5.6 AWS VPN CloudHub

- Can connect up to 10 customer gateway for each Virtual Private Gateway (VGW)
- Low cost hub-and-spoke model for primary or secondary network connectivity between locations
- Provide secure communication between sites, if you have multiple VPN connections
- It's a VPN connection so it goes over the public internet
- Can be a failover connection between your on-premises locations

15.5.7 VPN to multiple VPC

- For VPN-based customers, AWS recommends creating a separate VPN connection for each customer VPC
- Direct Connect is recommended because it has Direct Connect Gateway

15.5.8 Shared Services VPN

- Create a VPN connection between on-premises and shared service VPC
- Replicate services, applications, databases between on-premises and the Shared Services VPC or deploy proxies in the shared service VPC
- Do VPC peering between the VPC and the shared service VPC
- VPCs can directly access the Shared Service VPC services and do not need VPN connections to on-premises

15.6 AWS S2S VPN

15.7 AWS Client VPN

- Connect from your computer using OpenVPN to your private network in AWS and on-premises

15.7.1 Client VPN - Peered VPC

- Client VPN is compatible with VPC peering

15.7.2 Client VPN - Access to On-Premises

- Access On-Premises resources through AWS with ClientVPN

15.7.3 Client VPN - Internet Access

15.7.4 Client VPN - Transit Gateway

15.8 Direct Connect

- Provides a dedicated private connection from a remote network to your VPC
- Dedicated connection must be setup between you DC and AWS Direct Connection location
- More expensive than running a VPN solution
- Private access to AWS service through VIF
- Bypass ISP, reduce network cost, increase bandwidth and stability
- Not redundant by default (must setup a failover DX or VPN)

15.8.1 Direct Connect - Virtual Interfaces (VIF)

- Public VIF - connect to Public AWS Endpoints (S3 buckets, EC2 service, anything AWS ...)
- Private VIF - connect to resources in your VPC (EC2 instances, ALB, ...)
- Transit Virtual Interface - connect to resource in a VPC using a Transit Gateway
- VPC Interface Endpoints can be accessed through Private VIF

15.8.2 Direct Connect Diagram

15.8.3 Direct Connect - Connection Types

- Dedicated Connections: 1 Gbps, 10 Gbps, 100 Gbps capacity
- Physical ethernet port dedicated to a customer
- Request made to AWS first, the completed by AWS Direct Connect Partners
- Hosted Connections: 50Mbps, 500Mbps to 10Gbps
- Connection requests are made via AWS Direct Connect Partners
- Capacity can be added or removed on demand
- 1,2,5,10 Gbps available at select AWS Direct Connect Partners
- Lead times are often longer than 1 month to establish a new connection

15.8.4 Direct Connect - Encryption

- Data in transit is not encrypted but is private
- AWS Direct Connect + VPN provides an IPsec-encrypted private connection
- VPN over Direct Connect connection Uses Public VIF
- Good for an extra level of security, but slightly more complex to put in place

15.8.5 Direct Connect - Link Aggregation Groups (LAG)

- Get increased speed and failover by summing existing DX connections in a logical one
- Can aggregate up to 4 connections (active-active mode)
- Can add connections over time to the LAG
- All connection in the LAG:
- Must be dedicated connections
- Must have the same bandwidth
- Must terminate at the same AWS Direct Connect Endpoint
- Can set a minimum number of connections for the LAG to function

15.8.6 Direct Connect Gateway

- If you want to setup a Direct Connect to one or more VPC in many different regions (same/ cross account), you must use a Direct Connect Gateway

15.8.7 Site-to-site Active-Active Connection

15.8.8 Direct Connect - High Availability

15.8.9 Direct Connect Gateway - SiteLink

- Allows you to send data from on Direct Connect location to another
- Bypassing AWS regions
- Use cases: create private network connections between on-premises data centres by connecting them to Direct Connect locations
- Data is sent over the fastest path between Direct Connect locations

15.9 On-Premise Redundant Connections

15.10 VPC Flow Logs

- Capture information about IP traffic going into your interfaces:
- VPC flow logs
- Subnet flow logs
- Elastic Network Interface (ENI) Flow Logs
- Help to monitor and troubleshoot connectivity issues
- Flow logs data can go to S3, CloudWatch Logs and Kinesis Data Firehose
- Captures network information from AWS managed interfaces too: ELB, RDS, ElastiCache, Redshift, WorkSpaces, NATGW, Transit Gateway.....

15.10.1 VPC Flow Logs Syntax

- srcaddr and dstaddr - help identify problematic IP
- srcport and dstport - help identify problematics ports
- Action - success or failure of the request due to Security Group / NACL
- Can be used for analytics on usage patterns, or malicious behaviour
- Query VPC flow logs using Athena on S3 or CloudWatch logs Insights
- Flow logs examples: https://docs.aws.amazon.com/vpc/latest/userguide/flow-logsrecords- examples.html

15.10.2 VPC Flow Logs - Troubleshoot SG and NACL issues

Look at the ”Action” field

Incoming Requests

- Inbound REJECT =¿ NACL or SG
- Inbound ACCEPT, Outbound Reject =¿ NACL

Outgoing Requests

- Outbound REJECT =¿ NACL or SG
- Outbound ACCEPT, Inbound REJECT =¿ NACL

15.10.3 VPC Flow Logs - Architectures

15.10.4 VPC Flow Logs - with NAT Gateway

- My Virtual Private Cloud (VPC) flow logs show Action = ACCEPT for inbound traffic coming from public IP addresses. However, my understand of network address translation (NAT) gateway was that they don’t accept traffic from the internet. Is my NAT gateway accepting inbound traffic from the internet
- Inbound traffic is permitted by Security Group or NACLs
- Traffic? isn’t permitted by the NAT gateway, it’s dropped
- To confirm run the following query in CloudWatch Log Group
- Make ’xxx.xxx’ the first two octets of your VPC CIDR
- Replace Public IP with the IP you see in logs
- You will see traffic on the Private IP of the NAT Gateway but nowhere else: traffic was unsolicited and then dropped

15.11 AWS Network Firewall

15.11.1 Network Protection on AWS

- To protect network on AWS, we’ve seen
- Network Access Control Lists (NACLs)
- Amazon VPC security groups
- AWS WAF (protect against malicious requests)
- AWS Shield and AWS Shield Advanced
- AWS Firewall Manager (to manage them across accounts)
- But what if we want to protect in a sophisticated way out entire VPC?

15.11.2 AWS Network Firewall

- Protected your entire Amazon VPC
- From Layer 3 to Layer 7 protection
- Any direction, you can inspect
- VPC to VPC traffic
- Outbound to internet
- Inbound from internet
- To / from Direct Connect and Site-to-Site VPN
- Internally, the AWS Network Firewall uses the AWS Gateway Load Balancer
- Rules can be centrally managed cross-account by AWS Firewall Manager to apply to many VPCs

15.11.3 Network Firewall - Fine Grained Controls

- Supports 1000s of rules
- IP and port - example: 10,000s of IPs filtering
- Protocol - example: block the SMB protocol for outbound communications
- Stateful domain list rule groups, only allow outbound traffic to *.mycorp.com or third-party software repo
- General pattern matching using regex
- Traffic filtering: Allow drop or alert for the traffic that matches the rules
- Active flow inspection to protect against network threats with intrusion-prevention capabilities (like Gateway Load Balancer, but all managed by AWS)
- Send logs or rule matches to Amazon S3, CloudWatch Logs, Kinesis Data Firehose

15.11.4 AWS Network Firewall Architecture

Chapter 16

Machine Learning

16.1 Rekognition Overview

- Find objects, people, text, scenes in images and videos using ML
- Facial analysis and facial search to do user verification, people counting.
- Create a database of ”familiar faces” or compare against celebrities
- Use cases:
- Labeling
- Content Moderation
- Text Detection
- Face Detection and Analysis (gender, age range, emotions....)
- Face Search and Verification
- Celebrity Recognition
- Pathing (ex: for sports game analysis)

16.1.1 Amazon Rekognition - Content Moderation

- Detect content that is inappropriate, unwanted or offensive (image and videos)
- Used in social media, broadcast media, advertising and e-commerce situation to create a safer user experience
- Set a Minimum Confidence Threshold for items that will be flagged
- Flag sensitive content for manual review in Amazon Augmented AI (A2I)
- Help comply with regulations

16.2 Transcribe Overview

- Automatically convert speech to text.
- Uses a deep learning process called automatic speech recognition (ASR) to convert speech to text quickly and accurately.
- Automatically remove Personally Identifiable Information (PII) using Redaction.
- Supports Automatic Language Identification for multilingual audio
- Use cases:
  - transcribe customer service calls
  - automate closed captioning and subtitling
  - generate metadata for media assets to create a fully searchable archive

16.3 Polly Overview

- Turn text into lifelike speech using deep learning
- Allowing you to create applications that talk

16.3.1 Amazon Polly - Lexicon & SSML

- Customise the pronunciation of words with Pronunciation lexicons
- Stylized words: St3ph4ne =¿ ”Stephane”
- Acronyms: AWS =¿ ”Amazon Web Services”
- Upload the lexicons and use them in the SynthesizeSpeech operation.
- Generate speech from plain text or from documents marked up with Speech Synthesis Markup Language (SSML) - enables more customisation
- Emphasising specific words or phrases.
- Using phonetic pronunciation.
- Including breathing sounds, whispering.
- Using the Newscaster speaking style

16.4 Translate Overview

- Natural and accurate language translation
- Allows you to localise content - such as websites and applications - for international users and to easily translate large volumes of text efficiently.

16.5 Lex + Connect Overview

- Amazon Lex: (same technology that powers Alexa)
- Automatic Speech Recognition (ASR) to convert speech to text
- Natural Language Understanding to recognise the intent of text, callers
- Helps build chatbots and call centre bots
- Amazon Connect:
- Receive calls, create contact flows, cloud-based virtual contact centre
- Can integrate with other CRM systems of AWS
- No upfront payments, 80% cheaper than traditional contact center solutions *Hmmmm......*

16.6 Comprehend Overview

- For Natural Language Processing - NLP
- Fully managed and serverless service
- Uses machine learning to find insights and relationships in text
- Language of the text
- Extracts key phrases, places, people, brands or events
- Understands how positive or negative the text is
- Analyses text using tokenization and parts of speech
- Automatically organises a collection of text files by topic
- Sample use cases:
- Analyse customer interactions (emails) to find what leads to a positive or negative experience
- Create and groups articles by topics that Comprehend will uncover

16.7 Comprehend Medical Overview

- Amazon Comprehend Medical detects and returns useful information in unstructured clinical text:
- Physician's notes
- Discharge summaries
- Test results
- Case notes
- Uses NLP to detect Protected Health Information (PHI) - DetectPHI API
- Store your documents in Amazon S3, analyse real-time data with Kinesis Data Firehose or use Amazon Transcribe to transcribe patient narratives into text that can be analysed by Amazon Comprehend Medical.

16.8 SageMaker Overview

- Fully managed service for developers / data scientists to build ML models
- Typically difficult to do all processes in one place + provision servers

16.9 Kendra Overview

- Fully managed document search service powered by Machine Learning
- Extract answers from within a document (text, pdf, HTML, PowerPoint, MS Word, FAQs)
- Natural language search capabilities
- Learn from user interactions / feedback to promote preferred results (Incremental Learning)
- Ability to manually fine-tune search results (importance of data, freshness, customer)

16.10 Personalise Overview

- Fully managed ML-service to build apps with real-time personalised recommendations
- Example: personalised product recommendations/re-ranking, customised direct marketing
- Example: User bought gardening tools, provide recommendations on the next one to buy.
- Same technology used by Amazon.com
- Integrates into existing websites, applications, SMS, email marketing systems.........
- Implement in days, not months (don't need to build, train and deploy ML solutions)
- Use cases; retail stores, media and entertainment

16.11 Textract Overview

- Automatically extracts text, handwriting and data from any scanned documents using AI and ML
- Extract data from forms and tables
- Read and process any type of document (PDFs, images)
- Use cases:
  - Financial Services (e.g, invoices, financial reports)
  - Healthcare (e.g medical records, insurance claims)
  - Public Sector (e.g tax forms, ID documents, passports)

16.12 Machine Learning Summary

- Rekognition - face detection, labeling, celebrity recognition
- Transcribe - audio to text (ex: subtitles)
- Polly - text to audio

- Translate - translations
- Lex - build conversational bots - chatbots
- Comprehend - natural language processing
- SageMaker - Machine learning for every developer and data scientist
- Kendra - ML-powered search engine
- Personalise - real-time personalised recommendations
- Textract - detect text and data in documents

Chapter 17

Other Services

17.1 Other Services

17.2 IMPORTANT: CodeCommit Discontinuation

17.3 CI/CD

- Developer push the code to a commit repository
- A testing build server check the code as it's pushed (codebuild / jenkins CI/ etc.....)
- The developer gets feedback about the tests anc heck that have passed / failed
- Find bugs early fix bugs
- Deliver faster as the code is tested
- Deploy often
- Happier developers as they're unblocked
- Ensure that the software can be released reliablity whenever needed
- Ensures deployments happen often and are quick
- Shift ways from one release every 3 months to 5 relases a day
- That usually means automated deployment
  - CodeDeploy
  - Jenkins Cd
  - Spinnaker
  - Etc.....

17.3.1 Technology Stack for CICD

17.3.2 CICD Architecture

17.3.3 CodeCommit Trigger for AWS Lambda

- Every push to CodeCommit can trigger a Lambda function
- The Lambda function can scan for leaked AWS credential on every code push, and disable them automatically to remedy the issue

17.3.4 Good to know CICD

- You can use a manual approval stage in CodePipeline
- Running unit tests CodeCommit + CodeBuild + Code Pipeline
- Build and Store Docker Images: CodeBuild ECR
- Automated CloudFormation deployment: CodePipeline

17.3.5 CodePipeline - GitHub integration

17.4 Amazon CodeGuru

- An ML powered service for automated code reviews and application performance recommendations
- Provides two functionalities
  - CodeGuru Reviewer : automated code reviews for static code analysis (development)
  - CodeGuru Profiler : visibility / recommendations about application performance during runtime (production)

17.4.1 Amazon CodeGuru Reviewer

- Identify critical issues, security vulnerability and hard-to-find bugs
- Examples: common coding best practices, resource leaks, security detection, input validation
- Uses machine learning and automated reasoning
- Hard learned lessons across million of code reviews on 1000s of open source amazon repositories
- Supports Java and Python
- Integrations with GitHub, Bitbucket and AWS CodeCommit

17.4.2 Amazon CodeGuru Profiler

- Helps understand the runtime behaviour of your application
- Example: identify if you application is consuming excessive CPU capacity on a logging routine
- Features:
  - Identify and remove code inefficiencies
  - Improve application performance (e.g reduce CPU utilisation)
  - Decrease compute costs
  - Provides heap summary (identify which object using up memory)
  - Anomaly detection
- Support applications running on AWS or on-premise
- Minimal overhead on application

17.5 Alexa for Business, Lex and Connect

- Alexa for Business
  - Use Alexa to help employees be more productive in meeting rooms and their desk
  - Measure and increase the utilisation of meeting rooms in their workplace
- Amazon Lex: (Same technology that powers Alexa)
  - Automatic Speech Recognition (ASR) to convert speech to text
  - Natural language Understanding to recognise the intent of text, callers
  - Helps build chatbots, call centre bots
- Amazon Connect:
  - Receive calls, create contact flows, cloud-based virtual contact centre
  - Can integrate with other CRM systems or AWS

17.6 Kinesis Video Streams

- One video per streaming device (producers)
  - Security cameras,body worn camera, smartphone
  - Can use a kinesis video streams producer library
- Underlying data is stored in S3 (but we don't have access to it)
- Cannot output the stream data to S3 (must build customer solution)
- Consumers
  - Consumed by EC2 instances for real time analysis, or in batch
  - Can leverage the Kinesis Video Stream Parser Library
  - Integration with AWS Rekognition for facial detection

17.6.1 Video Streaming and Rekognition

17.7 AWS WorkSpaces

- Managed, Secure Cloud Desktop
- Great to eliminate management of on-premises VDI (Virtual Desktop Infrastructure)
- Pricing is either on-demand (pay per hour) or monthly subscription
- Secure, Encrypted, Network Isolation
- Integrated with Microsoft Active Directory
- Workspaces Application Manged (WAM)

- Deploy and Manage applications are virtualized application containers
- Provision at scale and keep the application updated using WAM

- Windows updates
  - By default, Amazon Workspaces are configured to install software updates
  - Amazon WorkSpaces with Windows will have Window Update turned on
  - You have full control over the Windows Update Frequency

- Maintenance Windows
  - Updates are installed during maintenance windows (you define them)
  - Always On WorkSpaces: default is from 00h00 to 04h00 in Sunday morning
  - AutoStop Workspaces: automatically start once a month to install updates
  - Manual maintenance: you define your windows and perform maintenance

17.7.1 Amazon Workspaces - Cross Region Redirection

17.7.2 Amazon WorkSpaces IP Access Control Groups

- Similar to security groups for Amazon WorkSpaces
- List of IP addresses / CIDR address ranges that users are authorized to connect from
- If users access WorkSpaces through VPN or NAT, the IP Access Control Group must authorise the public IP of these

17.8 Amazon AppStream 2.0

- Desktop Application Streaming Service
- Deliver to any computer without acquiring provisioning infrastructure
- The application is delivered from within a web browser

17.8.1 Amazon AppStream 2.0 vs WorkSpaces

- Workspaces
  - Fully managed VDI and desktop available
  - The users connect to VDI and open native or WAM applications
  - Workspaces are on-demand or always on

- AppStream 2.0
  - Stream a desktop application to web browsers (no need to connect to a VDI)
  - Works with any device (that has a browser)
  - Allow to configure an instance type per application type (CPU, RAM, GPU)

17.9 AWS Device Farm

- Application testing service for your mobile and web applications
- Test across real browsers and real mobile devices
- Fully automated using framework
- Improve the quality of web and mobile apps
- Generates video and logs to document the issues encountered
- Can remote log-in to device for debugging

17.10 Amazon Macie

- Amazon Macie is a fully managed data security and data privacy service that uses machine learning and pattern matching to discover and protect yoiur sensitive data in AWS
- Macie helps identify and alert you to sensitive daata such as pesonally identifiable PII

17.11 Amazon SES

17.11.1 Amazon Simple Email Service (Amazon SES)

- Fully managed service to send emails securely, globally and at scale
- Allows inbound / outbound emails
- Repution dashboard, performance insights, anti-spam feedback
- Provides statistics such as email deliveries, bounces, feedback loop results, email open
- Supports DomainKeys Identified Mail (DKIM) and Sender Policy Framework (SPF)
- Flexible IP deployment: shared dedicated and customer-owned IPs
- Send emails using your application using AWS console, APIs or SMTP
- Use cases: transactional, marketing and bulk email communications

17.11.2 Amazon SES - Configuration Sets

- Configure sets help you customise and analyse your email send events
- Event destination: Kinesis Data Firehose: receives metrics (number of sends, deliveries, opens, clicks, bounces and complaints) for each email - SNS: for immediate feedback on bounce and compliant information
- IP pool management: use IP pools to send particular types of emails

17.12 Amazon Pinpoint

- Scalable 2-way (outbound/ inbound) marketing communications service
- Supports email, SMS, push, voice and in-app messaging
- Ability to segment and personalise messages with the right content to customers
- Possibility to receive replies
- Scales to billions of messages per day
- Use cases: run campaigns by sending marketing, bulk, transactional SMS messages
- Versus Amazon SNS or Amazon SES
  - In SNS and SES you managed each messages' audience content and delivery schedule
  - In Amazon Pinpoint, you create message templates, delivery schedules, highly targeted segments and full campaigns

17.13 EC2 Image Builder

- Used to automate the creation of Virtual Machines or container images
- =¿ Automate the creation, maintain, validate and test EC2 AMIs
- Can be run on a schedule (weekly, whenever packages are updated etc,.......)
- Free service (only pay for the underlying resources)
- Can publish AMI to multiple regions and multiple accounts

17.13.1 EC2 Image Builder - CICD Architecture

17.14 Other Services Summary

17.14.1 AWS IoT Core

- IoT stands for "Internet of Things" - the network of internet-connected devices that are able to collect and transfer data
- AWS IoT Core allows you to easily connect IoT devices to the AWS cloud
- Serverless, secure and scalable to billions of devices and trillions of messages
- Integrates with a lot of AWS services (Lambda, S3, SageMaker, etc)
- Build IoT application that gather, process,analyse and act on data

17.14.2 IoT - Integrations

17.14.3 IoT - Kinesis Data Firehose